

# Backup — Z4 full results gallery

every diagnostic and result figure of the pipeline, in processing order

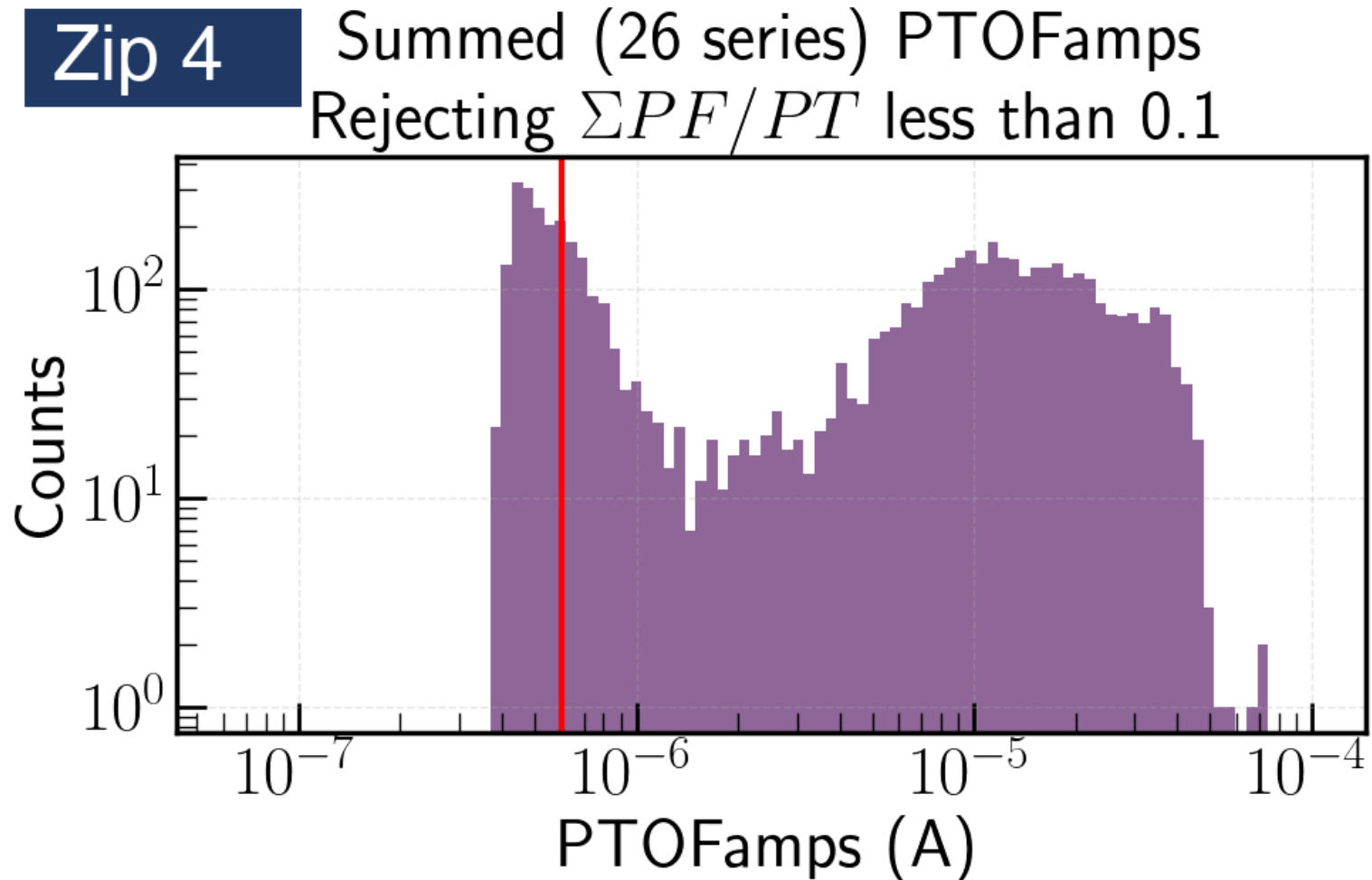
---

- Dataset: the K-line selection window
- Example grids: LP + fit and raw vs fit, event  $\times$  channel
- Fit diagnostics: NRMSE, fitted pretrigger,  $\tau_{\text{rise}}$  /  $\tau_{\text{fall}}$  distributions
- Aligned fans: no cut  $\rightarrow$   $\text{NRMSE} \leq 0.4 \rightarrow + \tau_{\text{rise}} \leq 0.3$  ms; kept vs rejected; measured traces
- Special populations: NRMSE-rejected events, well-fit slow-rise events
- Templates: PCA nxm0–4 per channel, summed PT / PS1 / PS2

# Z4 · K-line selection window

figure: kline\_zip4.png

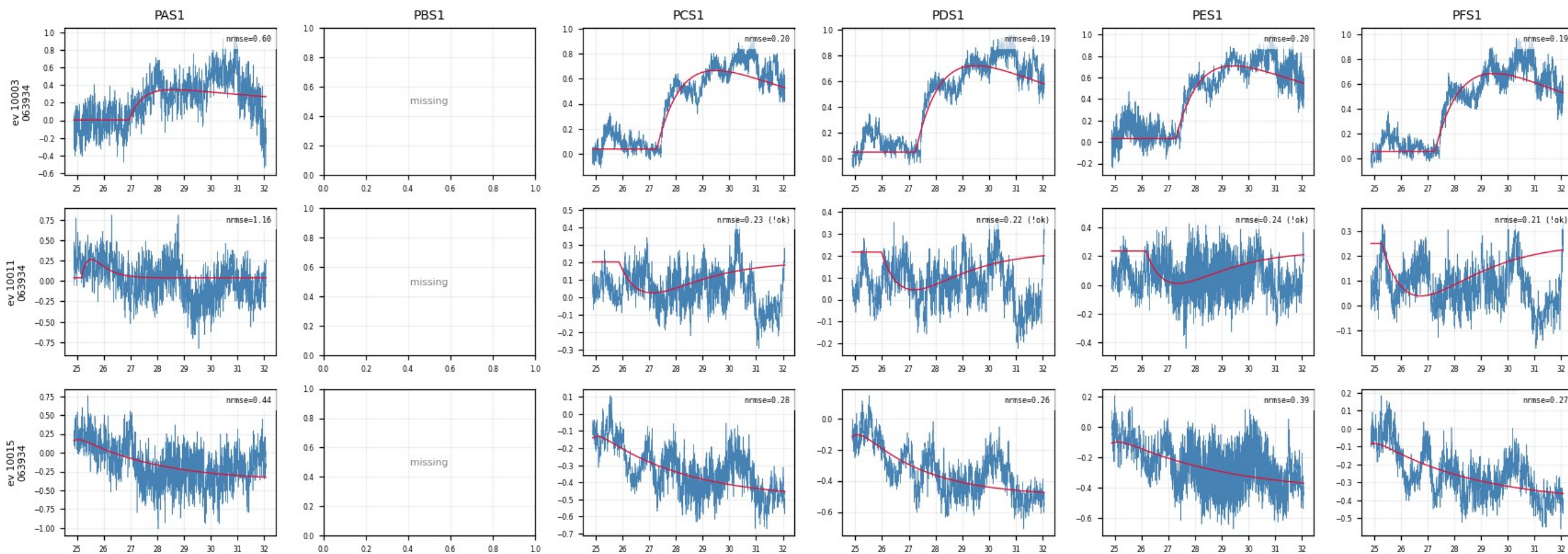
Summed PTOFamps spectrum of Z4; red line = Prof. Saab's 10.37 keV K-line position; the  $\times/\div 1.35$  window around it is the event selection; everything inside is cached raw.



# Z4 · fit\_examples — LP trace vs 2-exp fit (1/10)

figure: zip4\_fit\_examples.png — event rows 1–3 of 15, left half of the channels

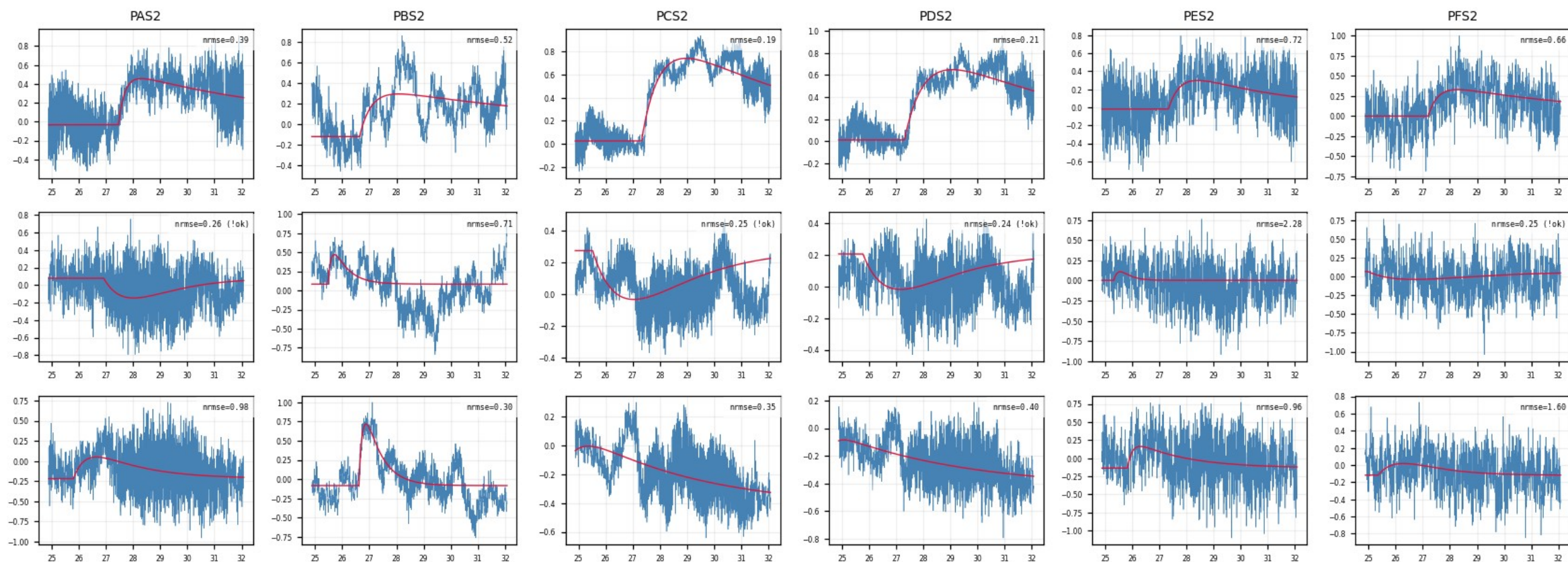
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



# Z4 · fit\_examples — LP trace vs 2-exp fit (2/10)

figure: zip4\_fit\_examples.png — event rows 1–3 of 15, right half of the channels

First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.

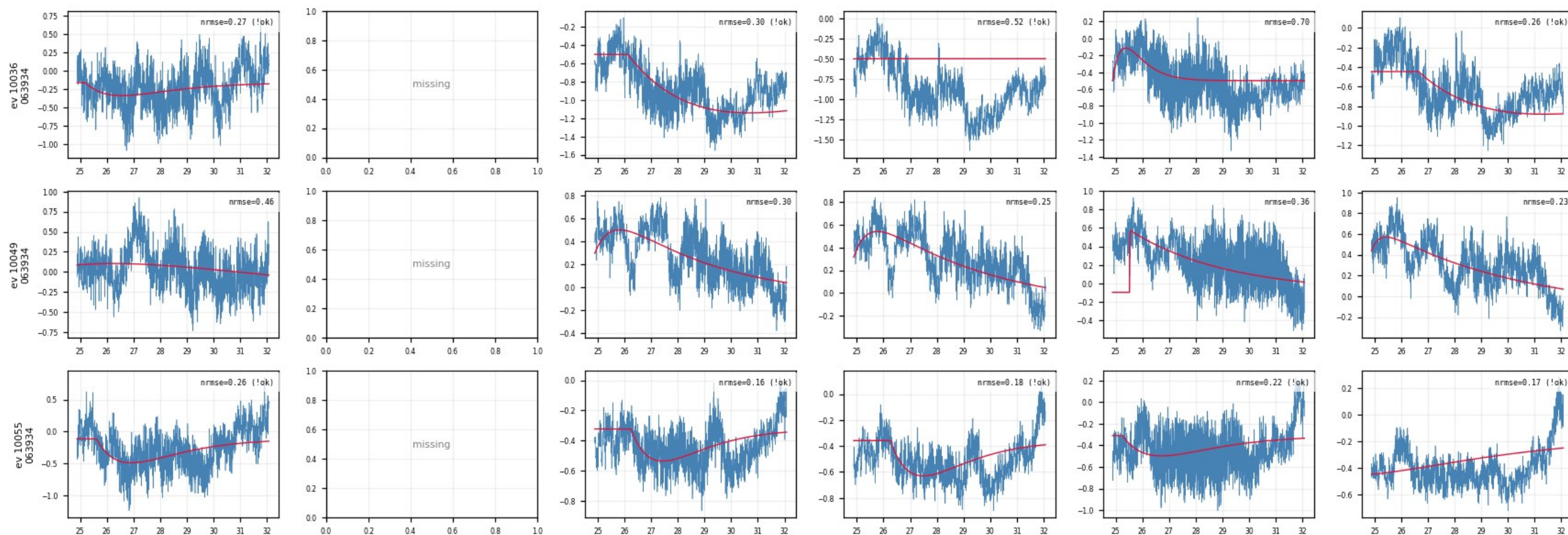




# Z4 · fit\_examples — LP trace vs 2-exp fit (3/10)

figure: zip4\_fit\_examples.png — event rows 4–6 of 15, left half of the channels

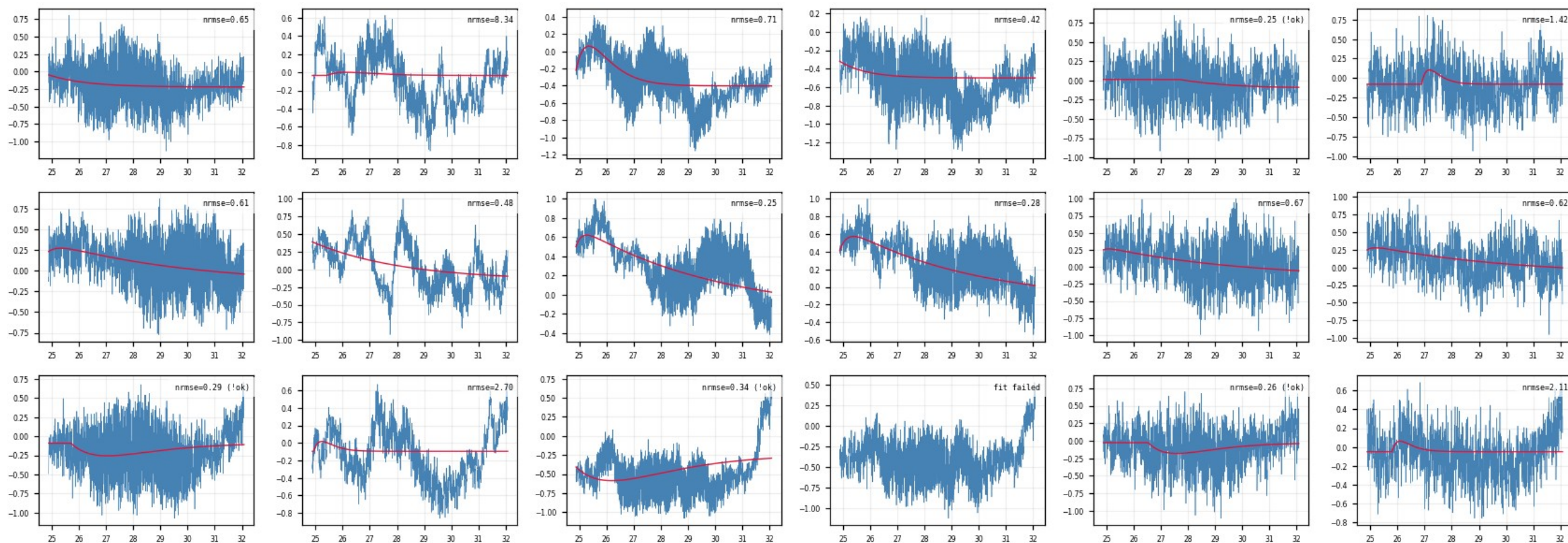
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



# Z4 · fit\_examples — LP trace vs 2-exp fit (4/10)

figure: zip4\_fit\_examples.png — event rows 4–6 of 15, right half of the channels

First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



## Z4 · fit\_examples — LP trace vs 2-exp fit (5/10)

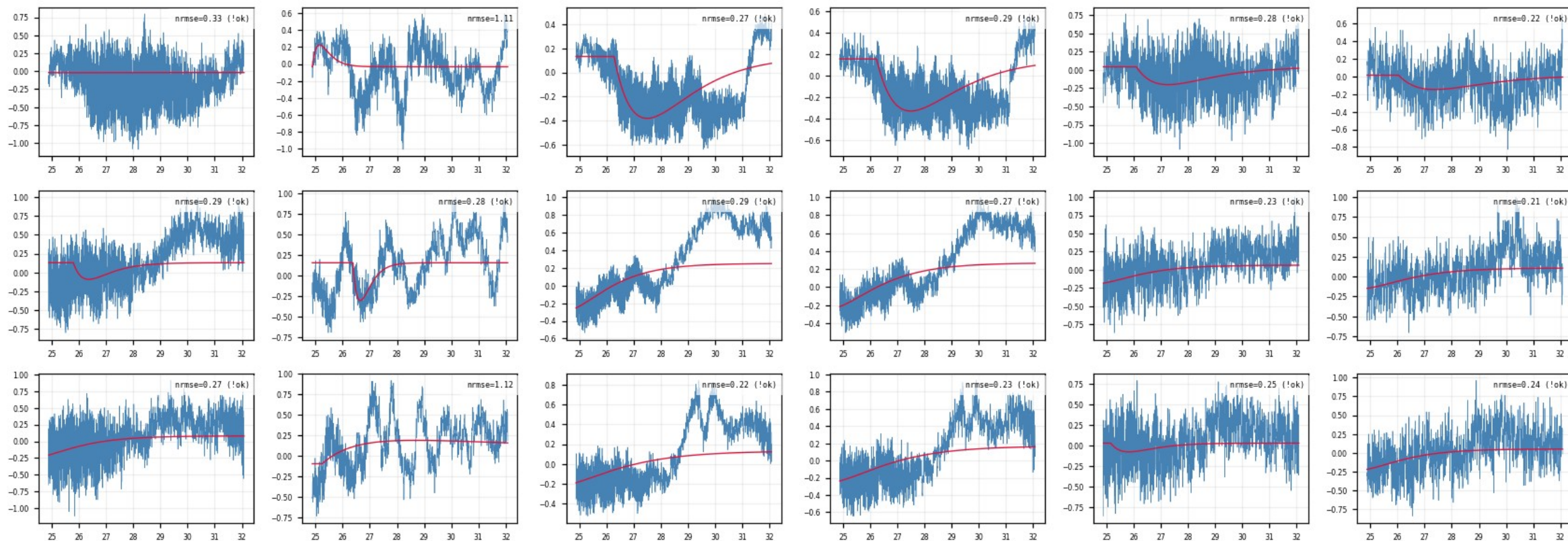
First 15 cached events  $\times$  all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



# Z4 · fit\_examples — LP trace vs 2-exp fit (6/10)

figure: zip4\_fit\_examples.png — event rows 7–9 of 15, right half of the channels

First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.

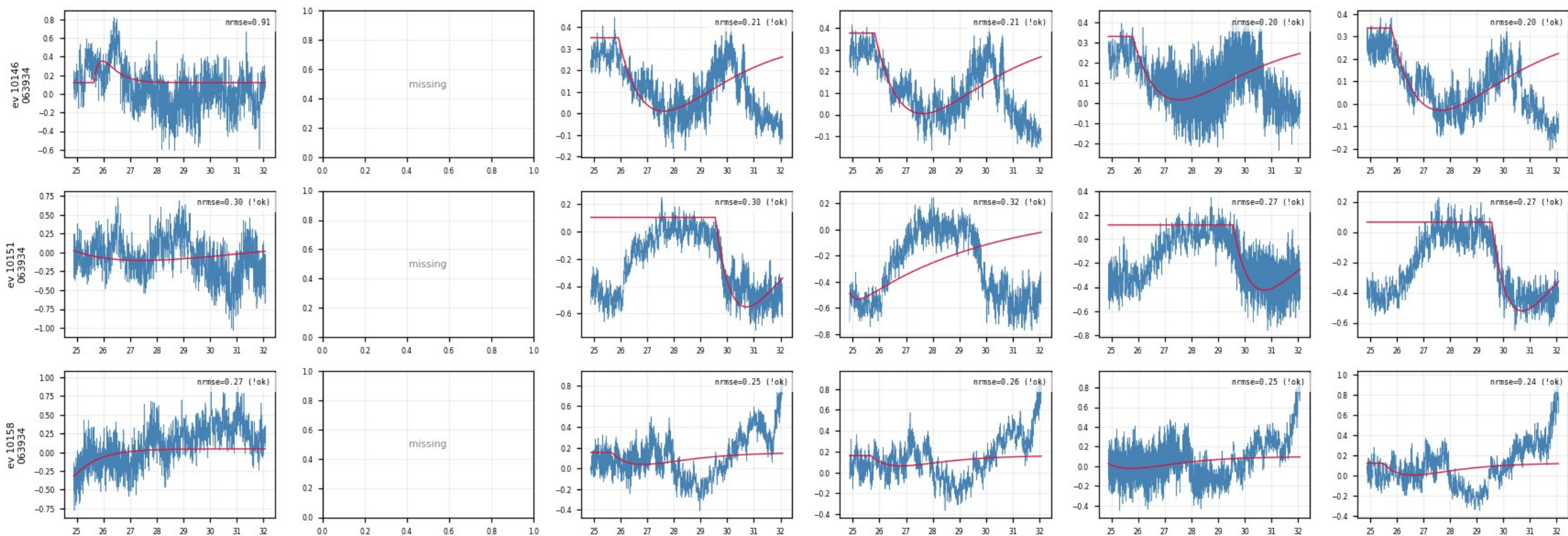




# Z4 · fit\_examples — LP trace vs 2-exp fit (7/10)

figure: zip4\_fit\_examples.png — event rows 10–12 of 15, left half of the channels

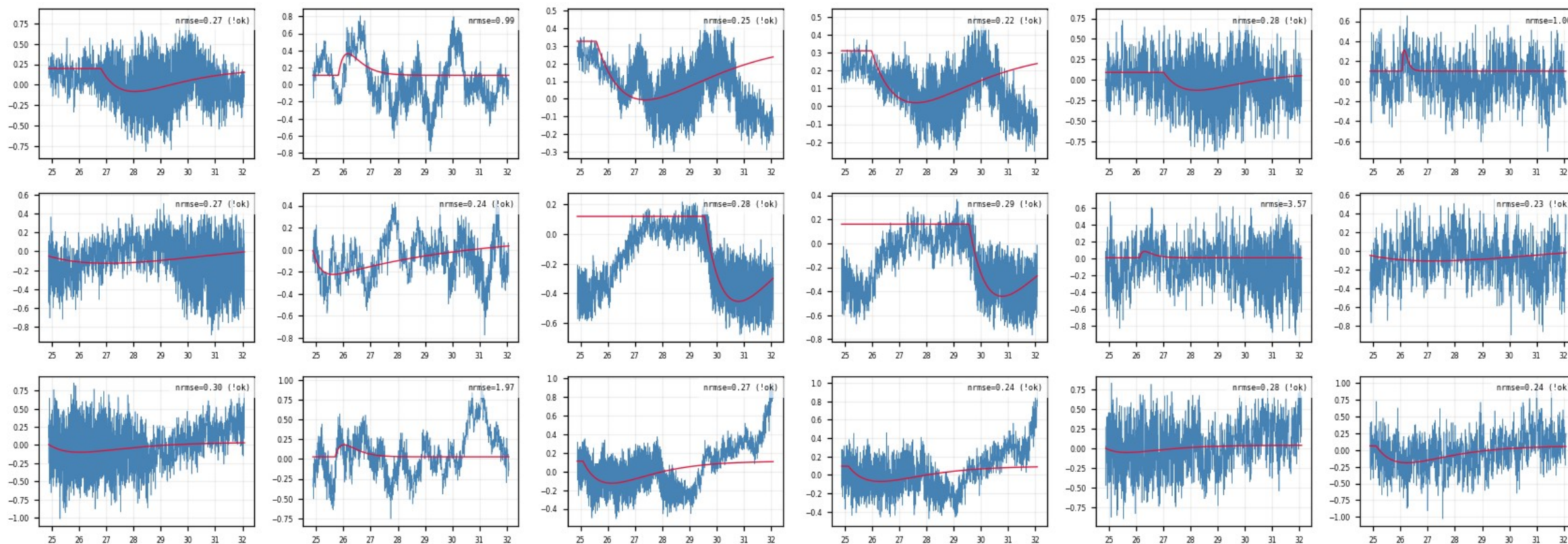
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



# Z4 · fit\_examples — LP trace vs 2-exp fit (8/10)

figure: zip4\_fit\_examples.png — event rows 10–12 of 15, right half of the channels

First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.





## Z4 · fit\_examples — LP trace vs 2-exp fit (9/10)

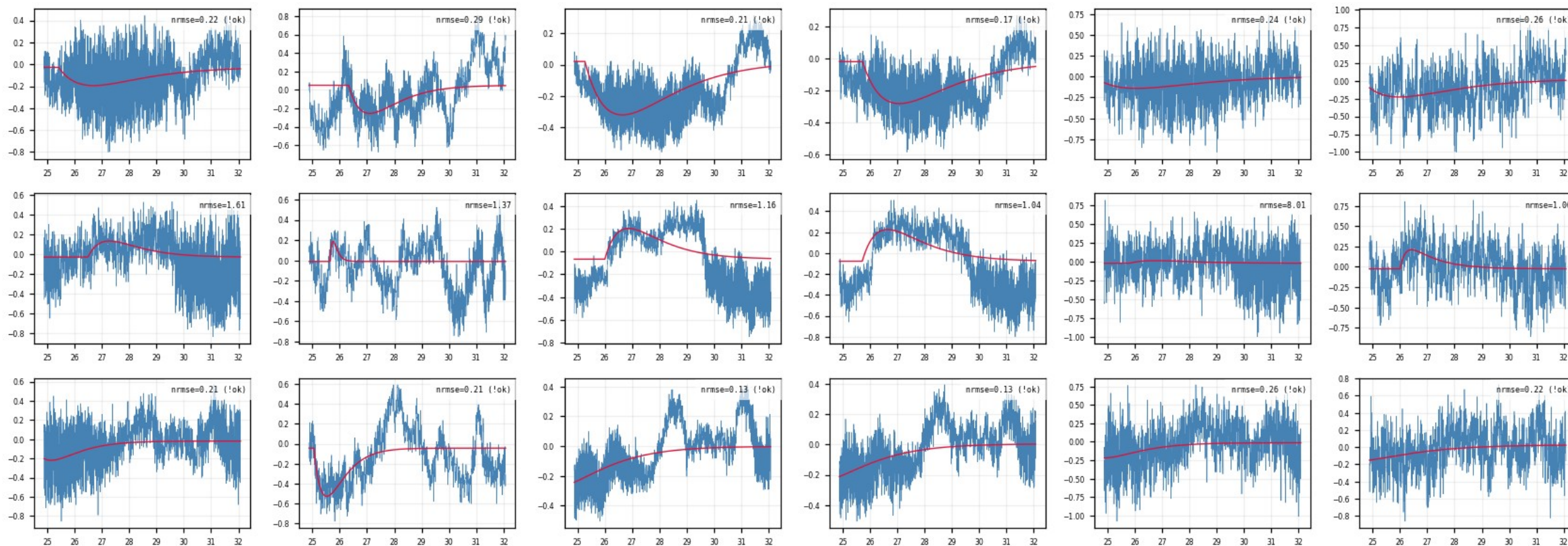
First 15 cached events  $\times$  all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



# Z4 · fit\_examples — LP trace vs 2-exp fit (10/10)

figure: zip4\_fit\_examples.png — event rows 13–15 of 15, right half of the channels

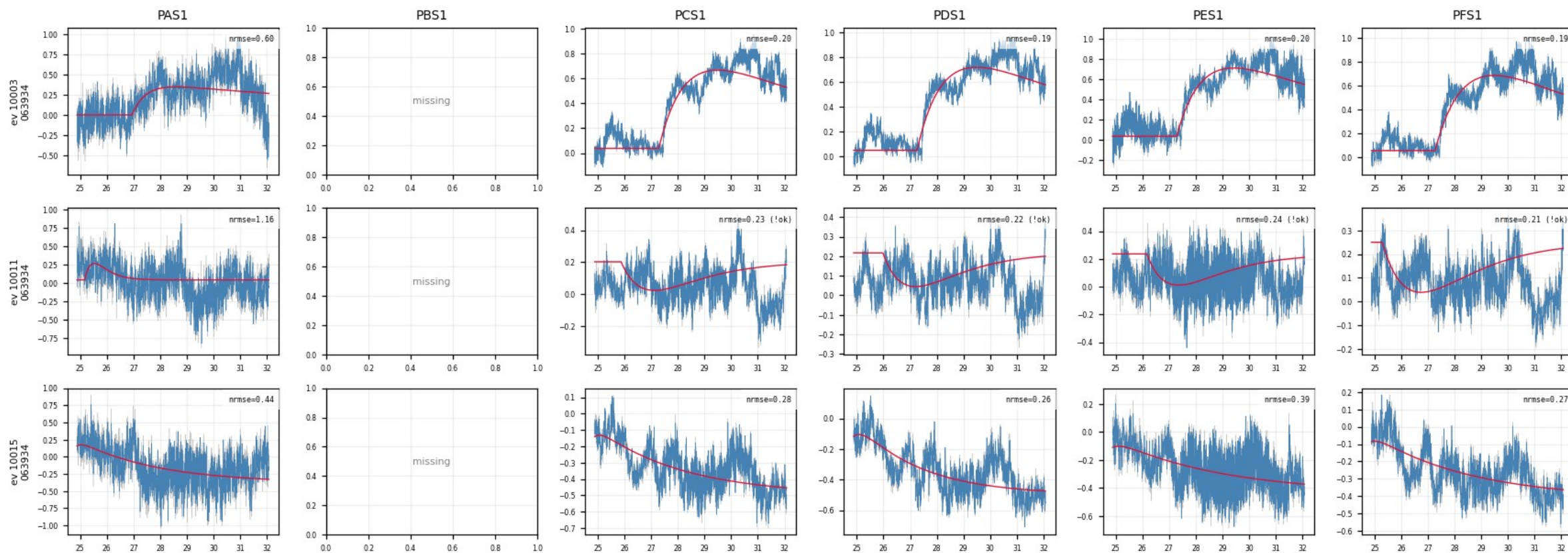
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



# Z4 · raw\_vs\_fit\_examples — raw trace vs fit (1/10)

figure: zip4\_raw\_vs\_fit\_examples.png — event rows 1–3 of 15, left half of the channels

Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.

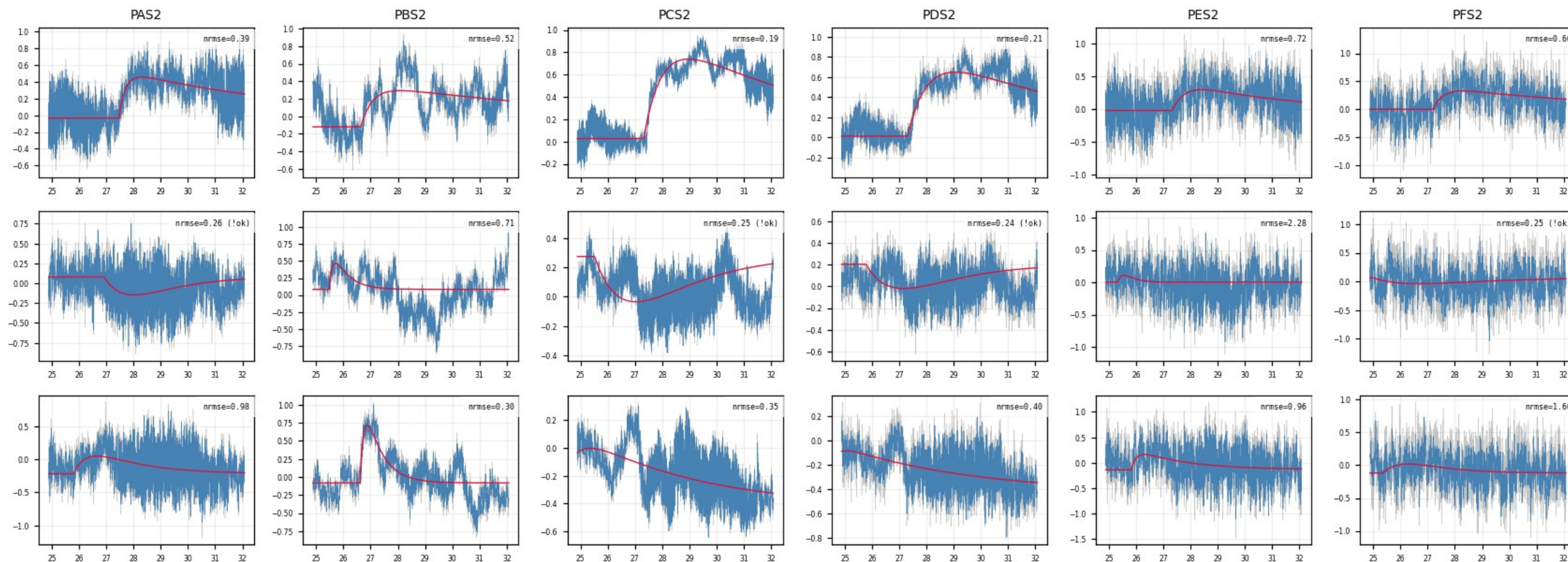




# Z4 · raw\_vs\_fit\_examples — raw trace vs fit (2/10)

figure: zip4\_raw\_vs\_fit\_examples.png — event rows 1–3 of 15, right half of the channels

Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.

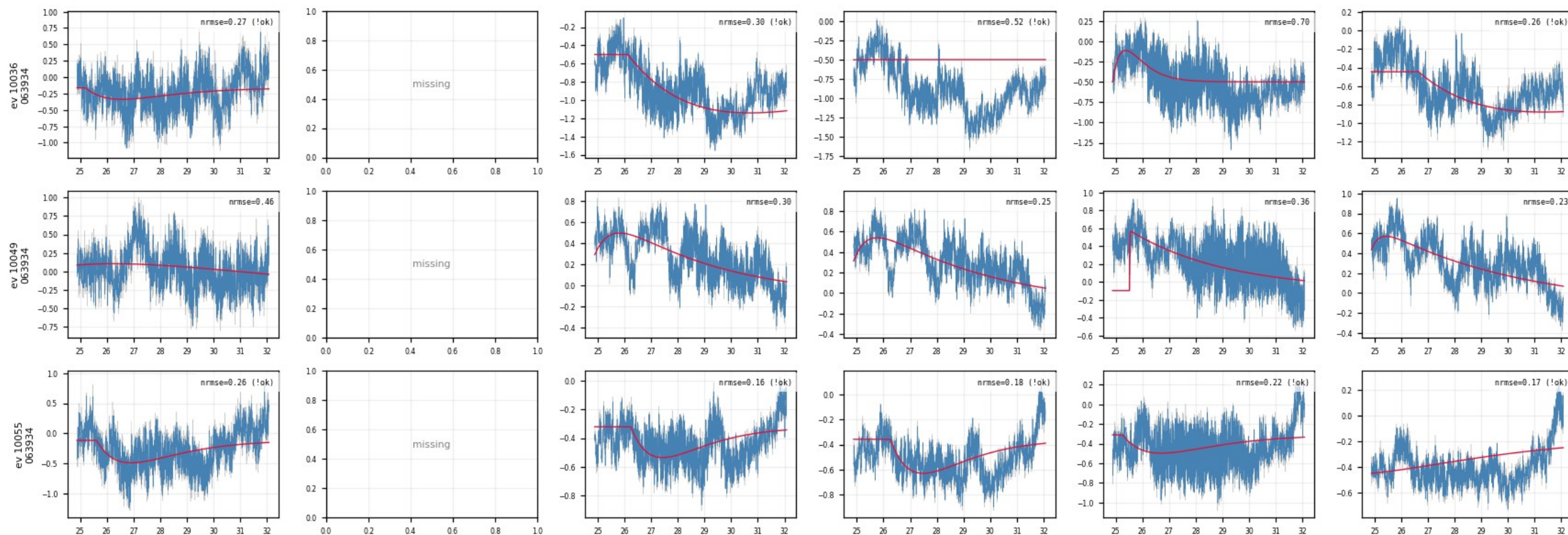




# Z4 · raw\_vs\_fit\_examples — raw trace vs fit (3/10)

figure: zip4\_raw\_vs\_fit\_examples.png — event rows 4–6 of 15, left half of the channels

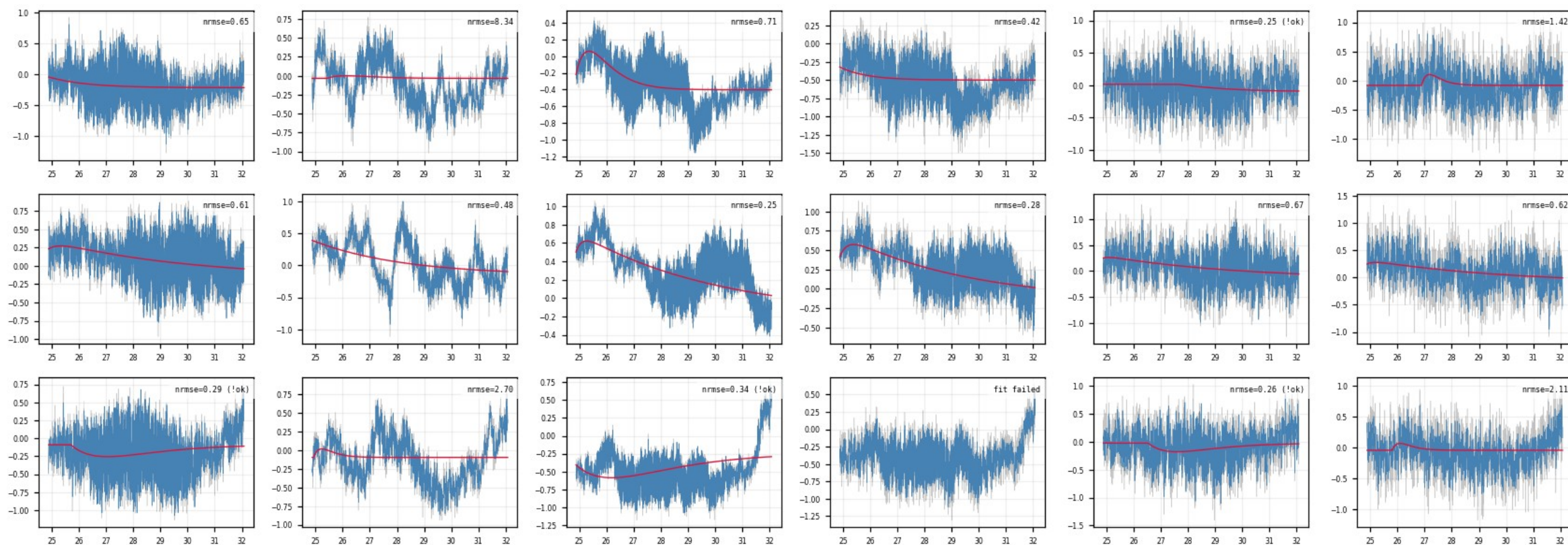
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



# Z4 · raw\_vs\_fit\_examples — raw trace vs fit (4/10)

figure: zip4\_raw\_vs\_fit\_examples.png — event rows 4–6 of 15, right half of the channels

Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.

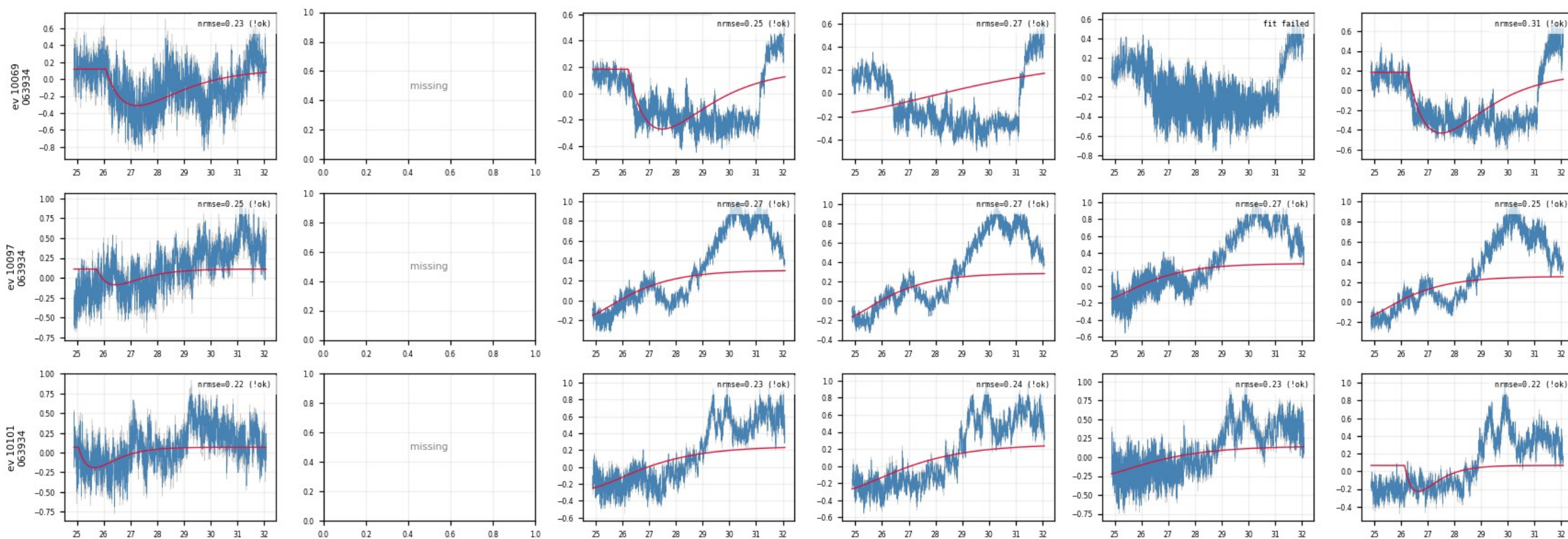




# Z4 · raw\_vs\_fit\_examples — raw trace vs fit (5/10)

figure: zip4\_raw\_vs\_fit\_examples.png — event rows 7–9 of 15, left half of the channels

Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.

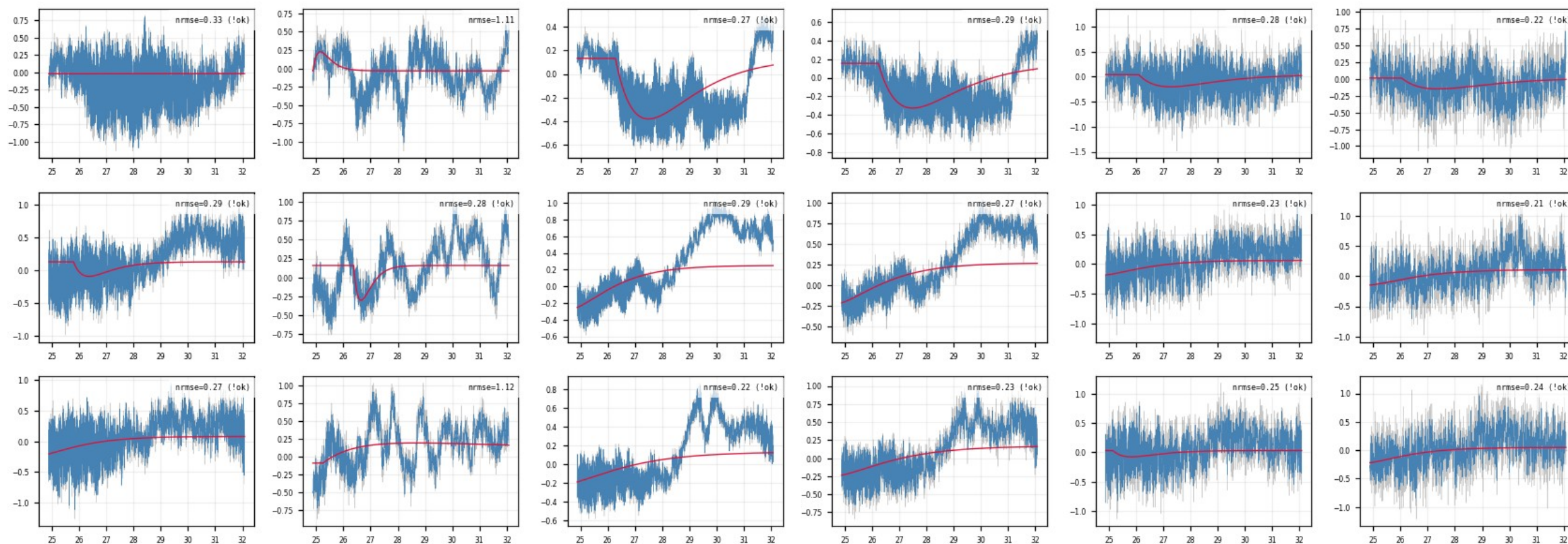




# Z4 · raw\_vs\_fit\_examples — raw trace vs fit (6/10)

figure: zip4\_raw\_vs\_fit\_examples.png — event rows 7–9 of 15, right half of the channels

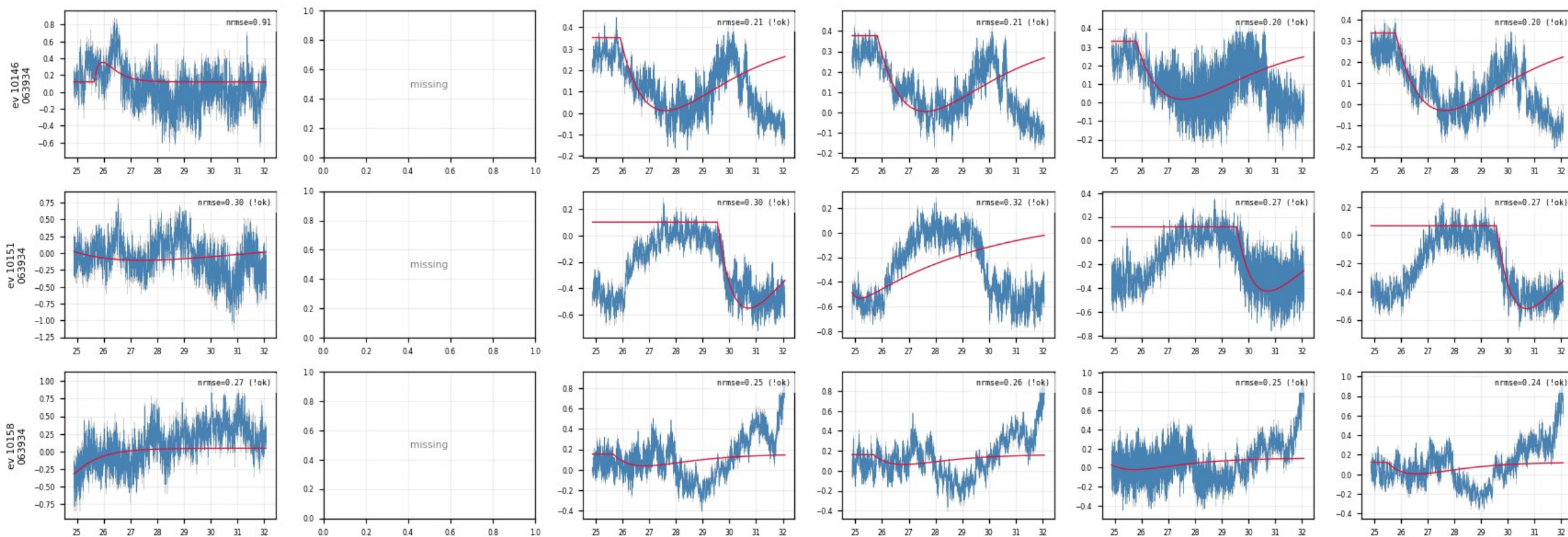
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



# Z4 · raw\_vs\_fit\_examples — raw trace vs fit (7/10)

figure: zip4\_raw\_vs\_fit\_examples.png — event rows 10–12 of 15, left half of the channels

Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.

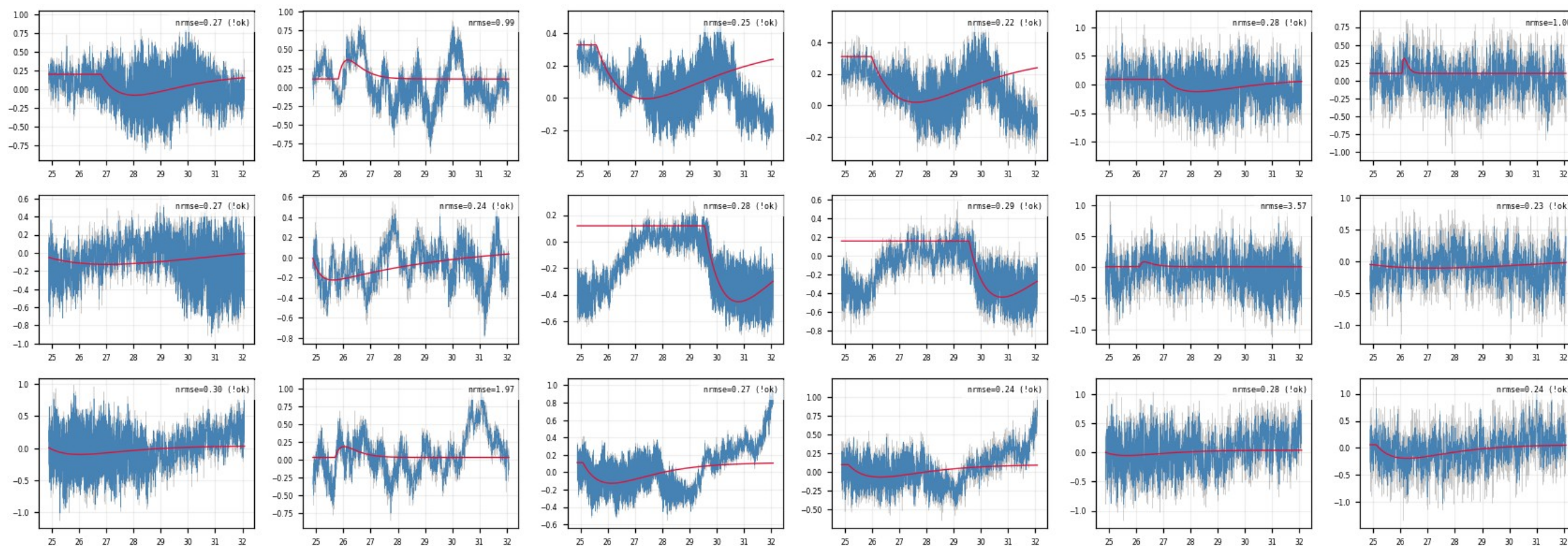




# Z4 · raw\_vs\_fit\_examples — raw trace vs fit (8/10)

figure: zip4\_raw\_vs\_fit\_examples.png — event rows 10–12 of 15, right half of the channels

Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.

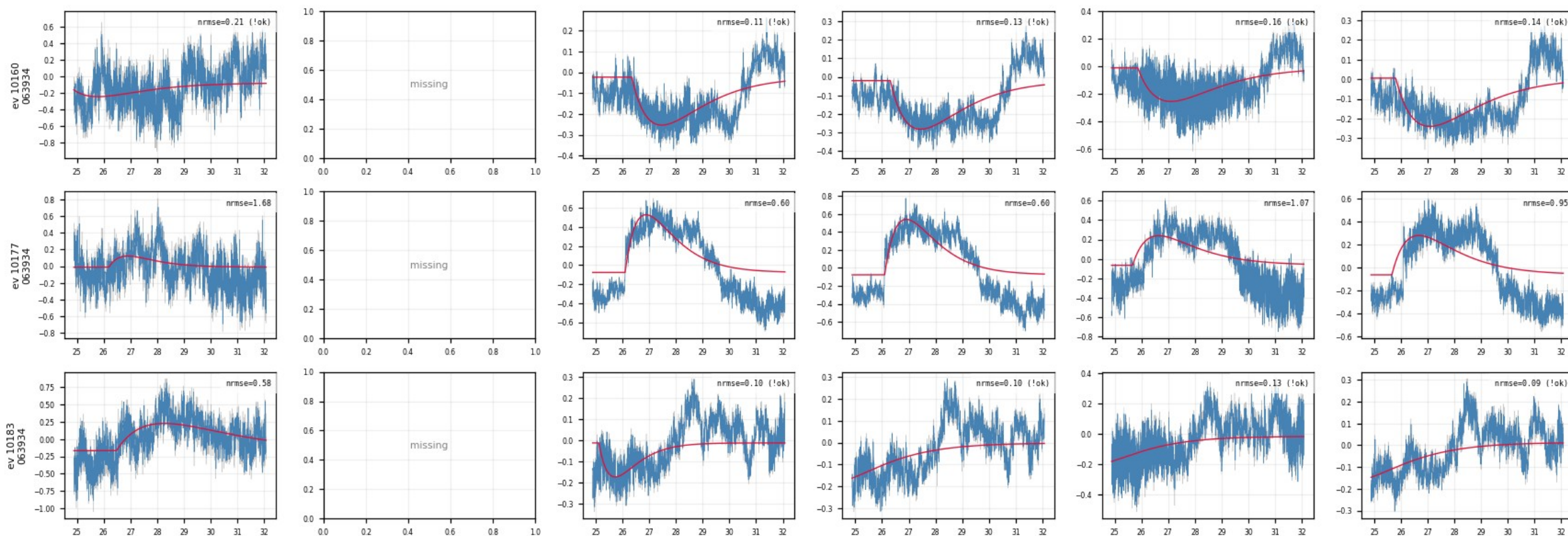




# Z4 · raw\_vs\_fit\_examples — raw trace vs fit (9/10)

figure: zip4\_raw\_vs\_fit\_examples.png — event rows 13–15 of 15, left half of the channels

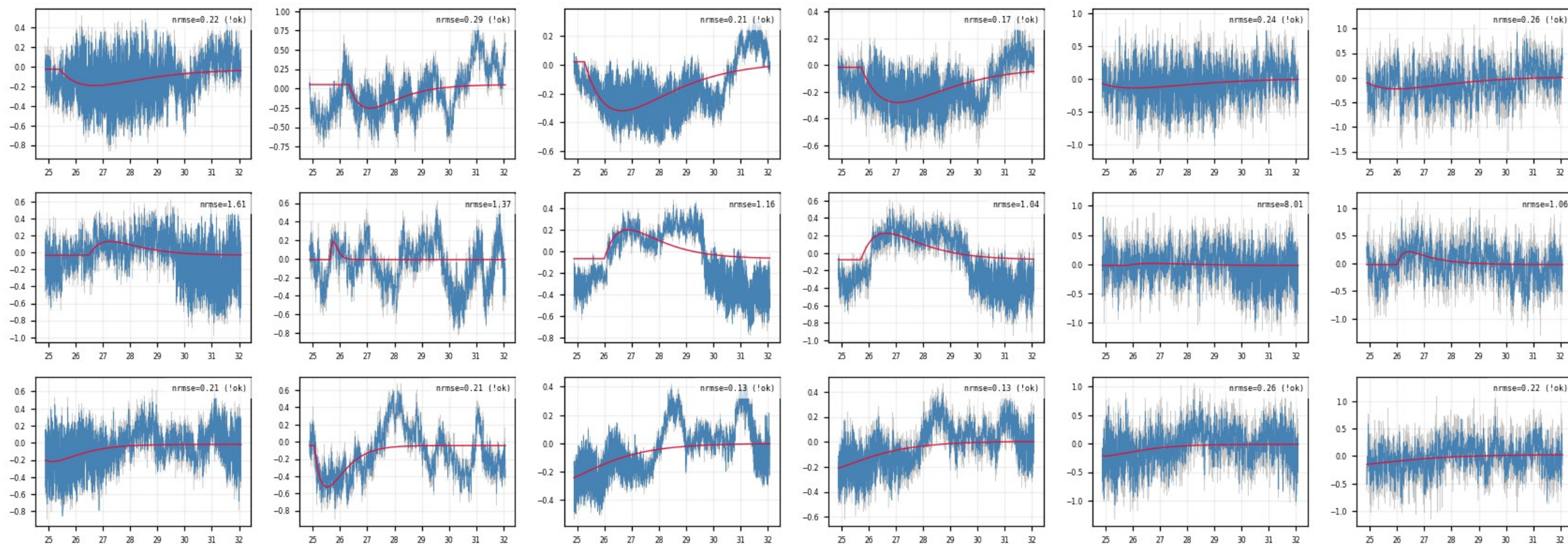
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



# Z4 · raw\_vs\_fit\_examples — raw trace vs fit (10/10)

figure: zip4\_raw\_vs\_fit\_examples.png — event rows 13–15 of 15, right half of the channels

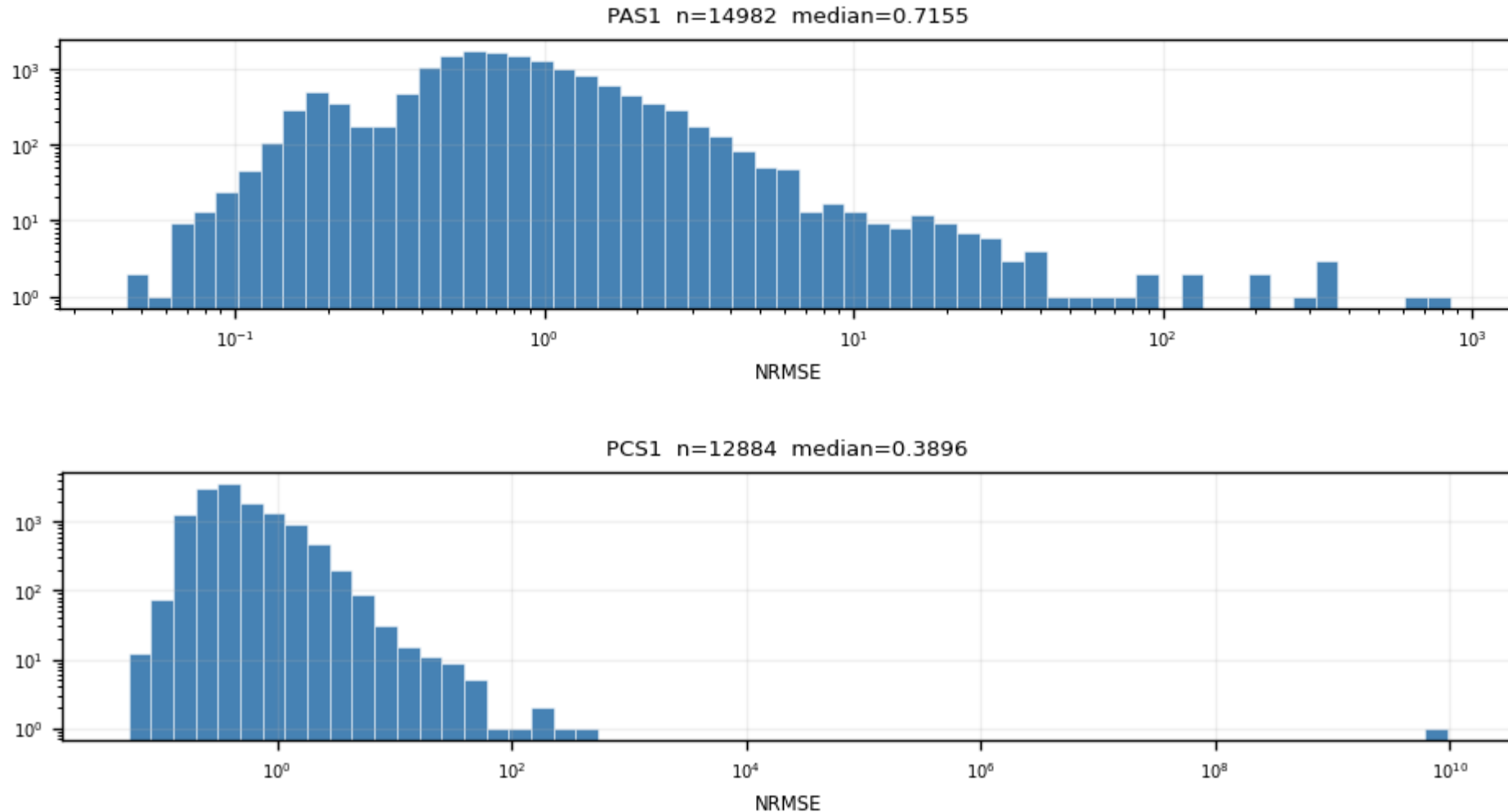
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



## Z4 · nrmse — NRMSE distribution per channel (1/6)

figure: zip4\_nrmse.png — channel panels 1–2 of 11, top to bottom

NRMSE =  $\text{RMS}(\text{fit residual}) / \text{fitted peak}$ , log-log. Bimodal: good fits vs noise triggers; the valley at  $\approx 0.4$  sets quality cut 1.

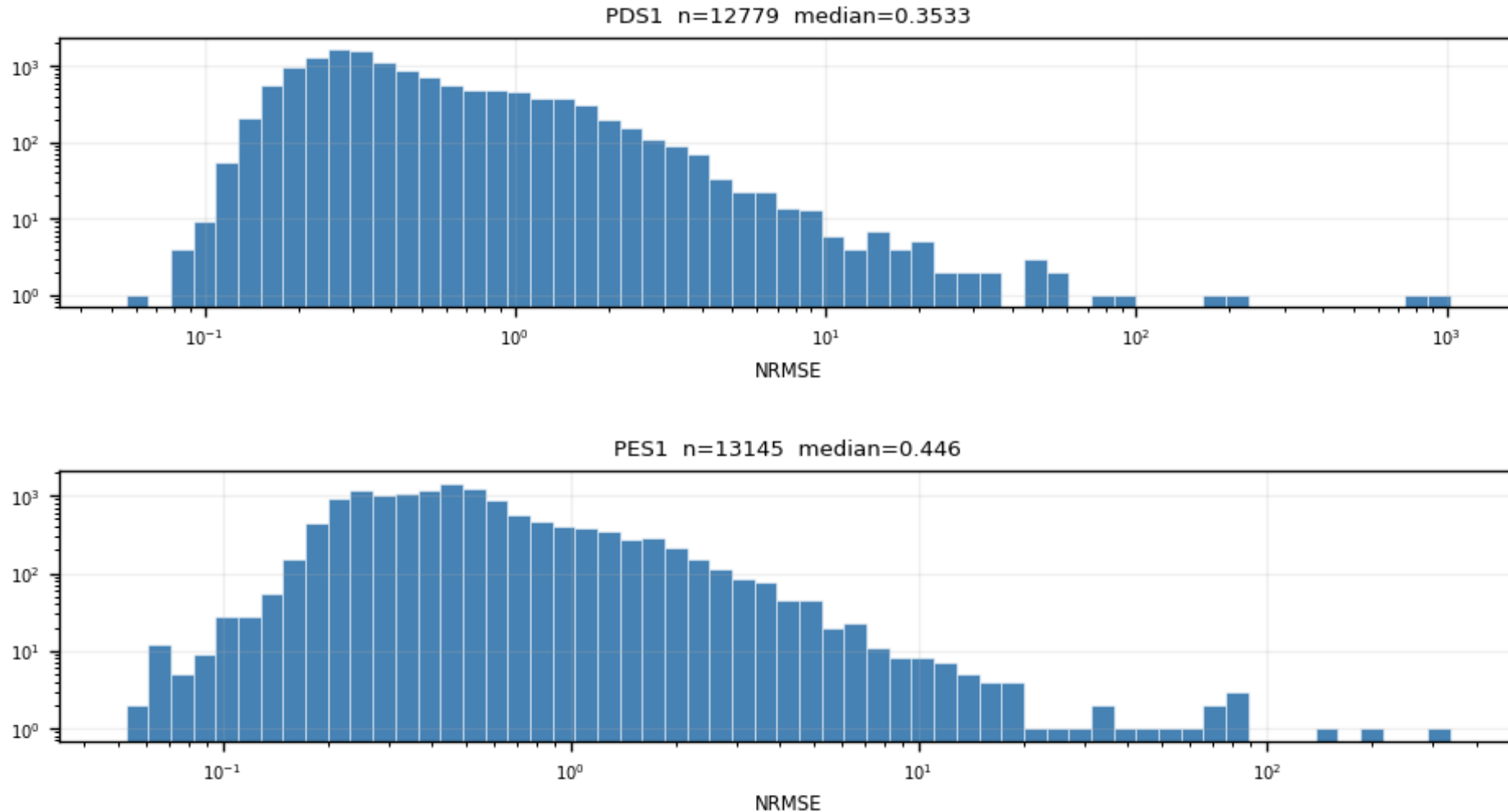




## Z4 · nrmse — NRMSE distribution per channel (2/6)

figure: zip4\_nrmse.png — channel panels 3–4 of 11, top to bottom

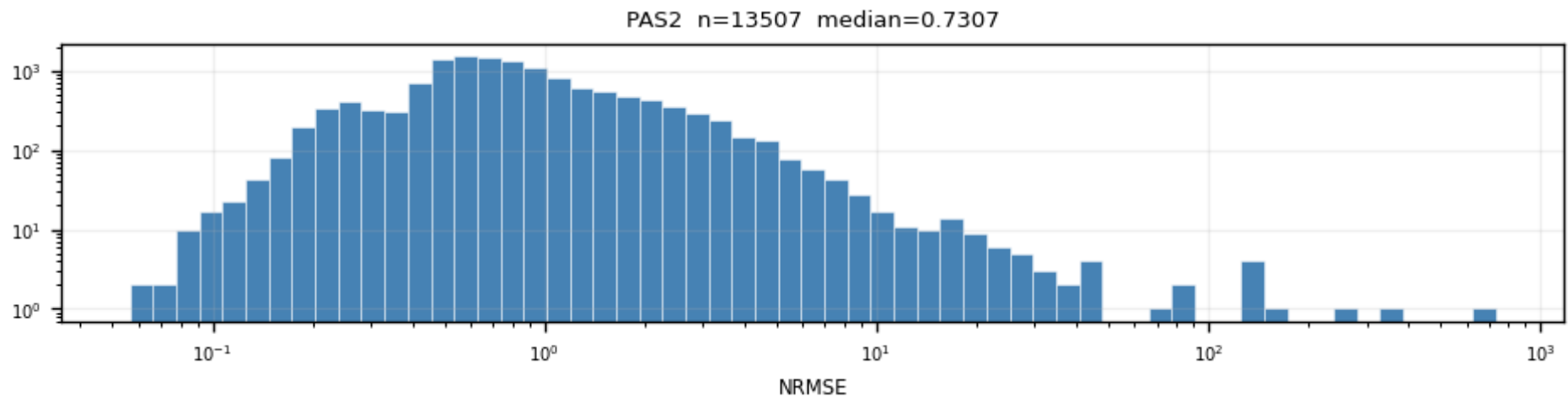
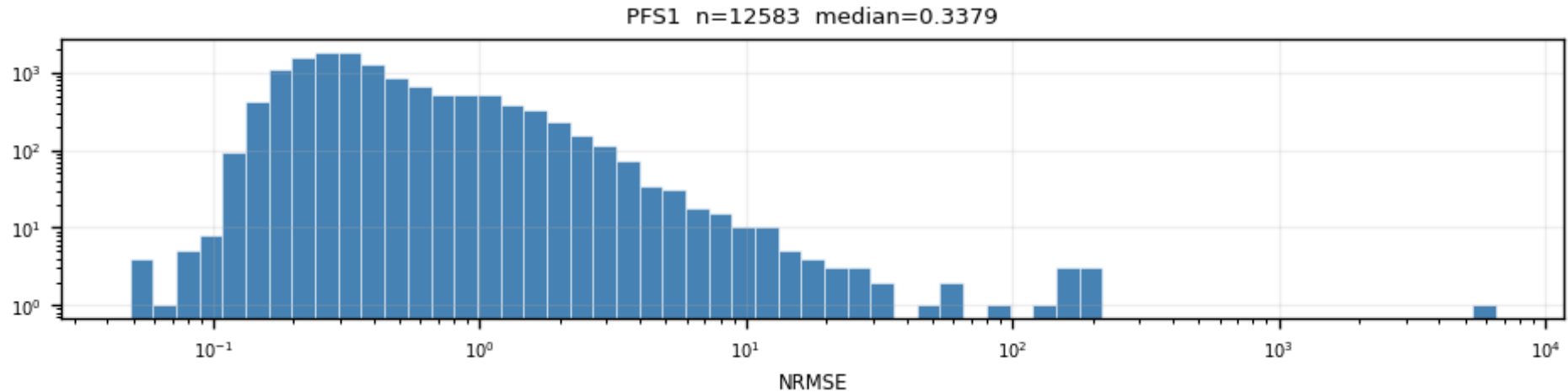
NRMSE =  $\text{RMS}(\text{fit residual}) / \text{fitted peak}$ , log-log. Bimodal: good fits vs noise triggers; the valley at  $\approx 0.4$  sets quality cut 1.



## Z4 · nrmse — NRMSE distribution per channel (3/6)

figure: zip4\_nrmse.png — channel panels 5–6 of 11, top to bottom

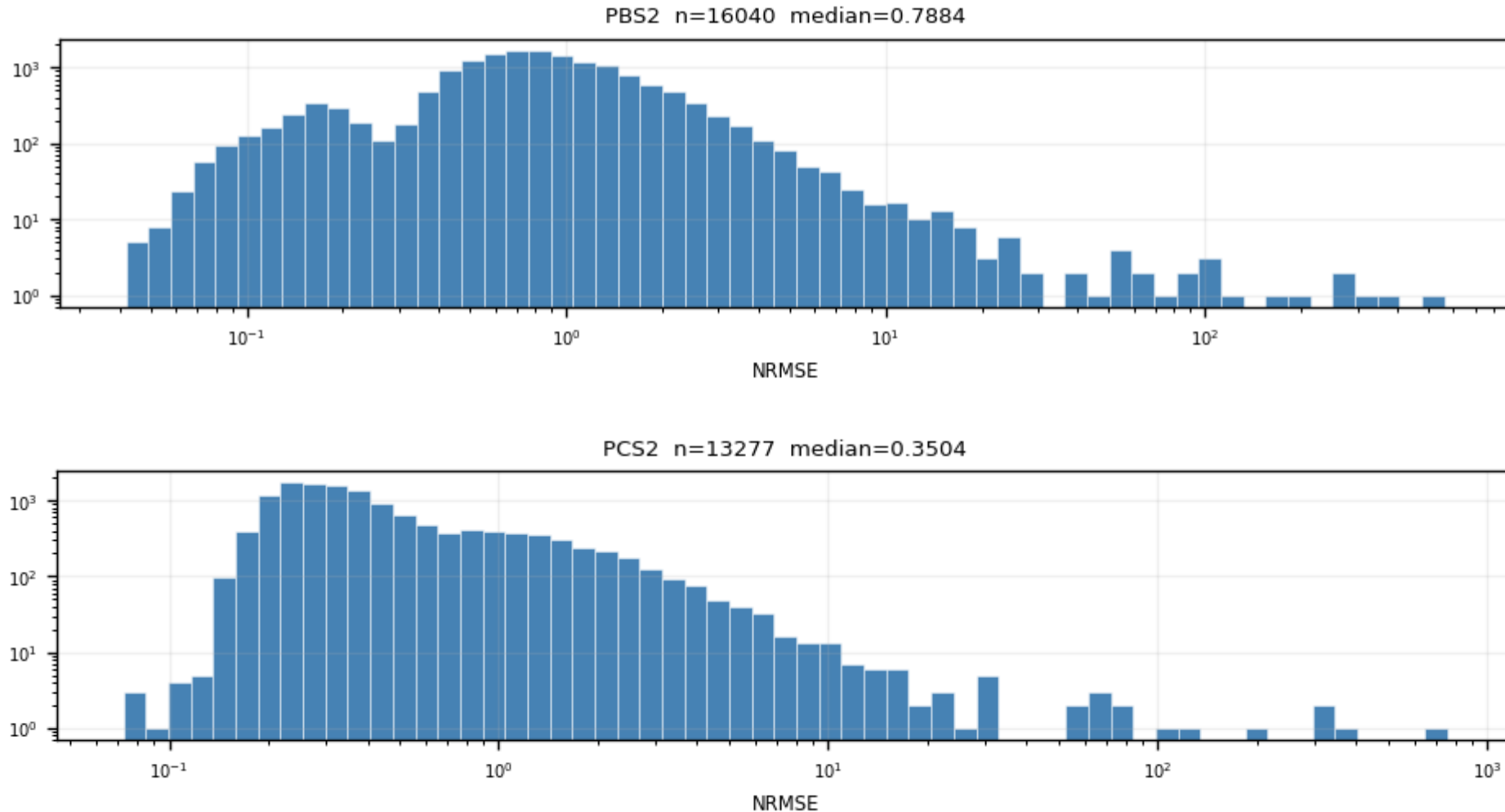
NRMSE =  $\text{RMS}(\text{fit residual}) / \text{fitted peak}$ , log-log. Bimodal: good fits vs noise triggers; the valley at  $\approx 0.4$  sets quality cut 1.



## Z4 · nrmse — NRMSE distribution per channel (4/6)

figure: zip4\_nrmse.png — channel panels 7–8 of 11, top to bottom

NRMSE =  $\text{RMS}(\text{fit residual}) / \text{fitted peak}$ , log-log. Bimodal: good fits vs noise triggers; the valley at  $\approx 0.4$  sets quality cut 1.

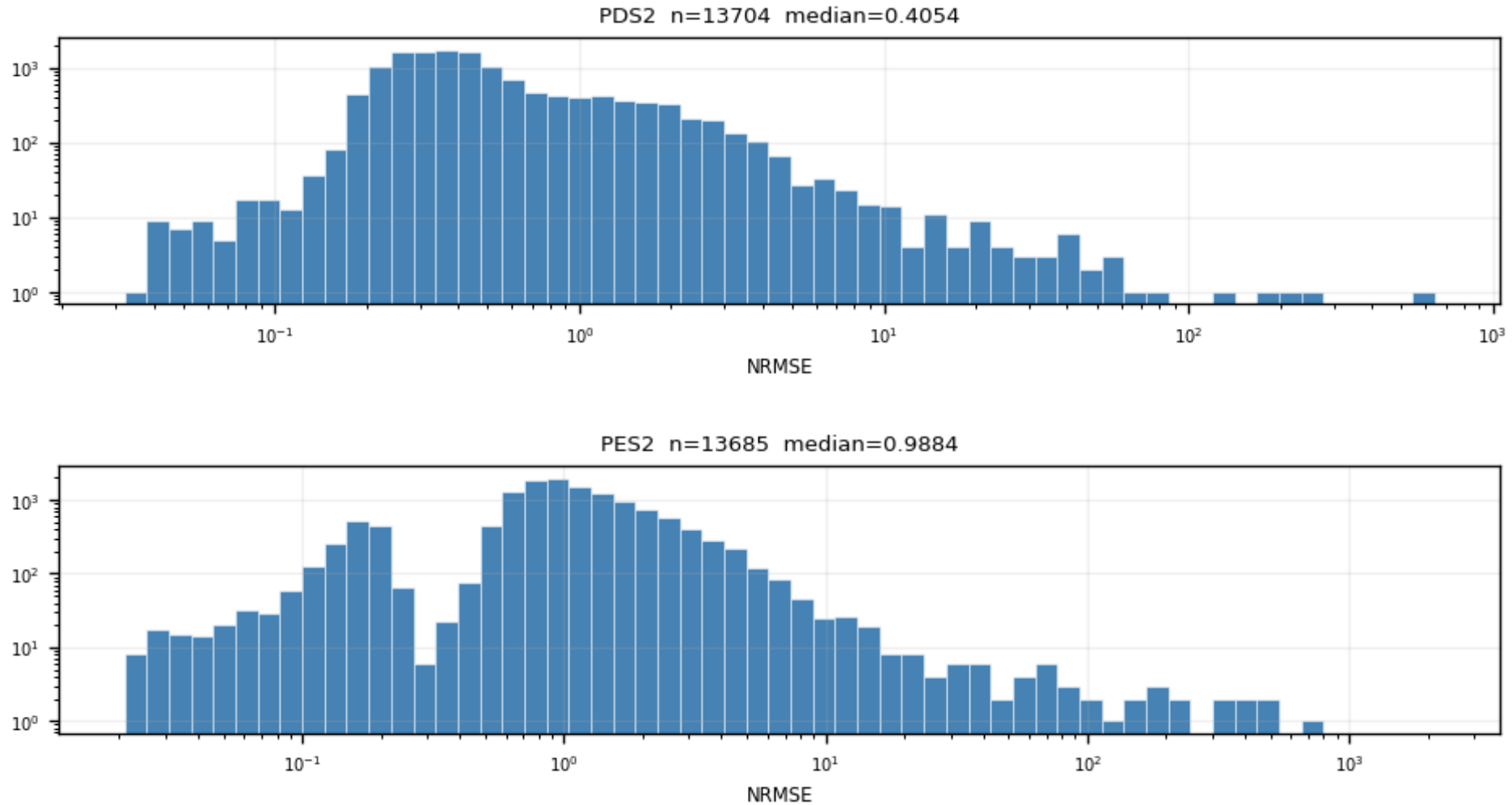




## Z4 · nrmse — NRMSE distribution per channel (5/6)

figure: zip4\_nrmse.png — channel panels 9–10 of 11, top to bottom

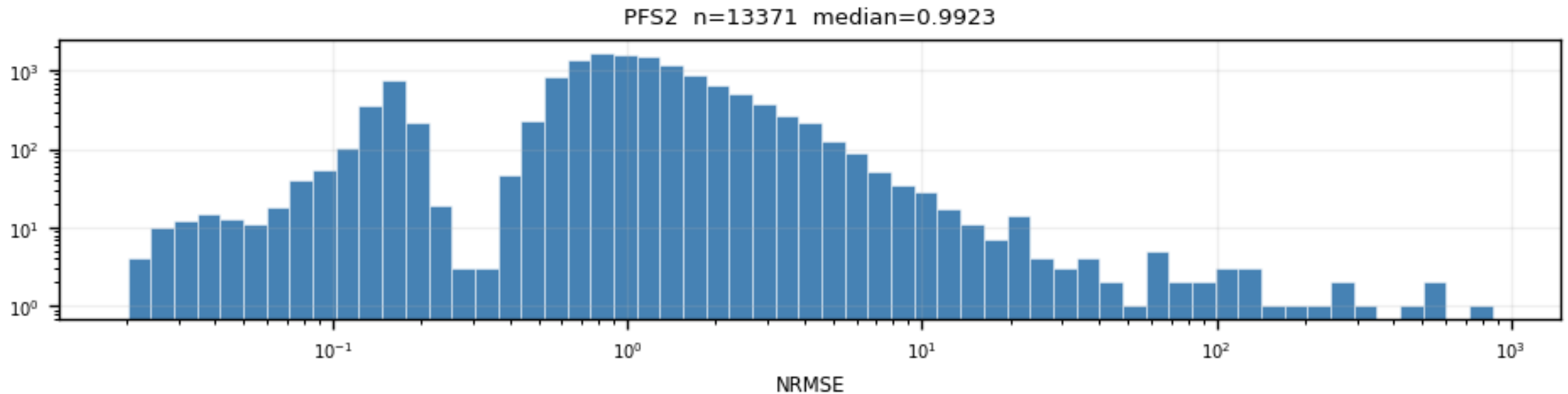
NRMSE =  $\text{RMS}(\text{fit residual}) / \text{fitted peak}$ , log-log. Bimodal: good fits vs noise triggers; the valley at  $\approx 0.4$  sets quality cut 1.



## Z4 · nrmse — NRMSE distribution per channel (6/6)

figure: zip4\_nrmse.png — channel panels 11–11 of 11, top to bottom

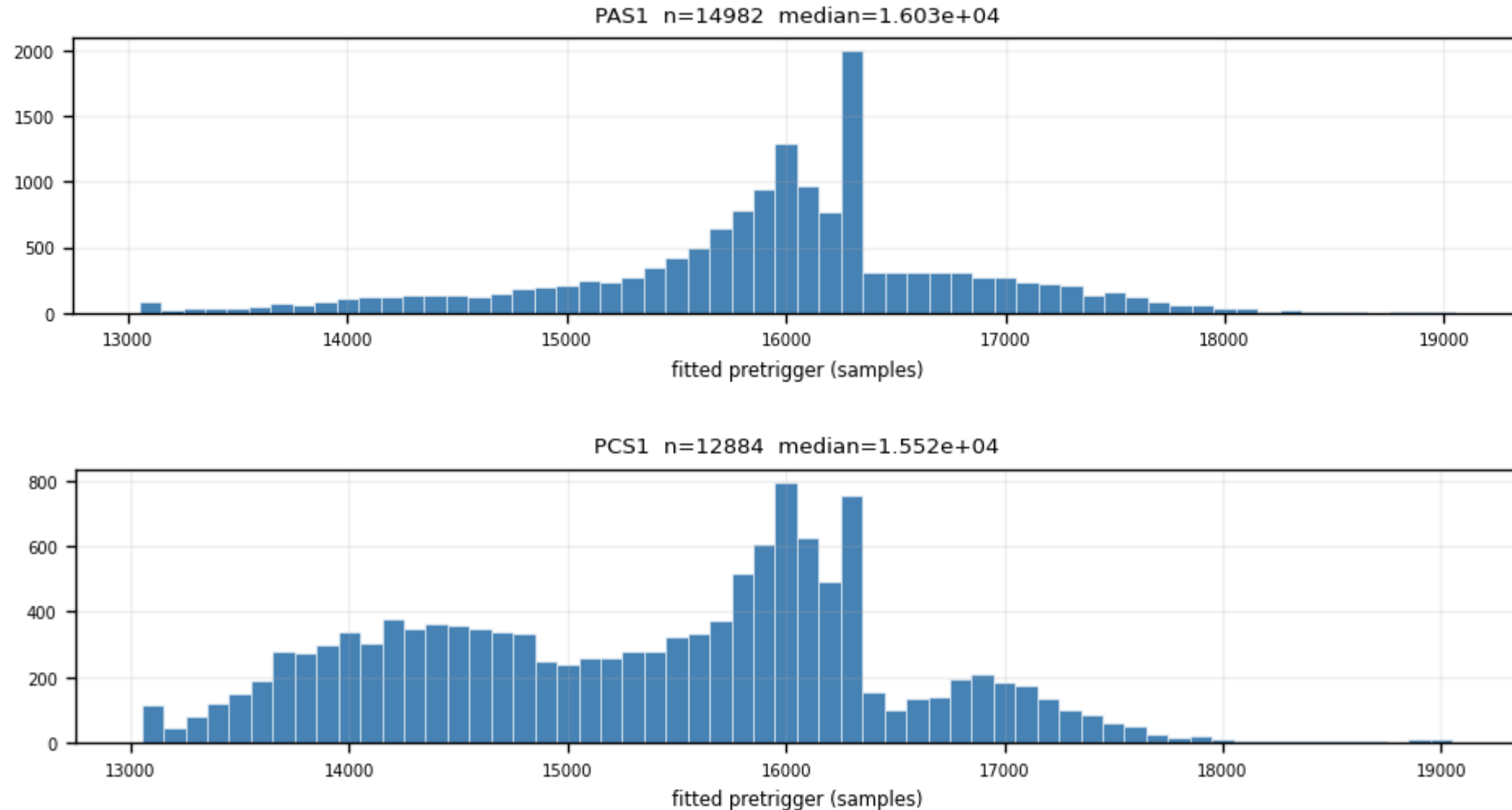
NRMSE =  $\text{RMS}(\text{fit residual}) / \text{fitted peak}$ , log-log. Bimodal: good fits vs noise triggers; the valley at  $\approx 0.4$  sets quality cut 1.



# Z4 · pretrigger — fitted onset per channel (1/6)

figure: zip4\_pretrigger.png — channel panels 1–2 of 11, top to bottom

The onset is a FREE fit parameter; pinning it inside the fit would bias every other parameter.

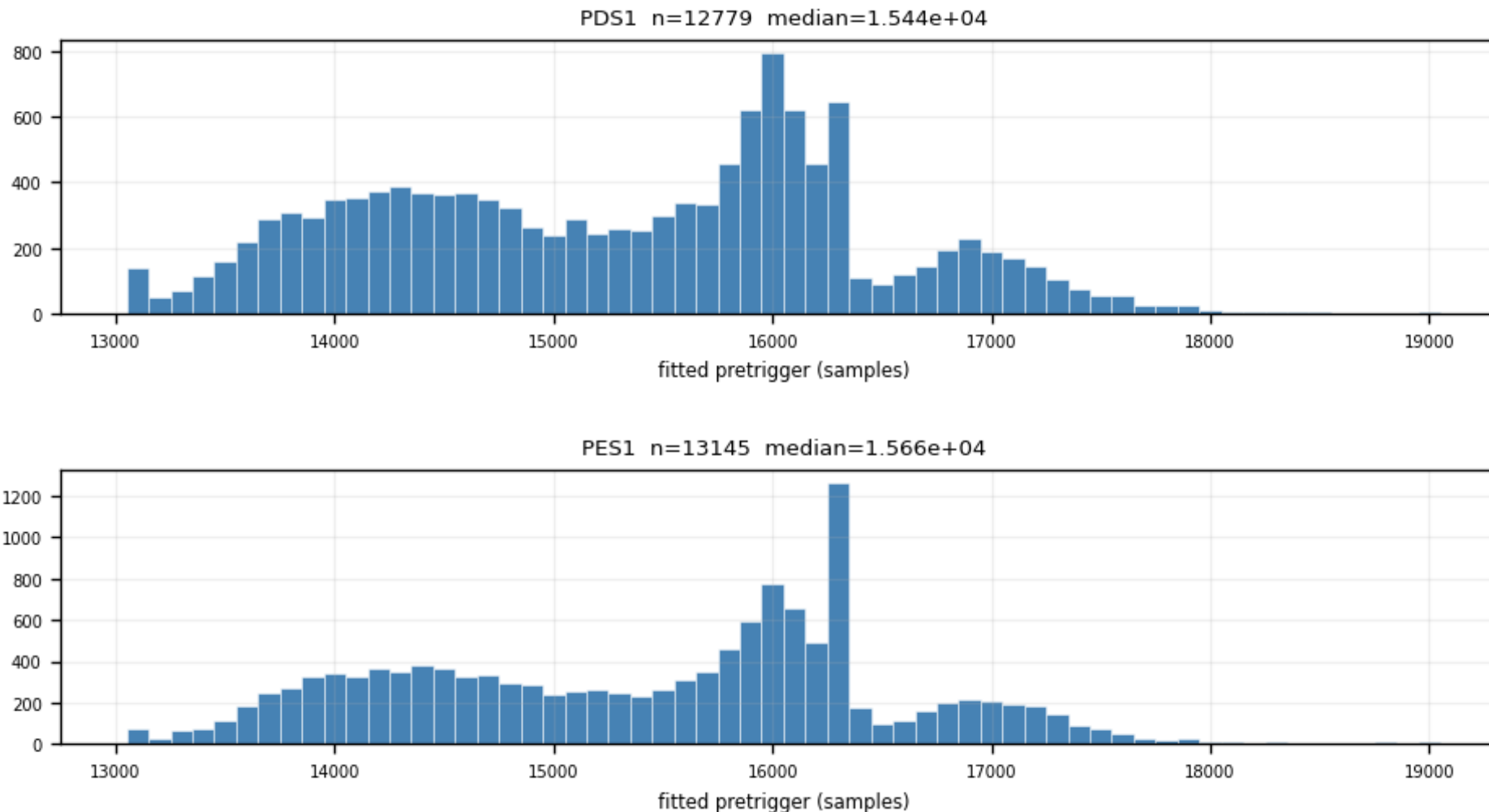




## Z4 · pretrigger — fitted onset per channel (2/6)

figure: zip4\_pretrigger.png — channel panels 3–4 of 11, top to bottom

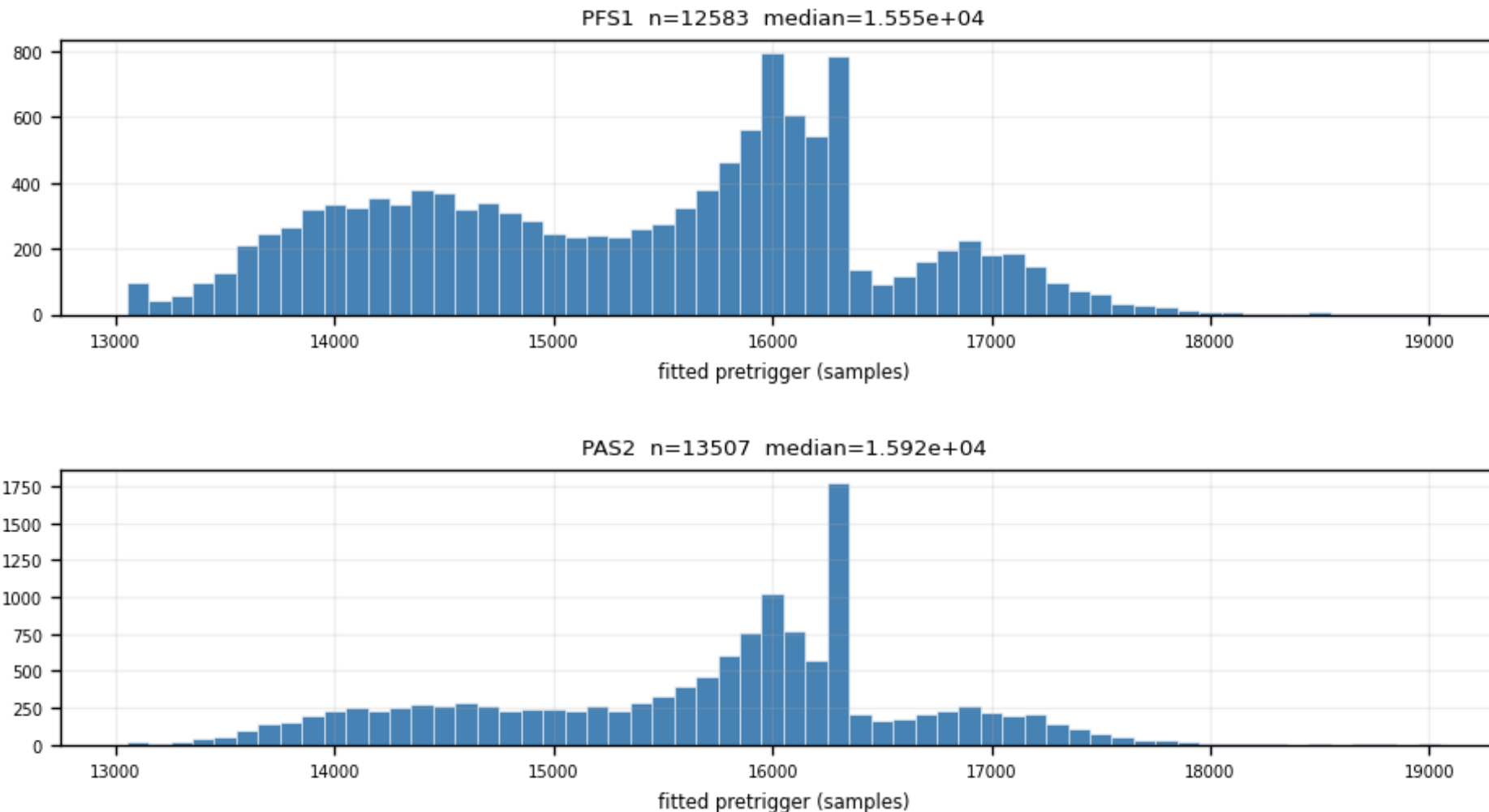
The onset is a FREE fit parameter; pinning it inside the fit would bias every other parameter.



# Z4 · pretrigger — fitted onset per channel (3/6)

figure: zip4\_pretrigger.png — channel panels 5–6 of 11, top to bottom

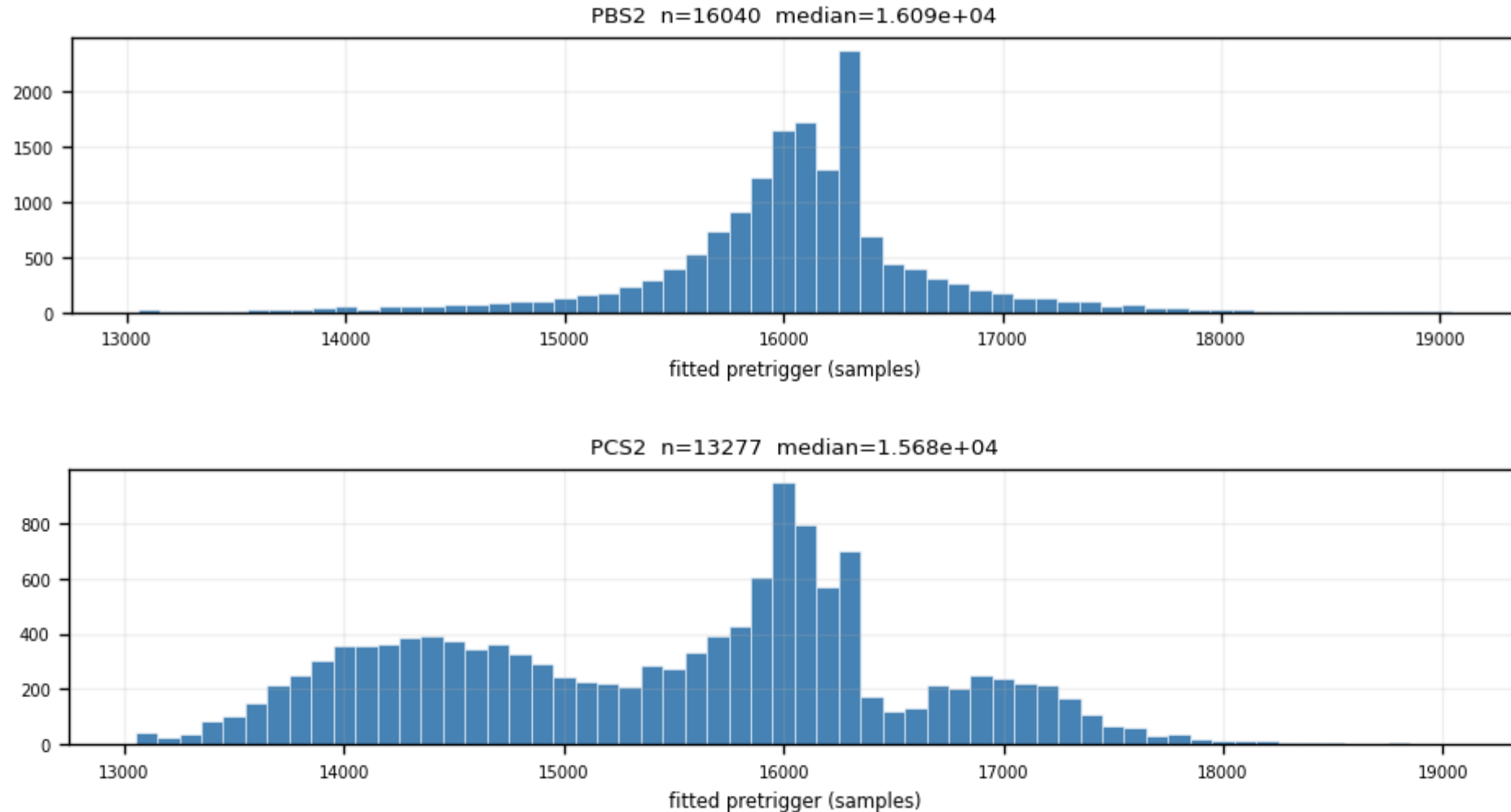
The onset is a FREE fit parameter; pinning it inside the fit would bias every other parameter.



# Z4 · pretrigger — fitted onset per channel (4/6)

figure: zip4\_pretrigger.png — channel panels 7–8 of 11, top to bottom

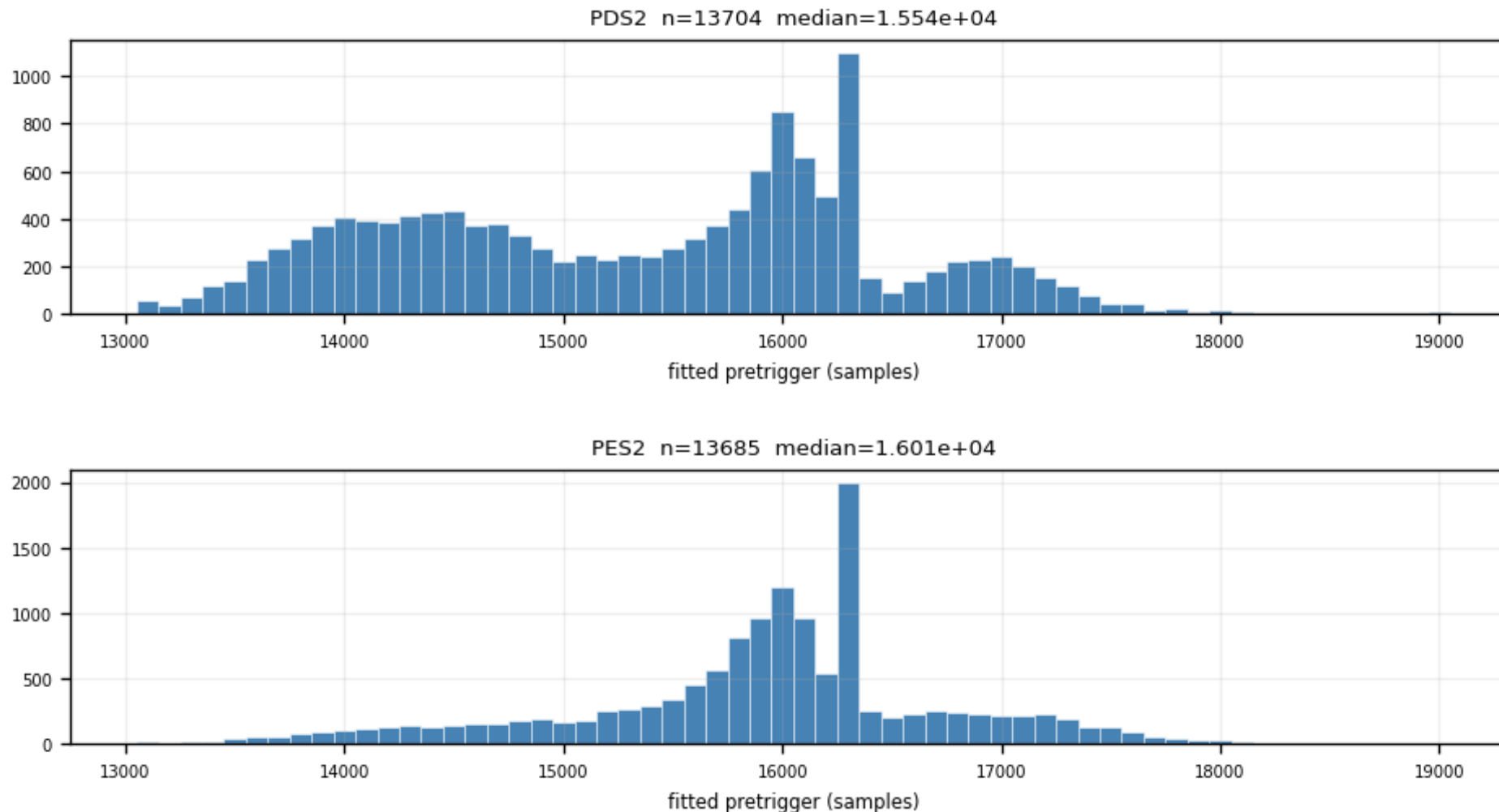
The onset is a FREE fit parameter; pinning it inside the fit would bias every other parameter.



## Z4 · pretrigger — fitted onset per channel (5/6)

figure: zip4\_pretrigger.png — channel panels 9–10 of 11, top to bottom

The onset is a FREE fit parameter; pinning it inside the fit would bias every other parameter.

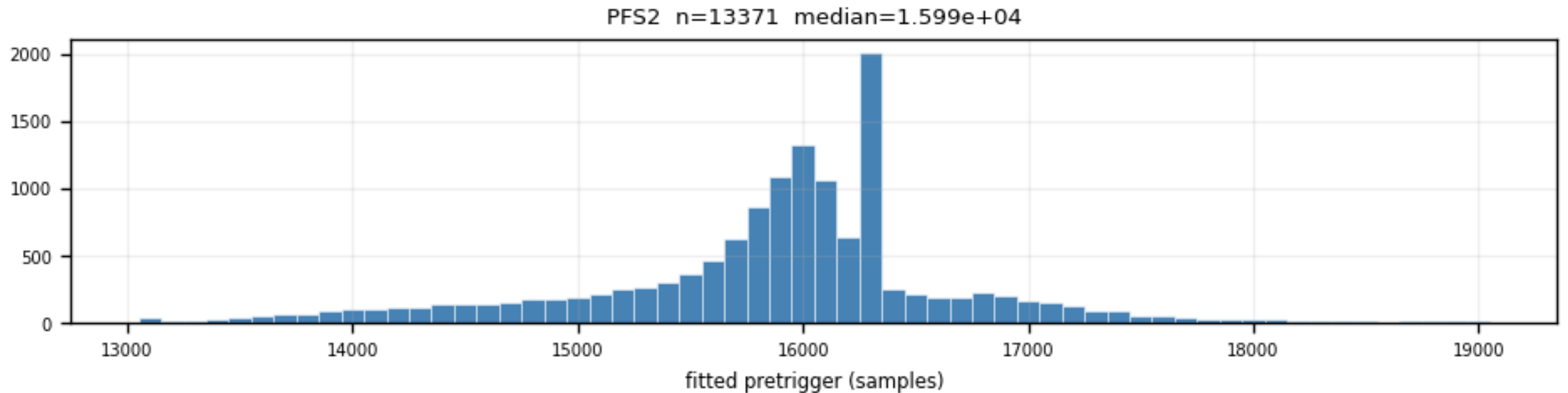




## Z4 · pretrigger — fitted onset per channel (6/6)

figure: zip4\_pretrigger.png — channel panels 11–11 of 11, top to bottom

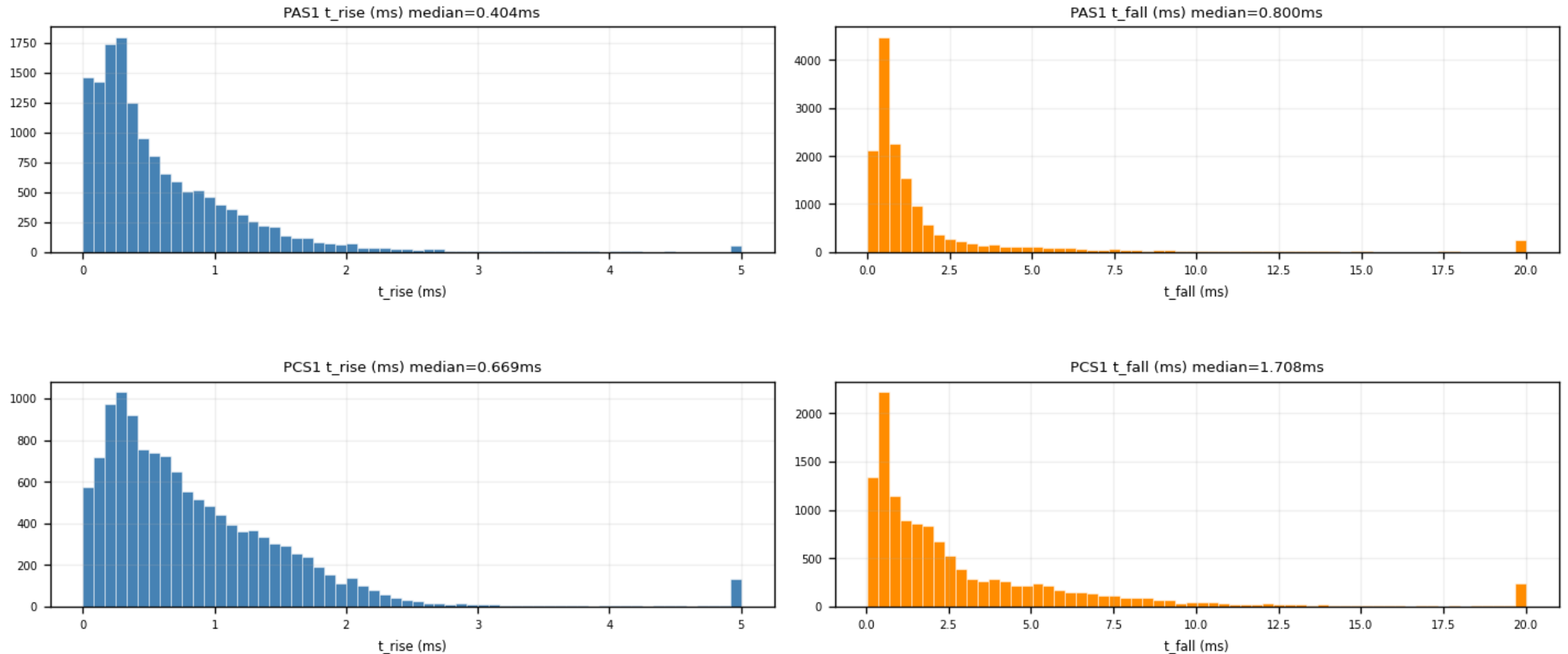
The onset is a FREE fit parameter; pinning it inside the fit would bias every other parameter.



## Z4 · time\_constants — $\tau_{\text{rise}}$ / $\tau_{\text{fall}}$ per channel (1/6)

figure: zip4\_time\_constants.png — channel panels 1–2 of 11, top to bottom

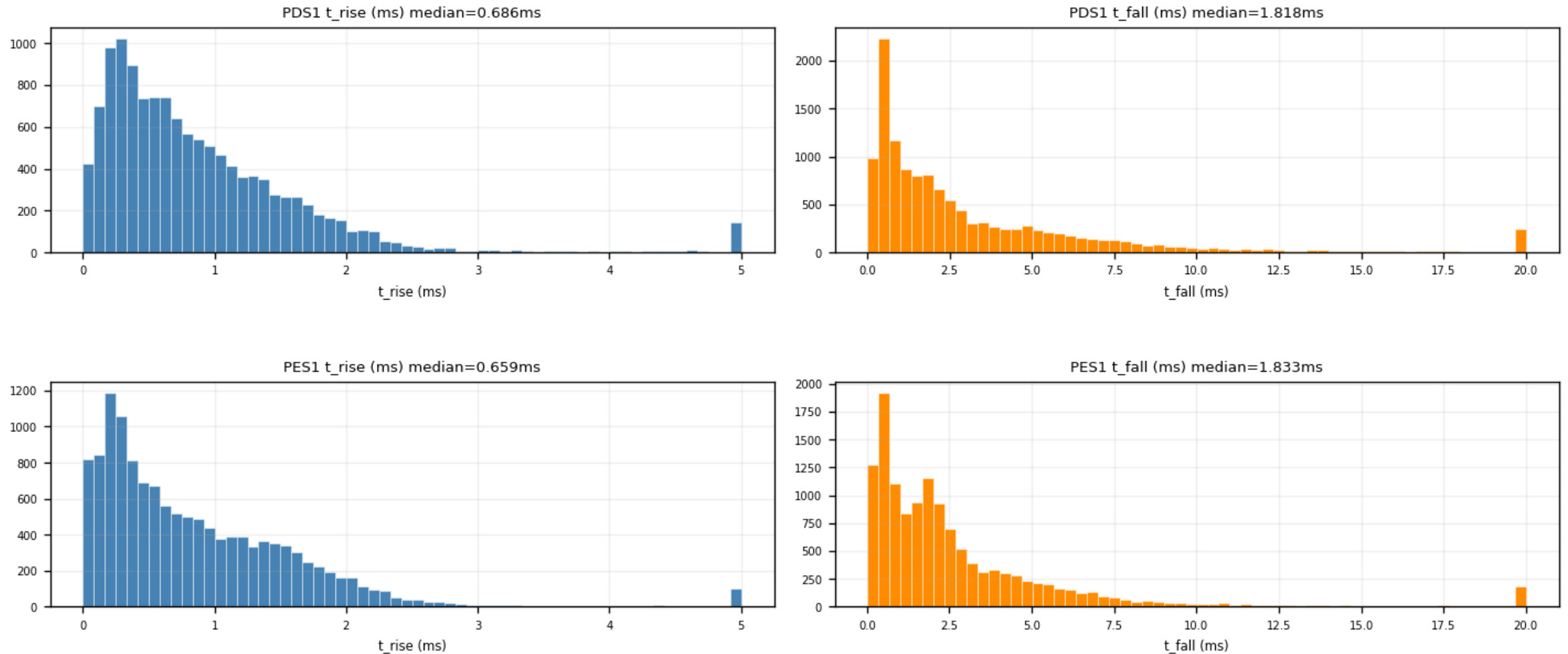
Fast-pulse population at  $\tau_{\text{rise}} \approx 0.1$  ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



## Z4 · time\_constants — $\tau_{\text{rise}}$ / $\tau_{\text{fall}}$ per channel (2/6)

figure: zip4\_time\_constants.png — channel panels 3–4 of 11, top to bottom

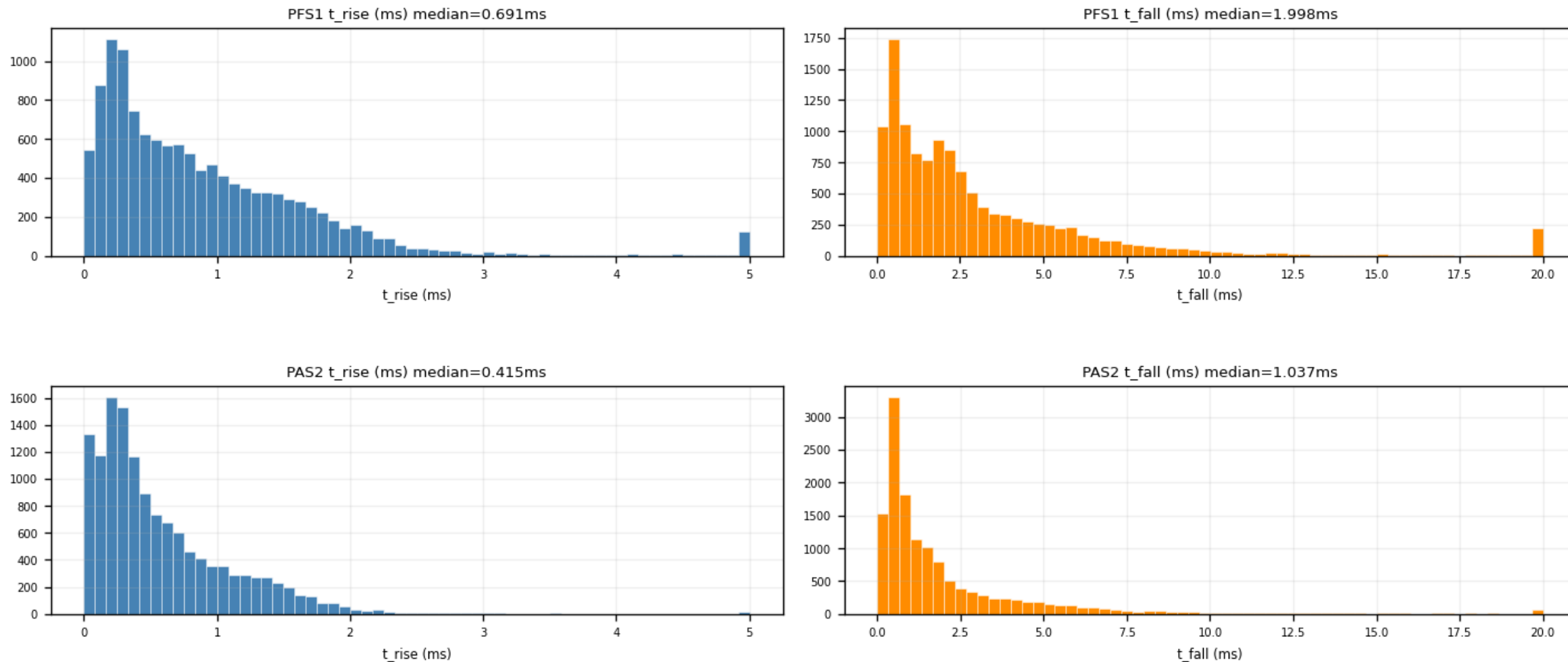
Fast-pulse population at  $\tau_{\text{rise}} \approx 0.1$  ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



## Z4 · time\_constants — $\tau_{\text{rise}}$ / $\tau_{\text{fall}}$ per channel (3/6)

figure: zip4\_time\_constants.png — channel panels 5–6 of 11, top to bottom

Fast-pulse population at  $\tau_{\text{rise}} \approx 0.1$  ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.

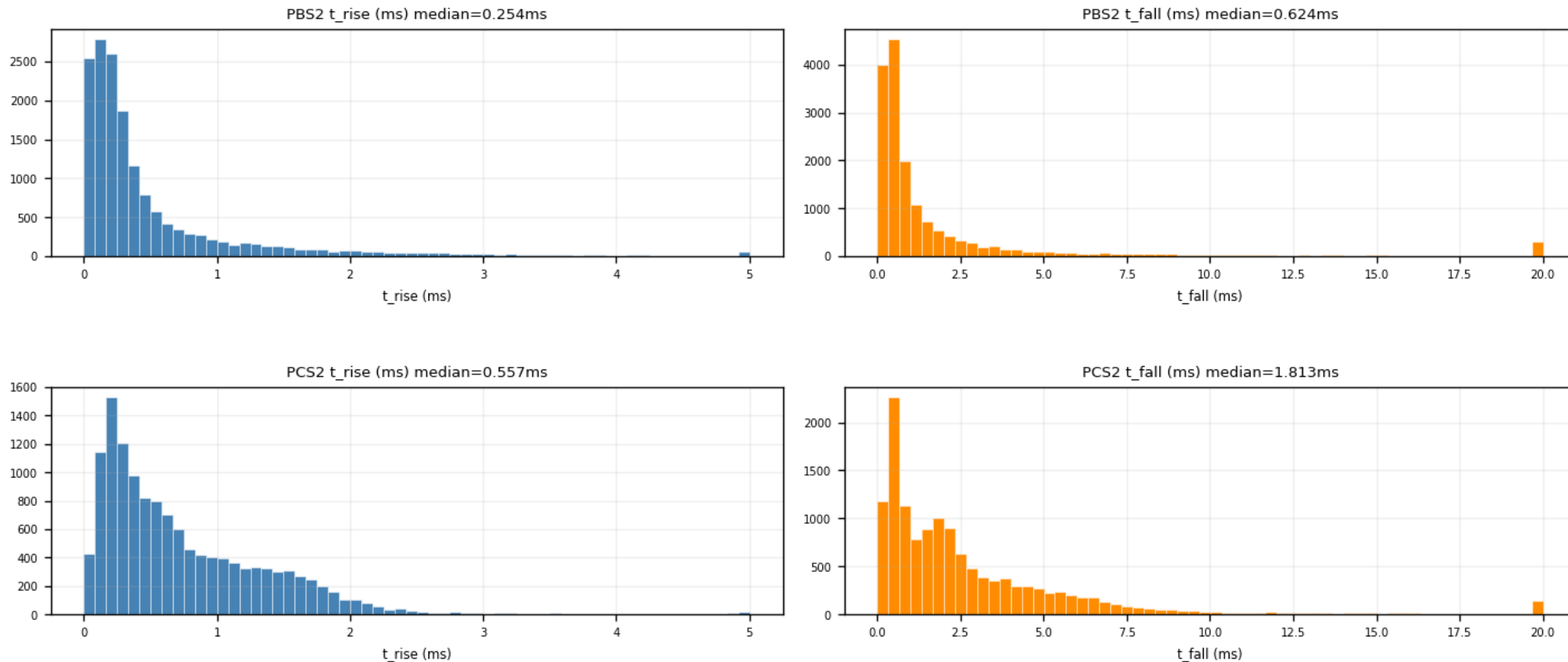




## Z4 · time\_constants — $\tau_{\text{rise}}$ / $\tau_{\text{fall}}$ per channel (4/6)

figure: zip4\_time\_constants.png — channel panels 7–8 of 11, top to bottom

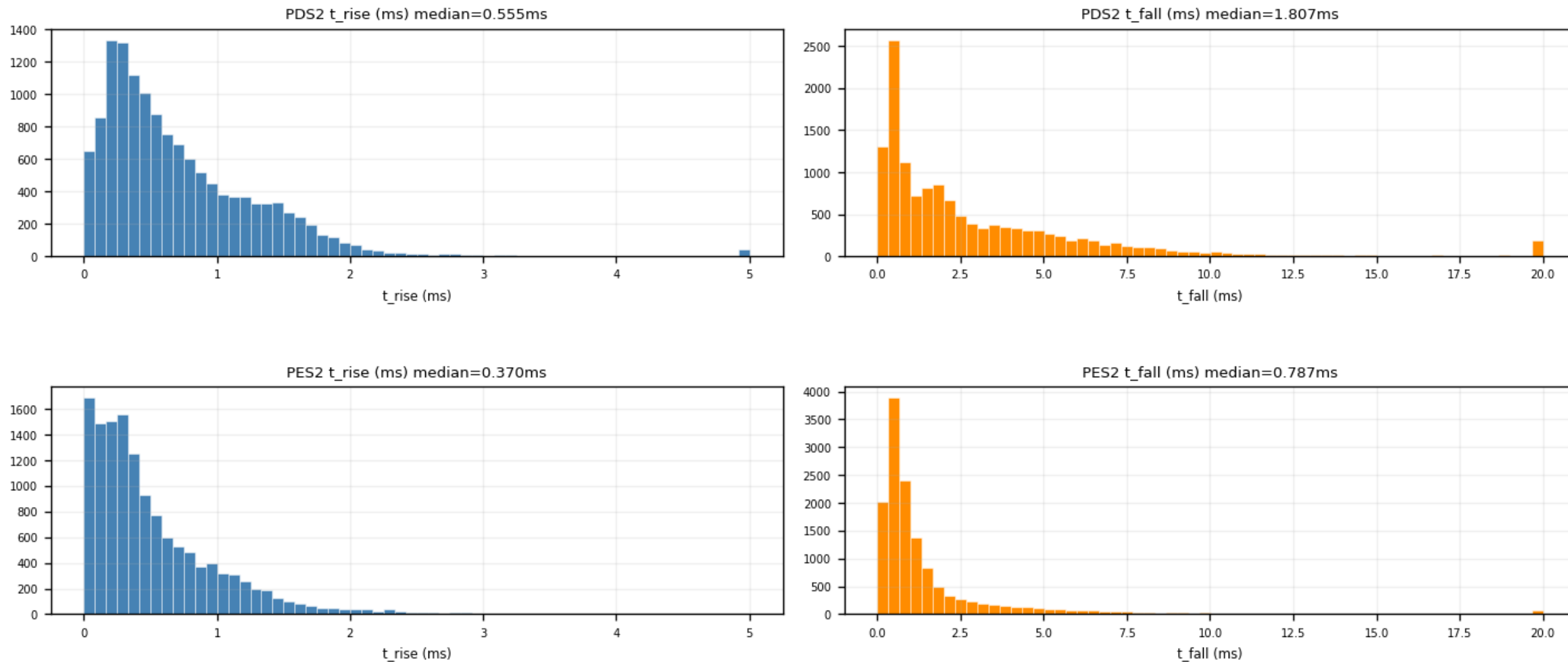
Fast-pulse population at  $\tau_{\text{rise}} \approx 0.1$  ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



## Z4 · time\_constants — $\tau_{\text{rise}}$ / $\tau_{\text{fall}}$ per channel (5/6)

figure: zip4\_time\_constants.png — channel panels 9–10 of 11, top to bottom

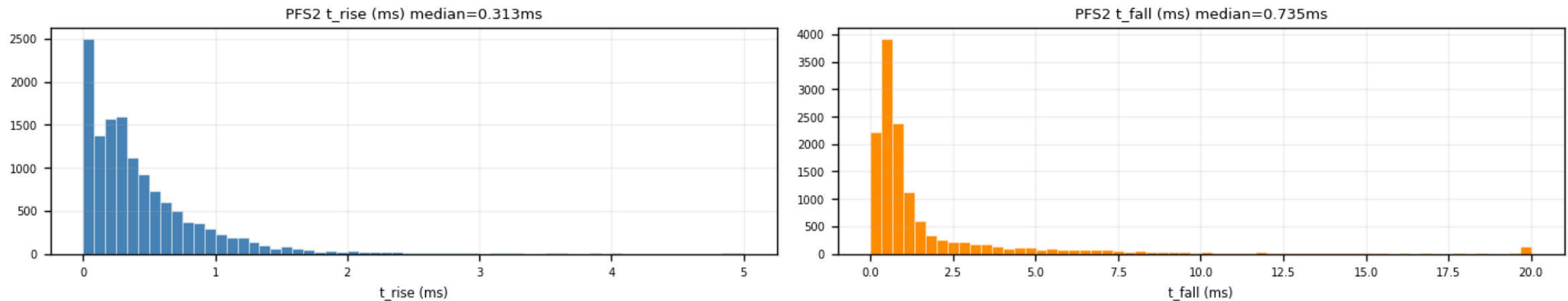
Fast-pulse population at  $\tau_{\text{rise}} \approx 0.1$  ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



## Z4 · time\_constants — $\tau_{\text{rise}}$ / $\tau_{\text{fall}}$ per channel (6/6)

figure: zip4\_time\_constants.png — channel panels 11–11 of 11, top to bottom

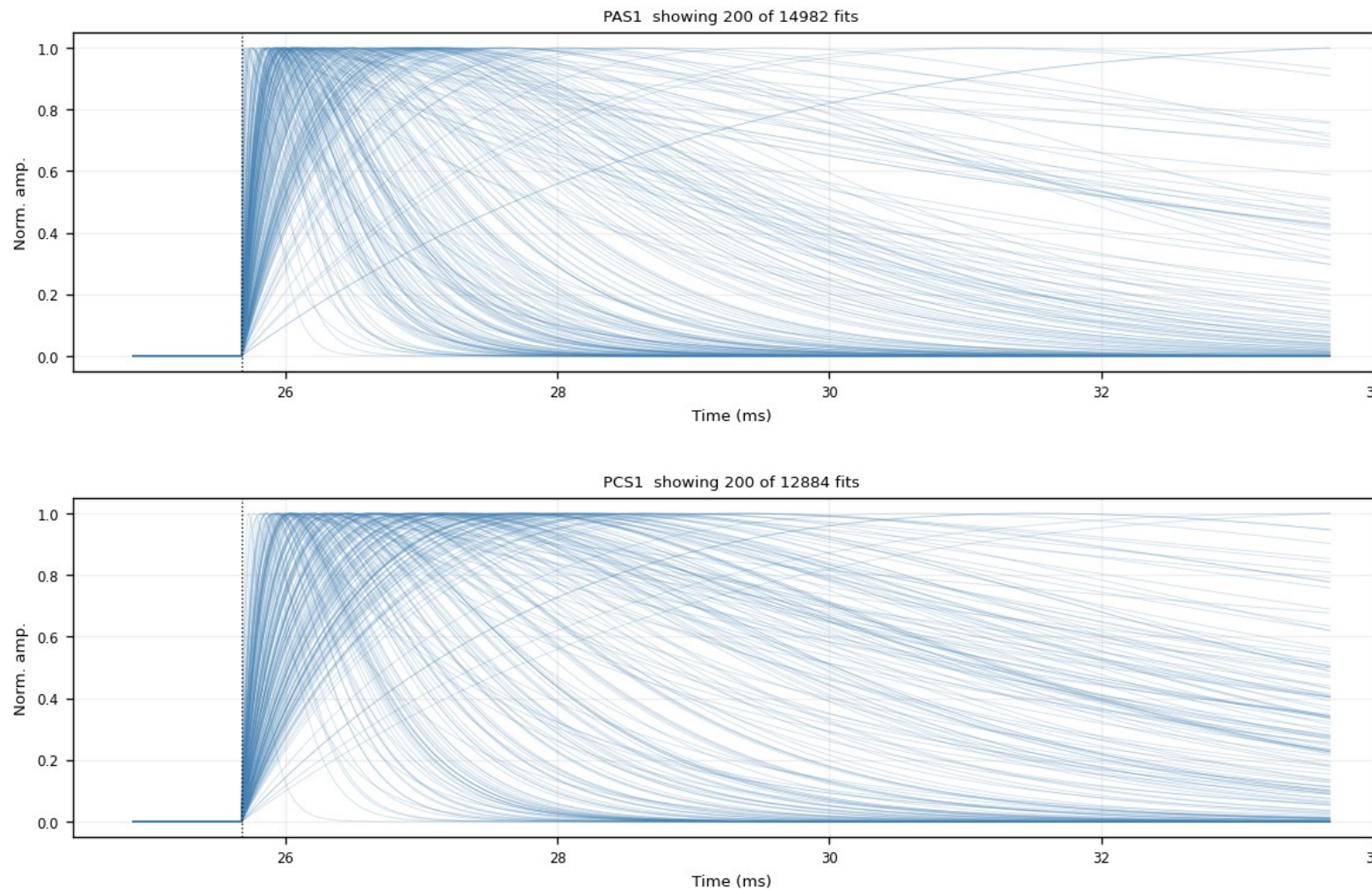
Fast-pulse population at  $\tau_{\text{rise}} \approx 0.1$  ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



## Z4 · fitted\_curves\_overlay — fan, no cut (1/6)

figure: zip4\_fitted\_curves\_overlay.png — channel panels 1–2 of 11, top to bottom

Every fit\_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.

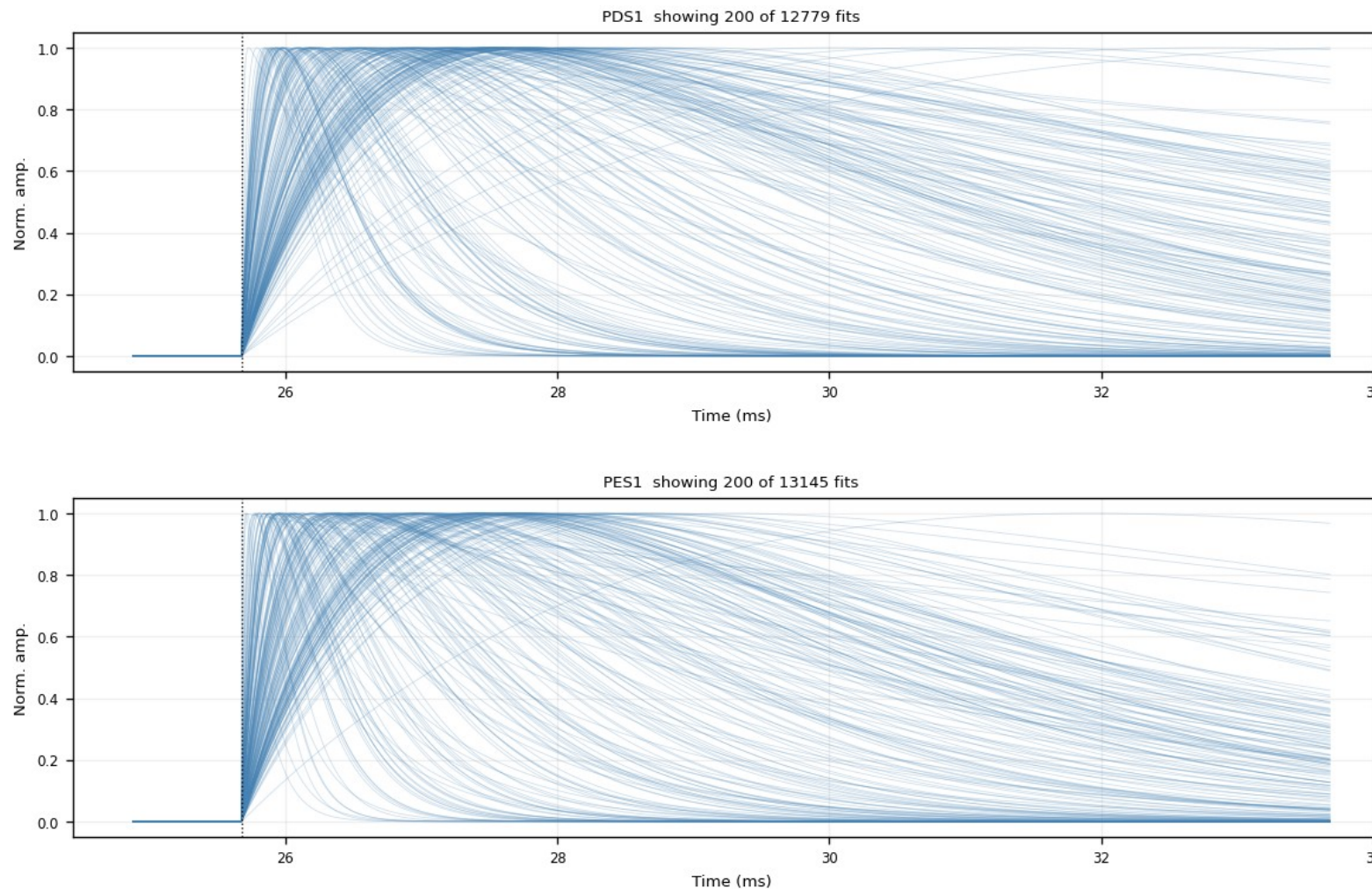




## Z4 · fitted\_curves\_overlay — fan, no cut (2/6)

figure: zip4\_fitted\_curves\_overlay.png — channel panels 3–4 of 11, top to bottom

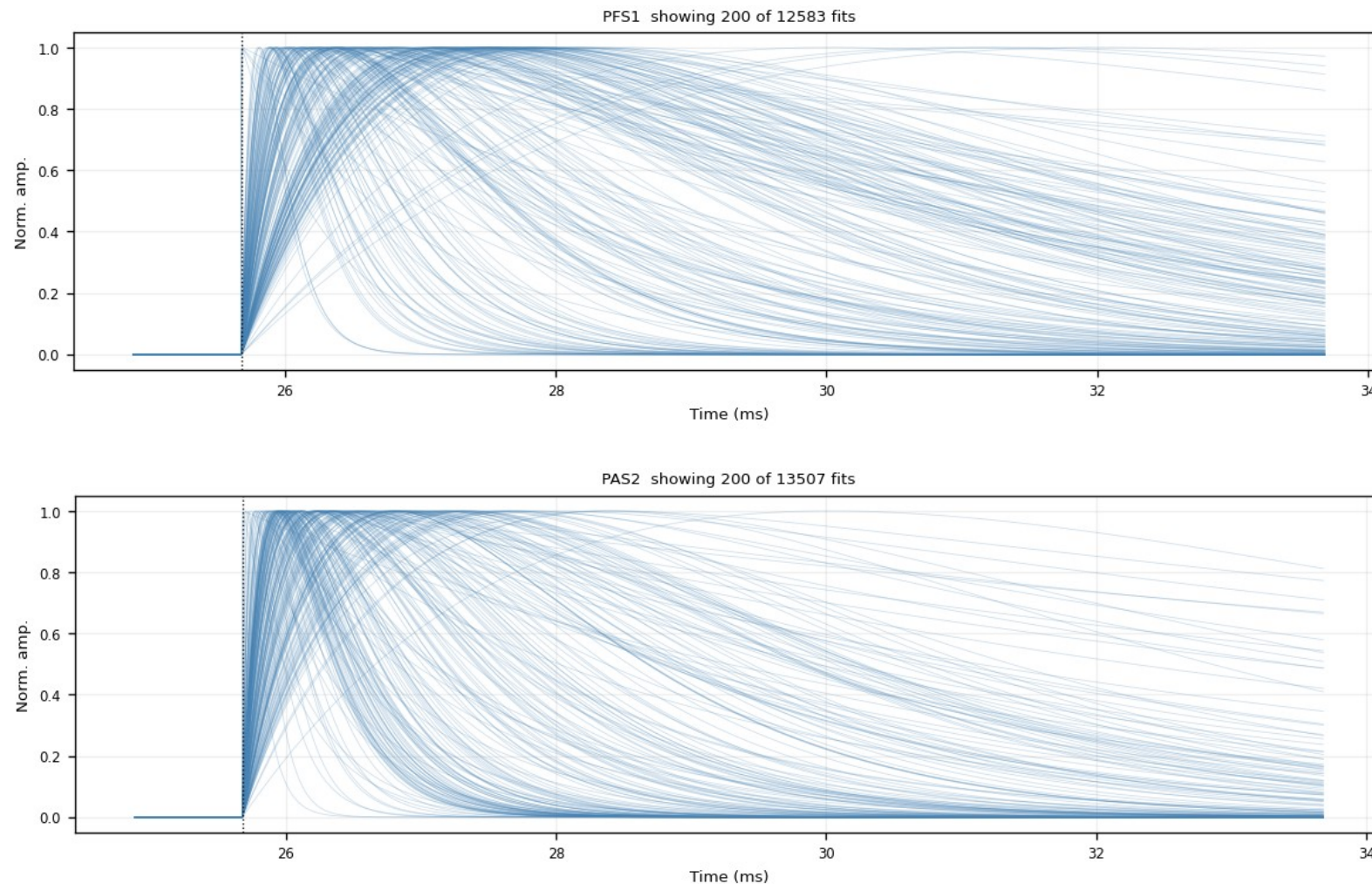
Every fit\_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



## Z4 · fitted\_curves\_overlay — fan, no cut (3/6)

figure: zip4\_fitted\_curves\_overlay.png — channel panels 5–6 of 11, top to bottom

Every fit\_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.

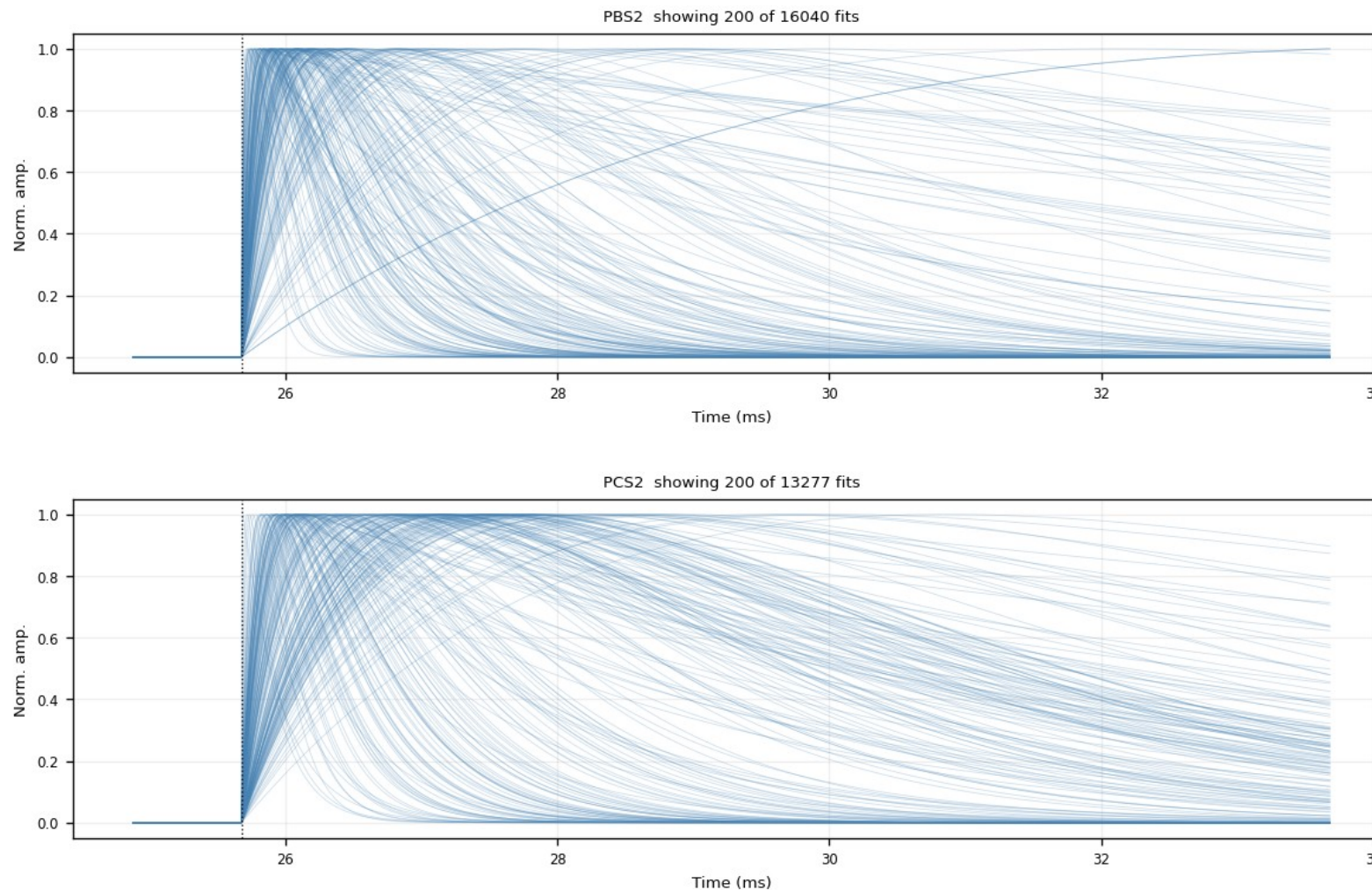




## Z4 · fitted\_curves\_overlay — fan, no cut (4/6)

figure: zip4\_fitted\_curves\_overlay.png — channel panels 7–8 of 11, top to bottom

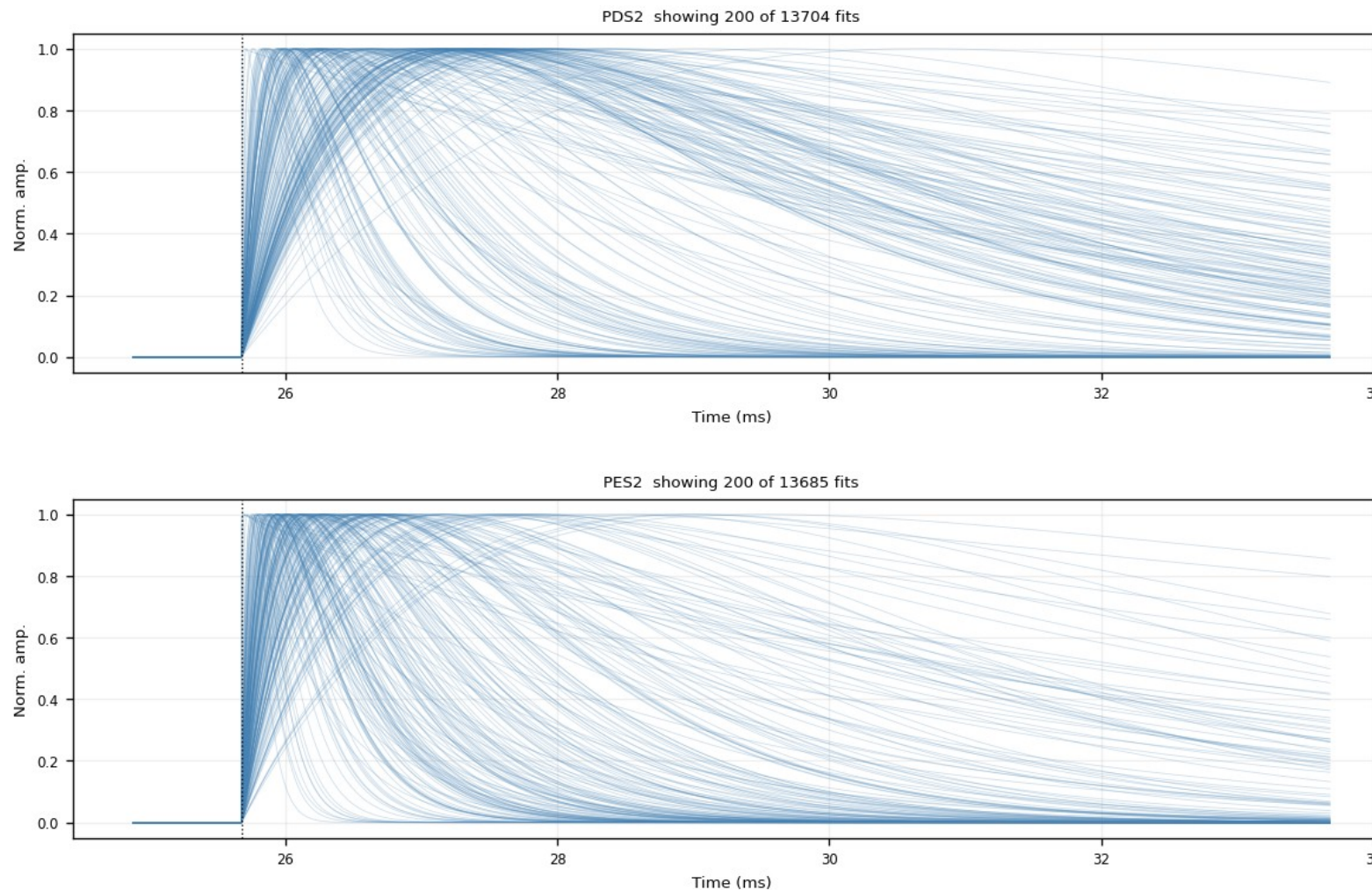
Every fit\_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



## Z4 · fitted\_curves\_overlay — fan, no cut (5/6)

figure: zip4\_fitted\_curves\_overlay.png — channel panels 9–10 of 11, top to bottom

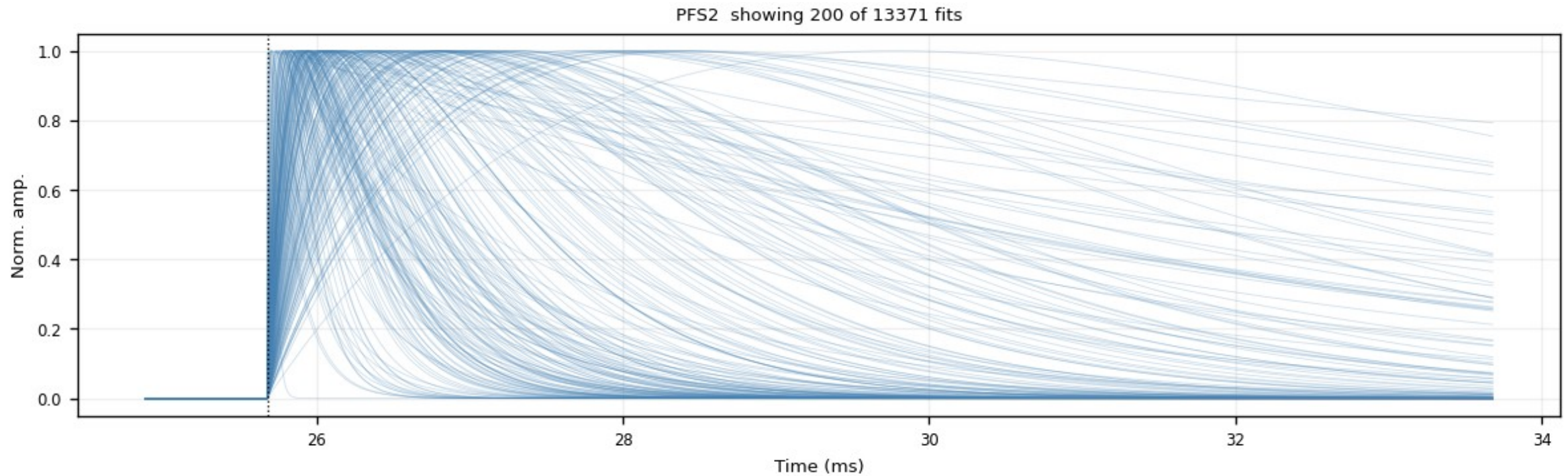
Every fit\_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



## Z4 · fitted\_curves\_overlay — fan, no cut (6/6)

figure: zip4\_fitted\_curves\_overlay.png — channel panels 11–11 of 11, top to bottom

Every fit\_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.

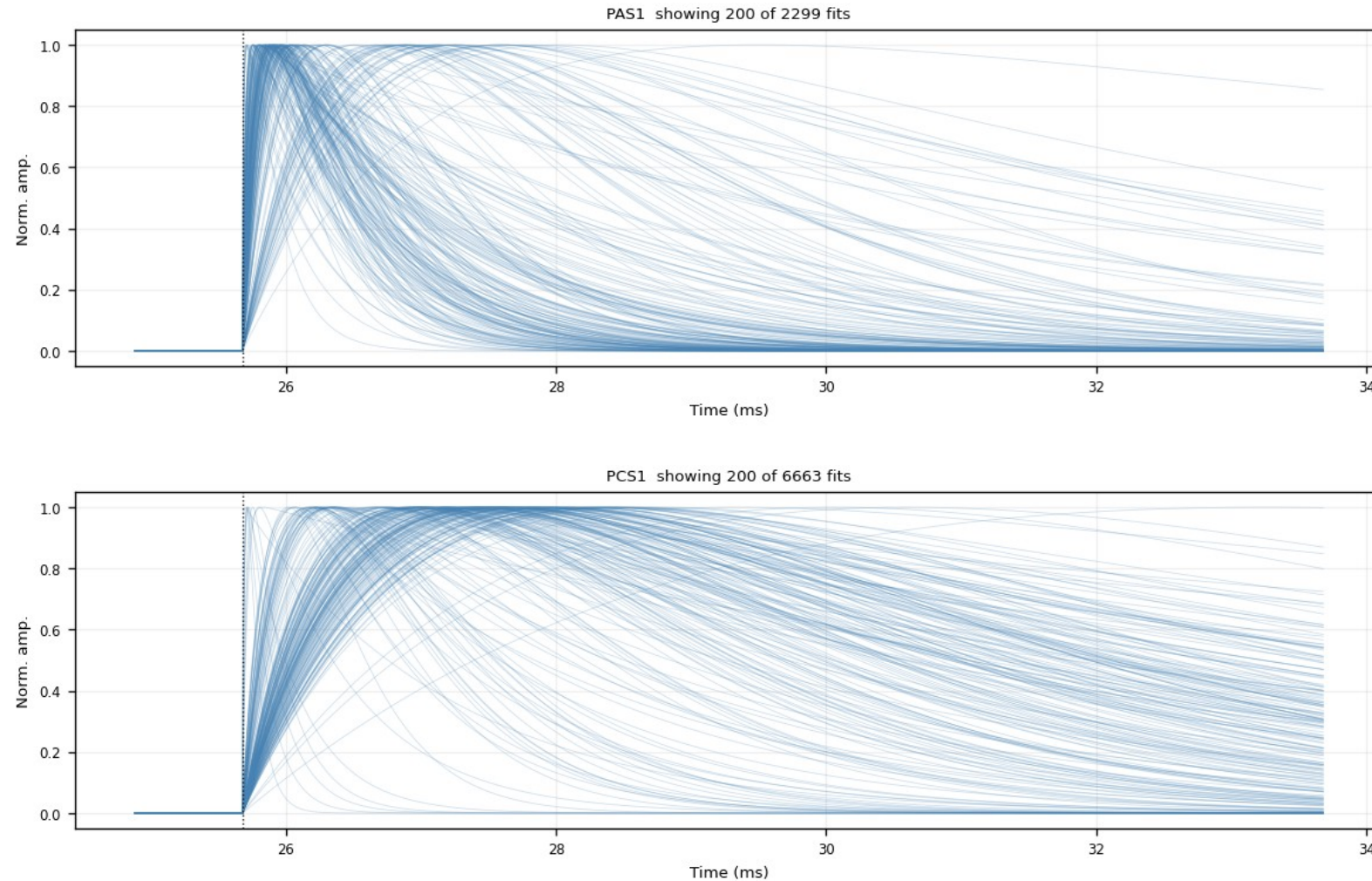




## Z4 · fitted\_curves\_overlay — after $\text{NRMSE} \leq 0.4$ (1/6)

figure: zip4\_fitted\_curves\_overlay\_nrmse0.4.png — channel panels 1–2 of 11, top to bottom

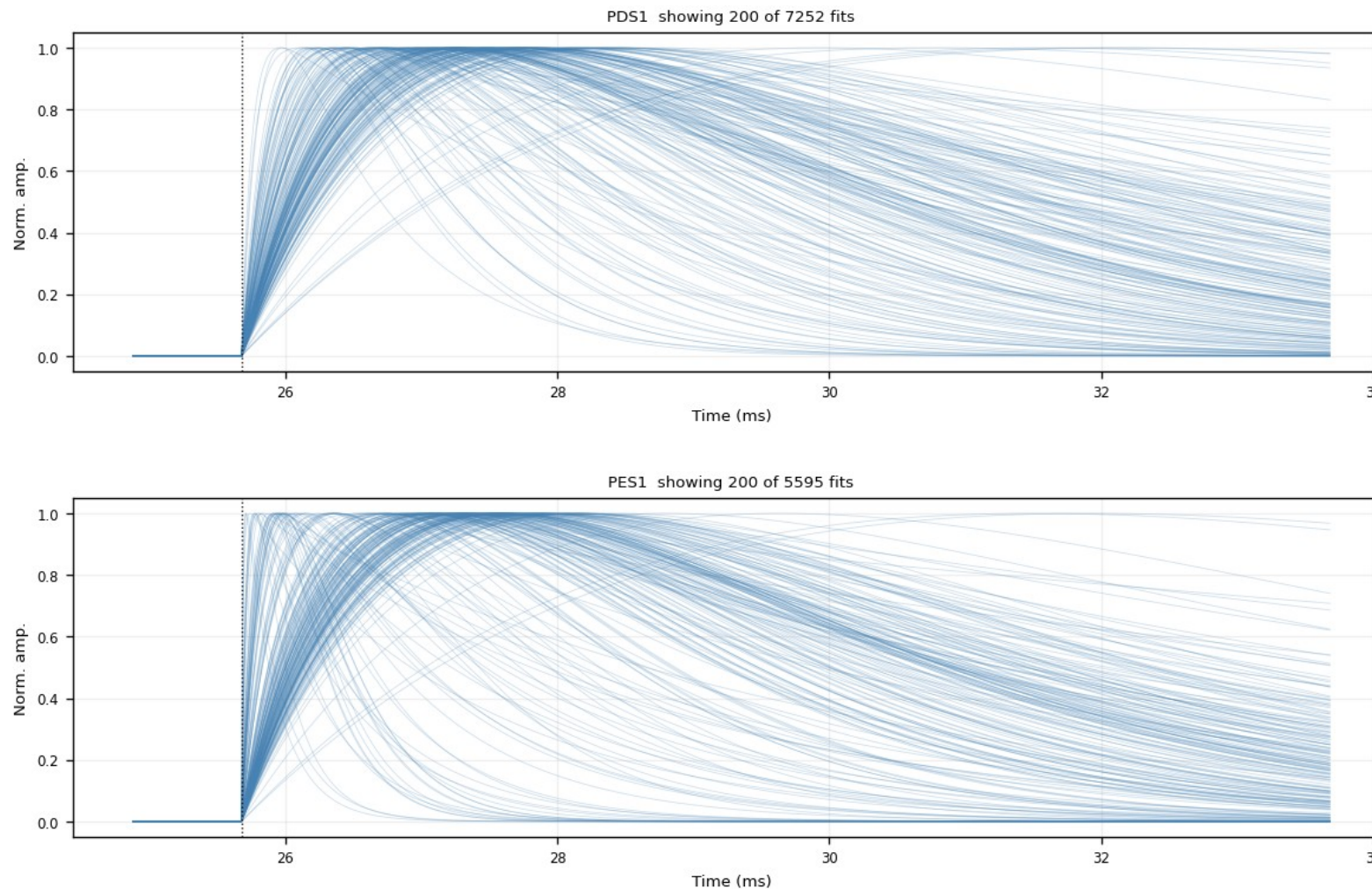
Same fan after quality cut 1: the noise fan collapses, the fast-pulse bundle remains.



## Z4 · fitted\_curves\_overlay — after $\text{NRMSE} \leq 0.4$ (2/6)

figure: zip4\_fitted\_curves\_overlay\_nrmse0.4.png — channel panels 3–4 of 11, top to bottom

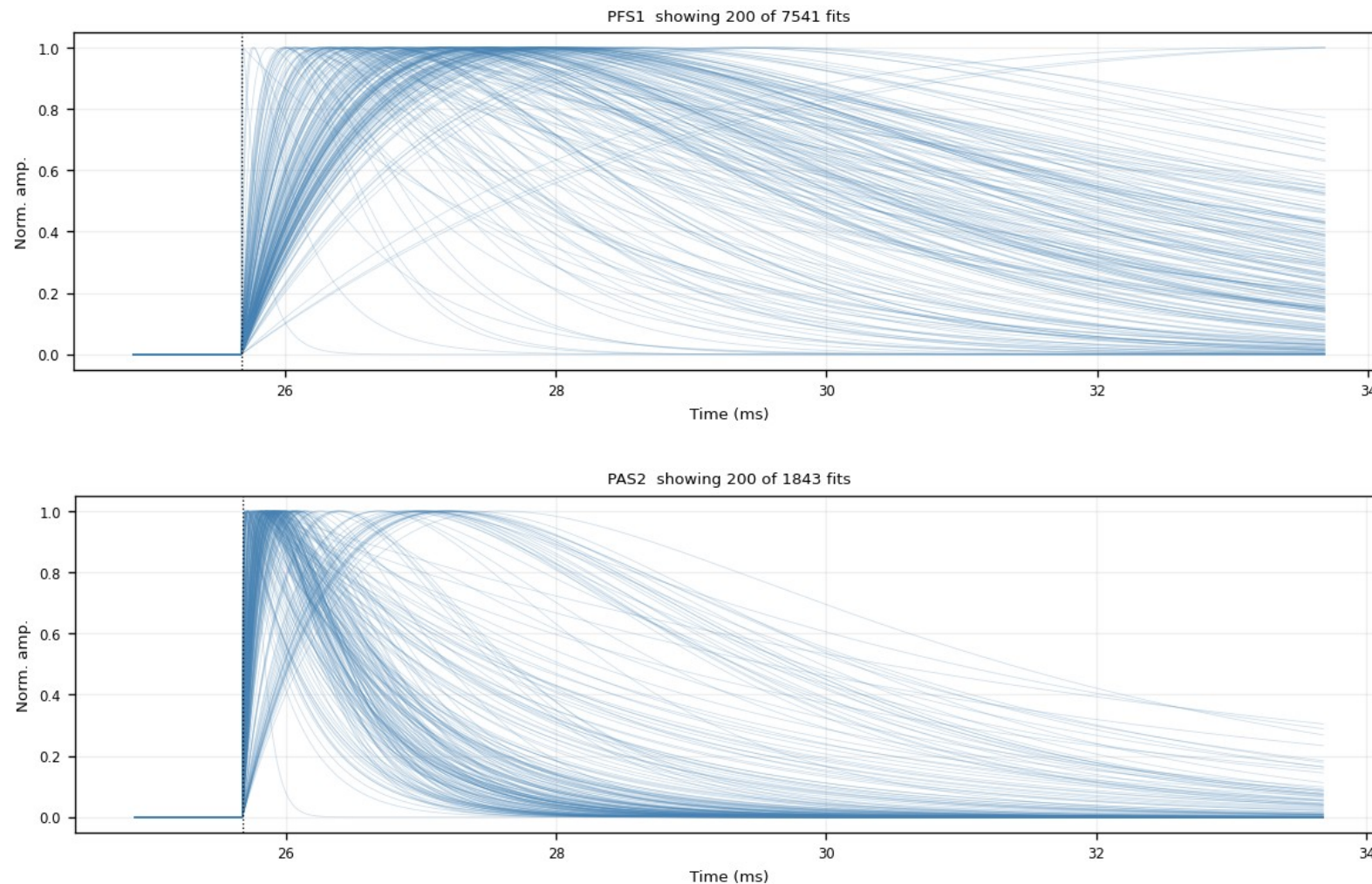
Same fan after quality cut 1: the noise fan collapses, the fast-pulse bundle remains.



## Z4 · fitted\_curves\_overlay — after $\text{NRMSE} \leq 0.4$ (3/6)

figure: zip4\_fitted\_curves\_overlay\_nrmse0.4.png — channel panels 5–6 of 11, top to bottom

Same fan after quality cut 1: the noise fan collapses, the fast-pulse bundle remains.

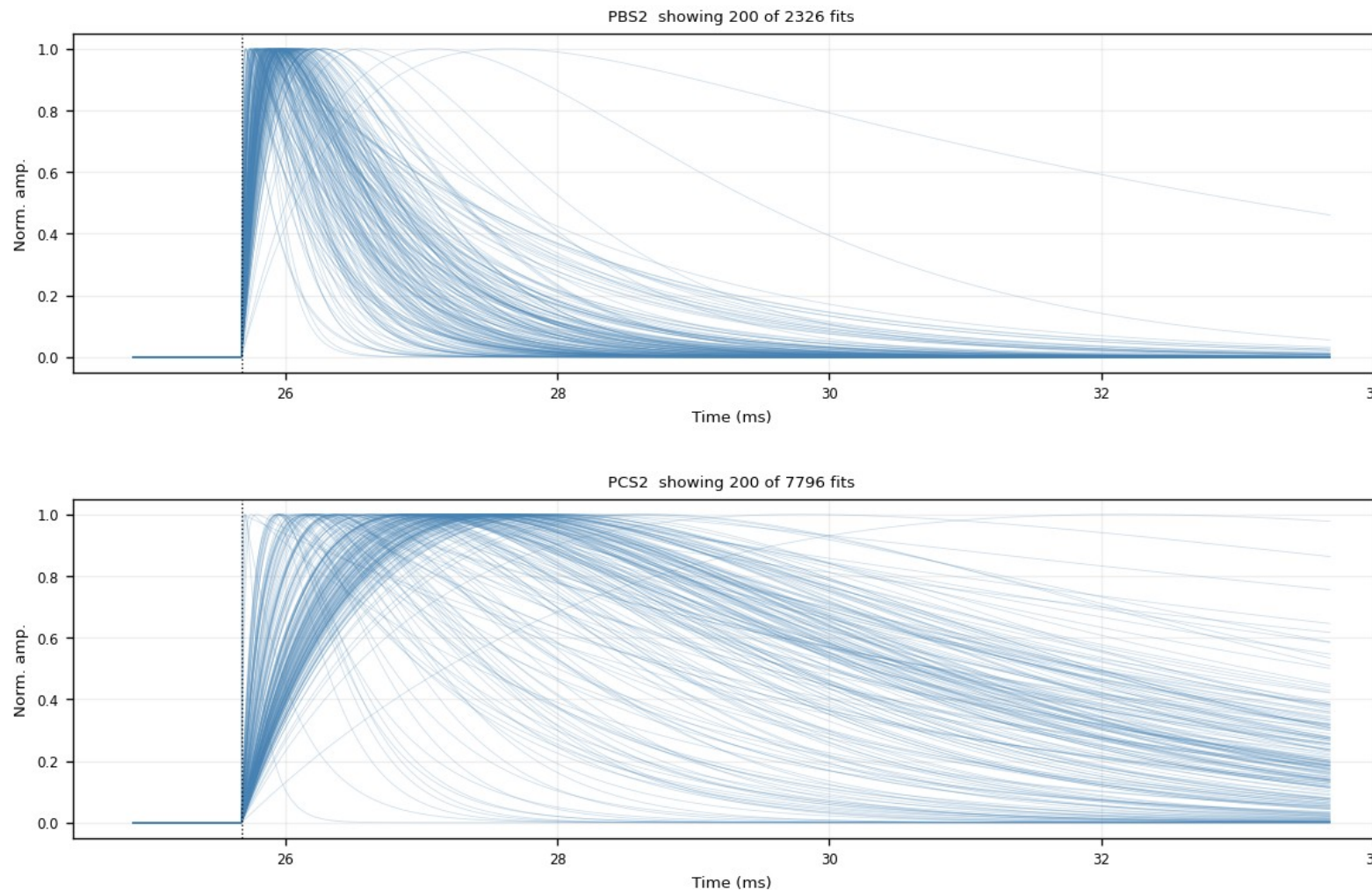




## Z4 · fitted\_curves\_overlay — after $\text{NRMSE} \leq 0.4$ (4/6)

figure: zip4\_fitted\_curves\_overlay\_nrmse0.4.png — channel panels 7–8 of 11, top to bottom

Same fan after quality cut 1: the noise fan collapses, the fast-pulse bundle remains.

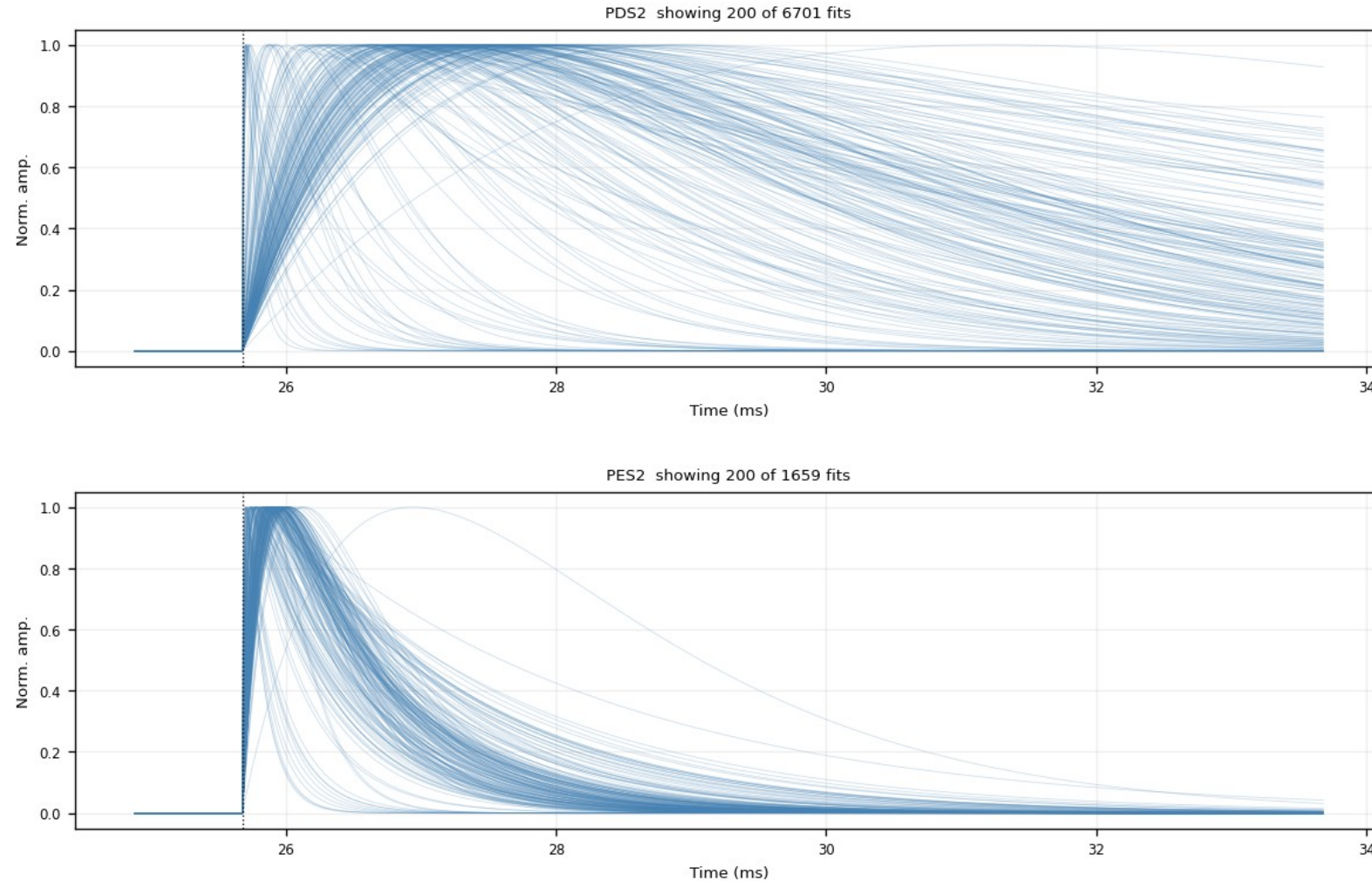




## Z4 · fitted\_curves\_overlay — after $\text{NRMSE} \leq 0.4$ (5/6)

figure: zip4\_fitted\_curves\_overlay\_nrmse0.4.png — channel panels 9–10 of 11, top to bottom

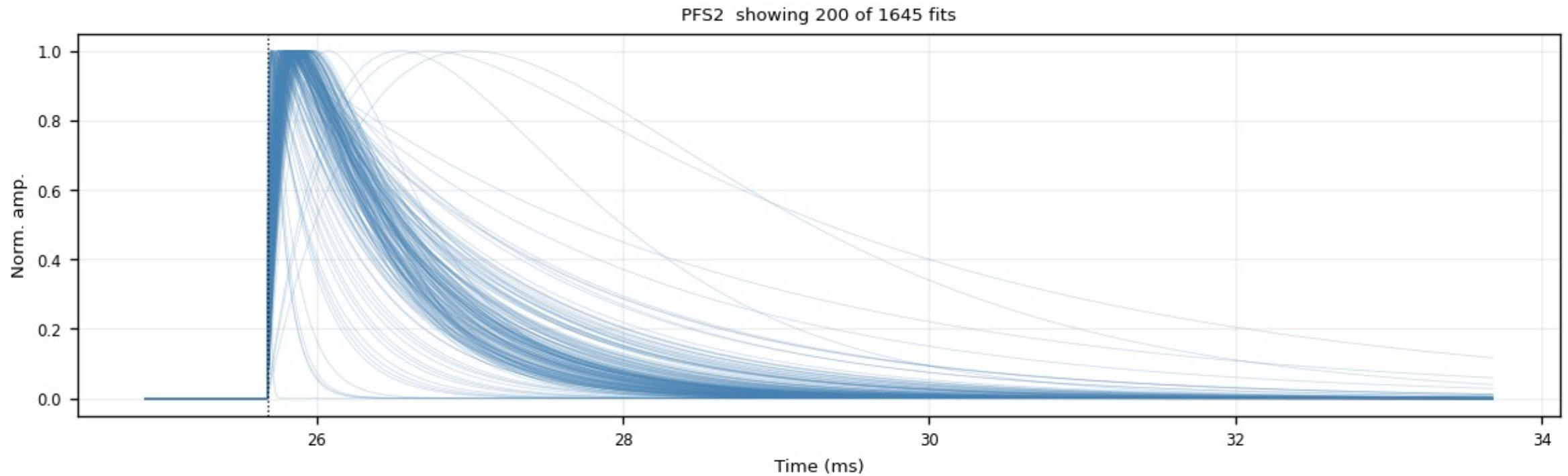
Same fan after quality cut 1: the noise fan collapses, the fast-pulse bundle remains.



## Z4 · fitted\_curves\_overlay — after $\text{NRMSE} \leq 0.4$ (6/6)

figure: zip4\_fitted\_curves\_overlay\_nrmse0.4.png — channel panels 11–11 of 11, top to bottom

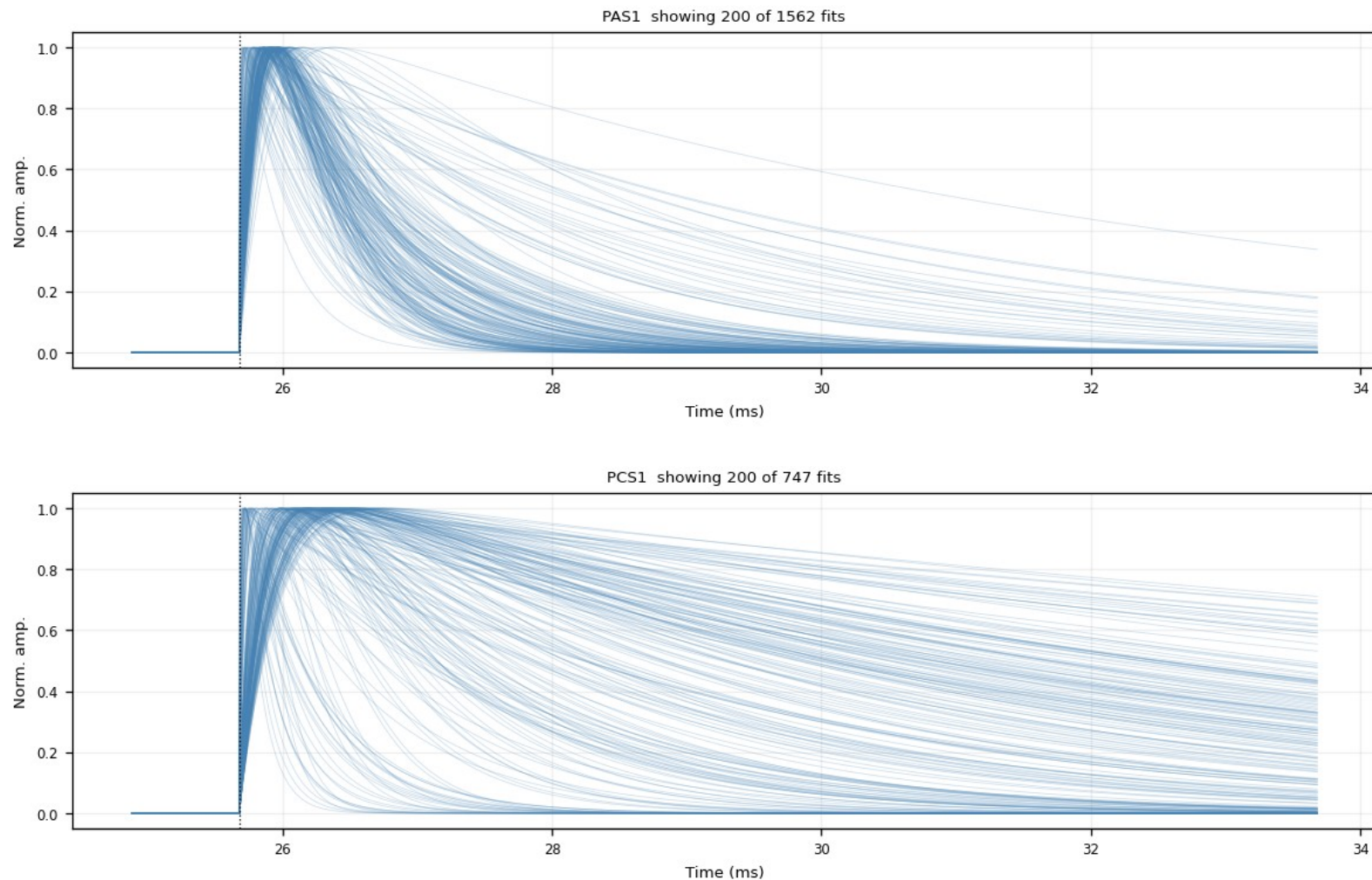
Same fan after quality cut 1: the noise fan collapses, the fast-pulse bundle remains.



## Z4 · fitted\_curves\_overlay — after both cuts (1/6)

figure: zip4\_fitted\_curves\_overlay\_nrmse0.4\_trise0.30ms.png — channel panels 1–2 of 11, top to bottom

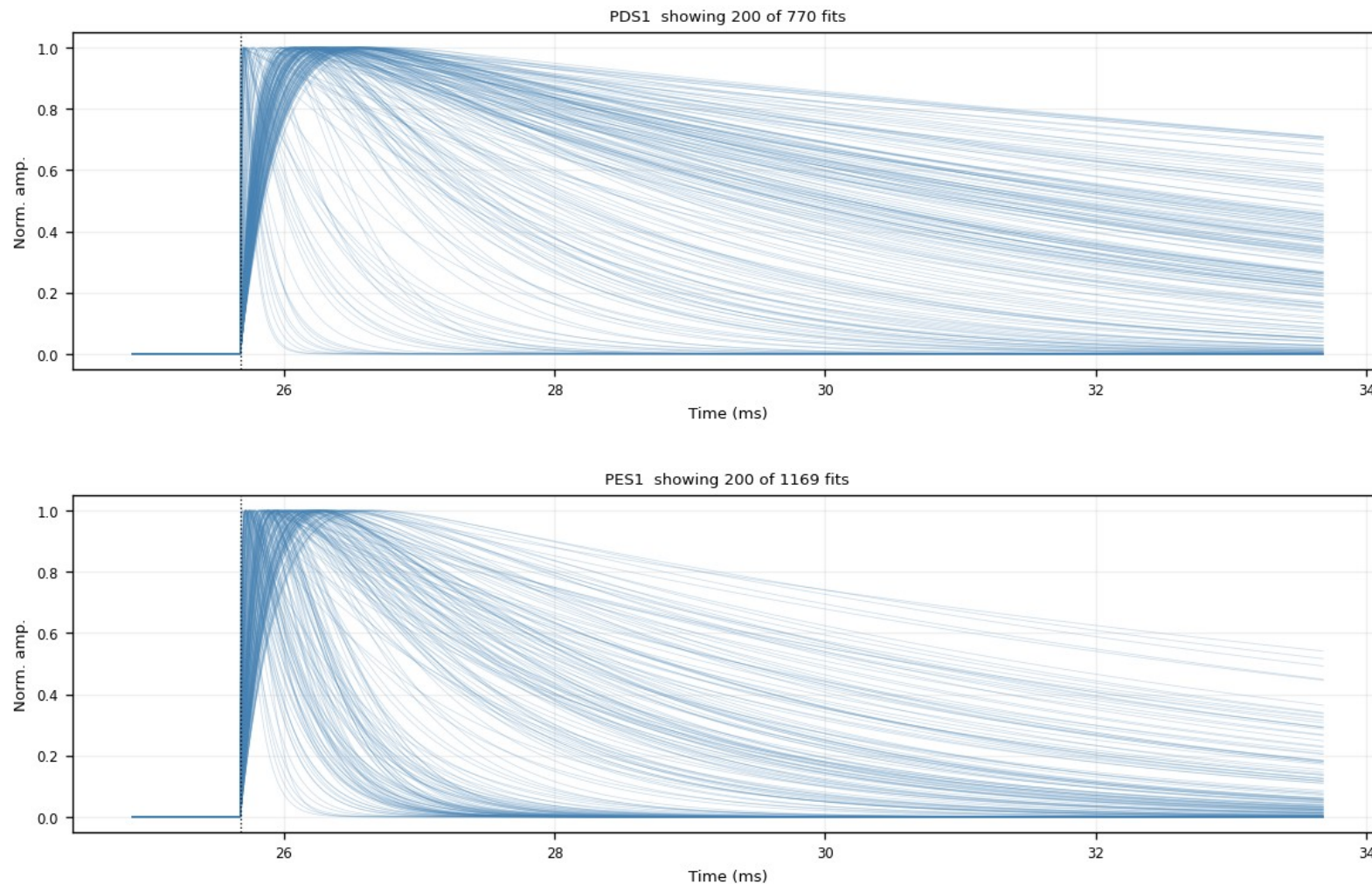
After  $\text{NRMSE} \leq 0.4$  AND  $\tau_{\text{rise}} \leq 0.3$  ms — the exact PCA input population: the clean fast-pulse bundle only.



## Z4 · fitted\_curves\_overlay — after both cuts (2/6)

figure: zip4\_fitted\_curves\_overlay\_nrmse0.4\_trise0.30ms.png — channel panels 3–4 of 11, top to bottom

After  $\text{NRMSE} \leq 0.4$  AND  $\tau_{\text{rise}} \leq 0.3$  ms — the exact PCA input population: the clean fast-pulse bundle only.

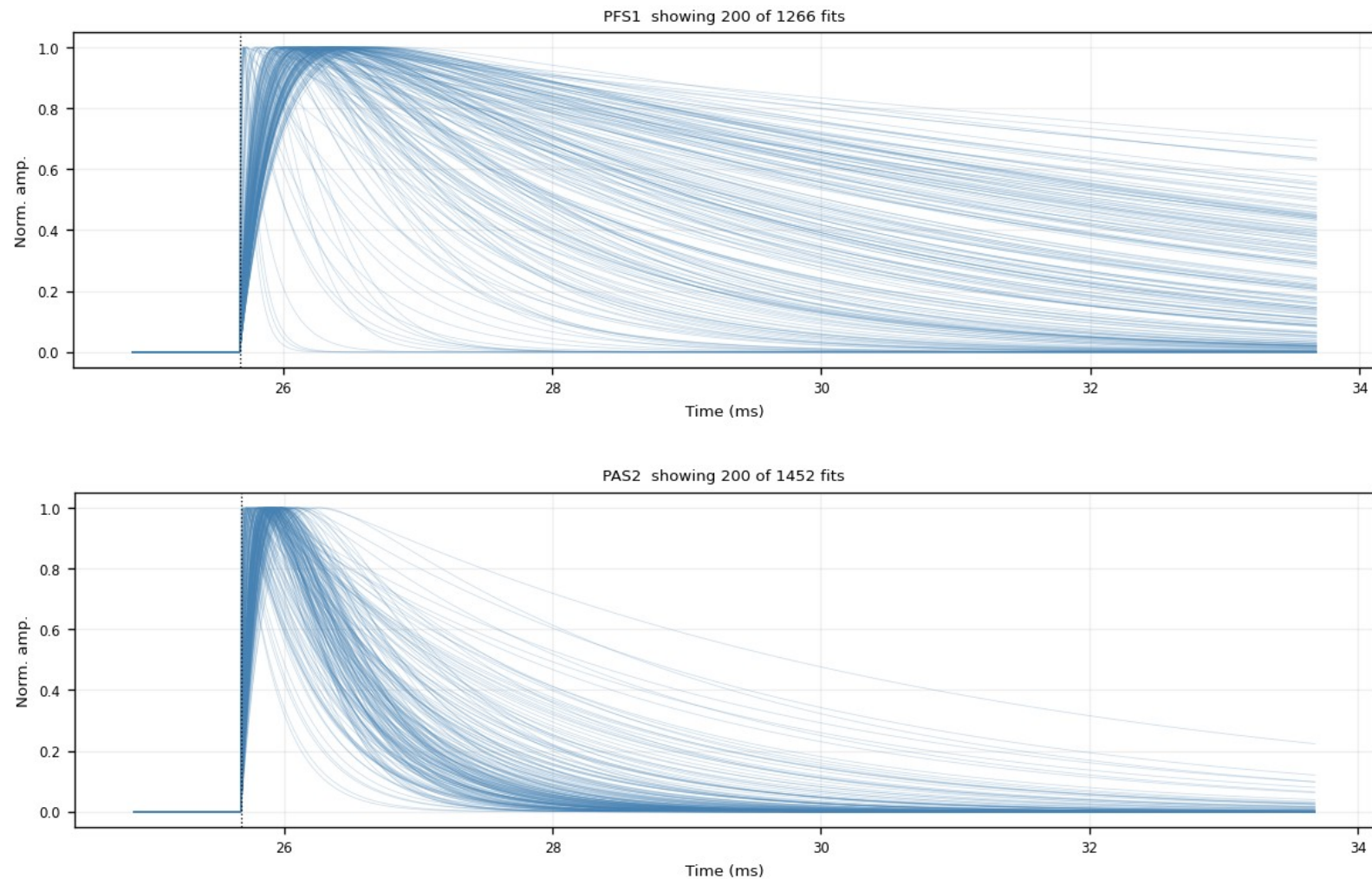




## Z4 · fitted\_curves\_overlay — after both cuts (3/6)

figure: zip4\_fitted\_curves\_overlay\_nrmse0.4\_trise0.30ms.png — channel panels 5–6 of 11, top to bottom

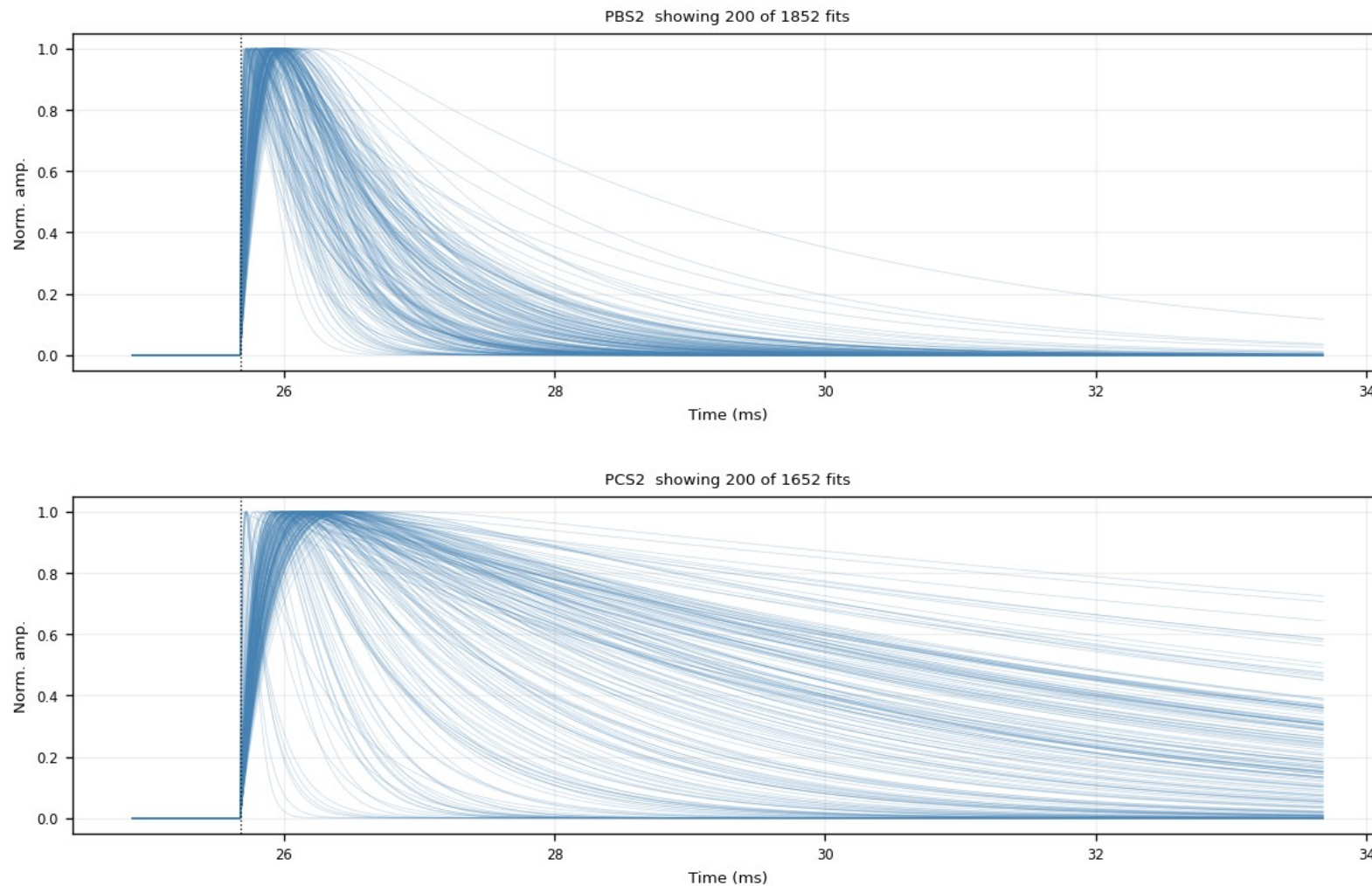
After  $\text{NRMSE} \leq 0.4$  AND  $\tau_{\text{rise}} \leq 0.3$  ms — the exact PCA input population: the clean fast-pulse bundle only.



## Z4 · fitted\_curves\_overlay — after both cuts (4/6)

figure: zip4\_fitted\_curves\_overlay\_nrmse0.4\_trise0.30ms.png — channel panels 7–8 of 11, top to bottom

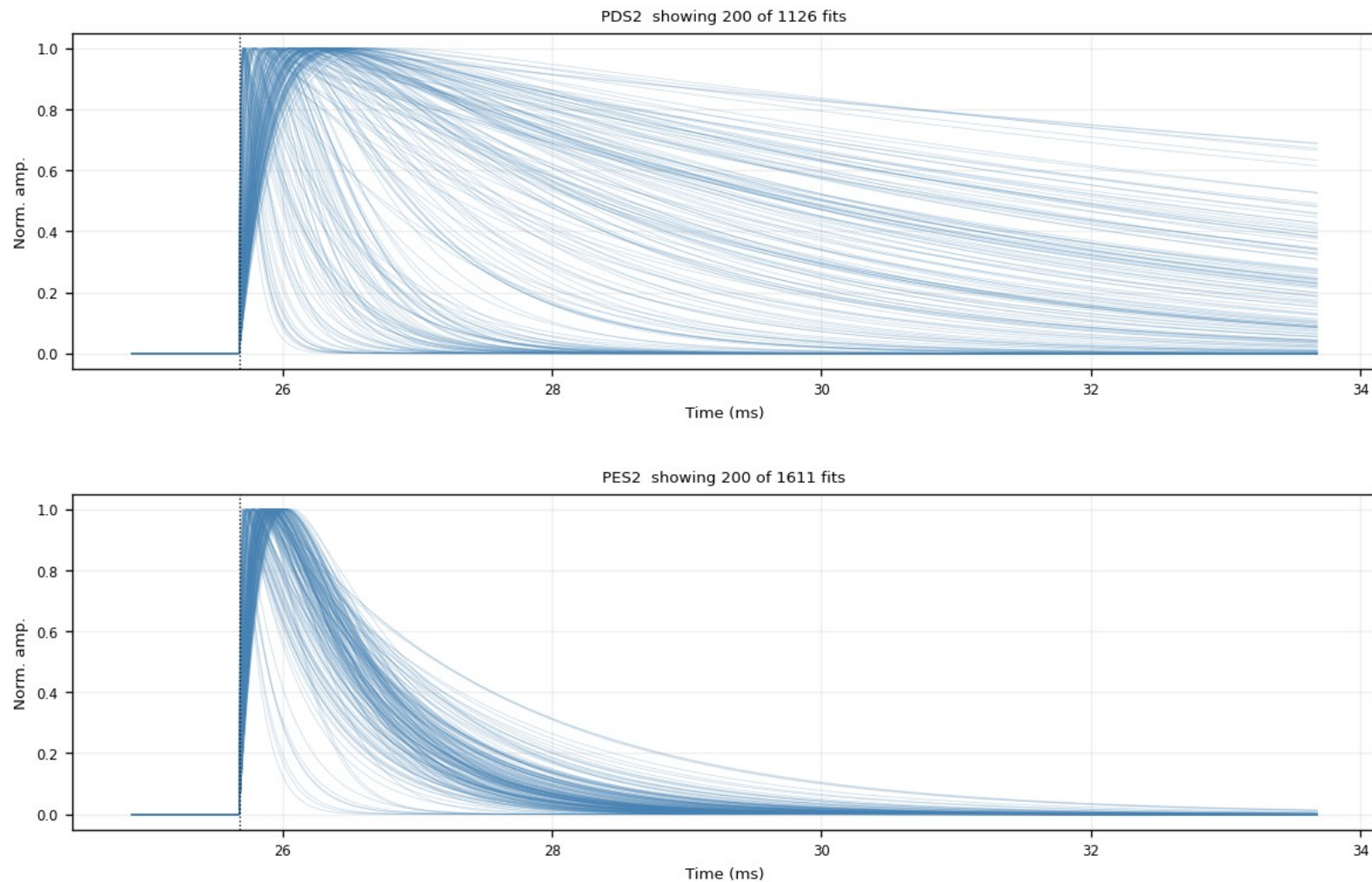
After  $\text{NRMSE} \leq 0.4$  AND  $\tau_{\text{rise}} \leq 0.3$  ms — the exact PCA input population: the clean fast-pulse bundle only.



## Z4 · fitted\_curves\_overlay — after both cuts (5/6)

figure: zip4\_fitted\_curves\_overlay\_nrmse0.4\_trise0.30ms.png — channel panels 9–10 of 11, top to bottom

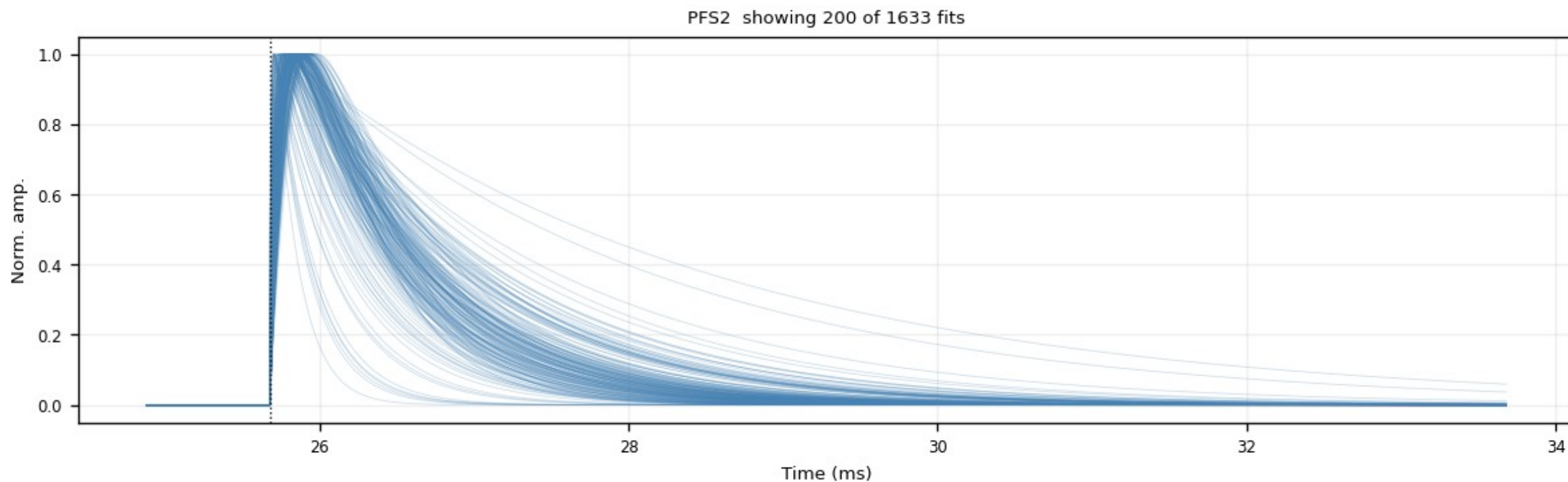
After  $\text{NRMSE} \leq 0.4$  AND  $\tau_{\text{rise}} \leq 0.3$  ms — the exact PCA input population: the clean fast-pulse bundle only.



## Z4 · fitted\_curves\_overlay — after both cuts (6/6)

figure: zip4\_fitted\_curves\_overlay\_nrmse0.4\_trise0.30ms.png — channel panels 11–11 of 11, top to bottom

After  $\text{NRMSE} \leq 0.4$  AND  $\tau_{\text{rise}} \leq 0.3$  ms — the exact PCA input population: the clean fast-pulse bundle only.

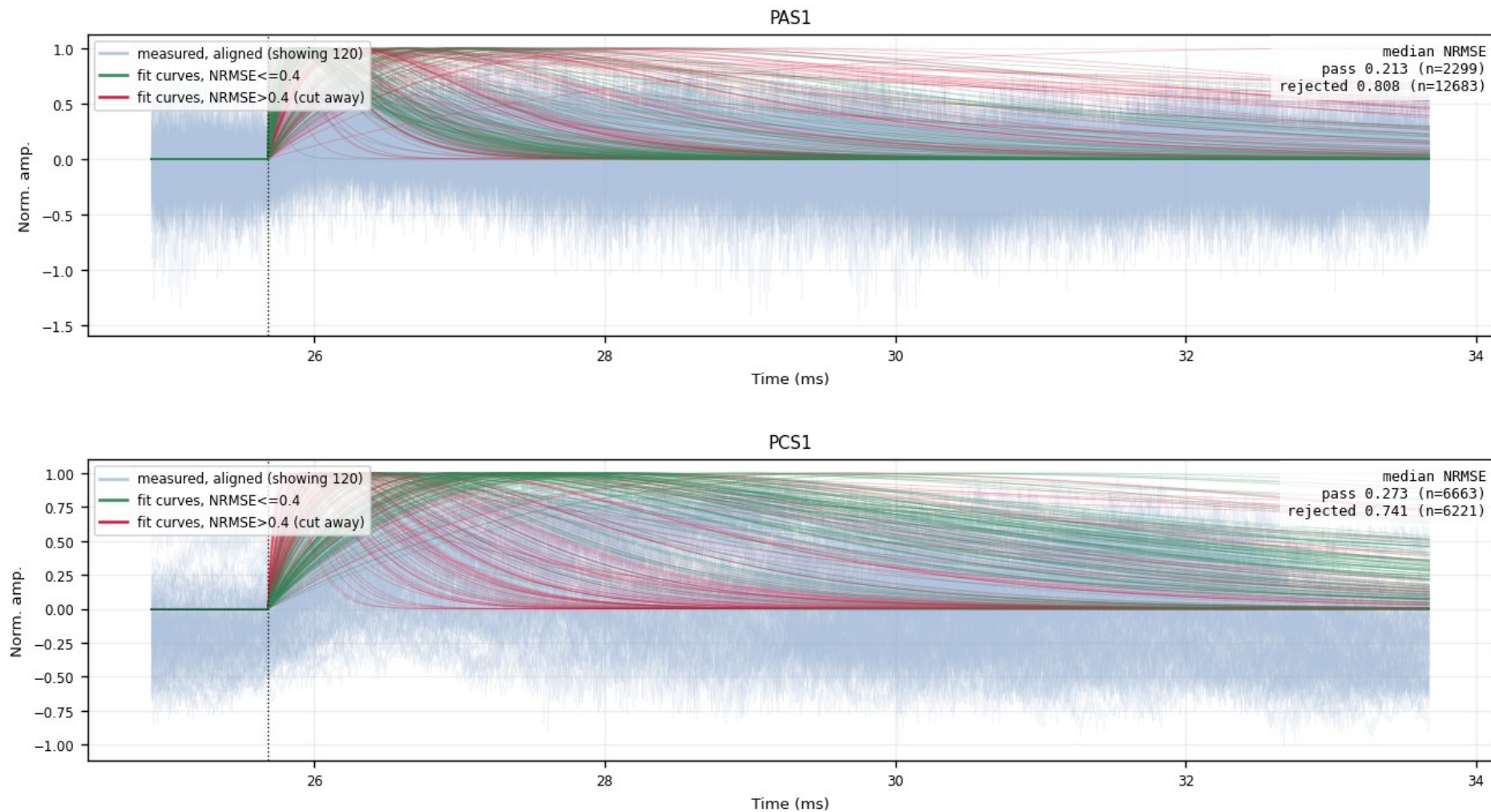




# Z4 · overlay\_fan\_cut — kept vs rejected fits (1/6)

figure: zip4\_overlay\_fan\_cut\_nrmse0.4.png — channel panels 1–2 of 11, top to bottom

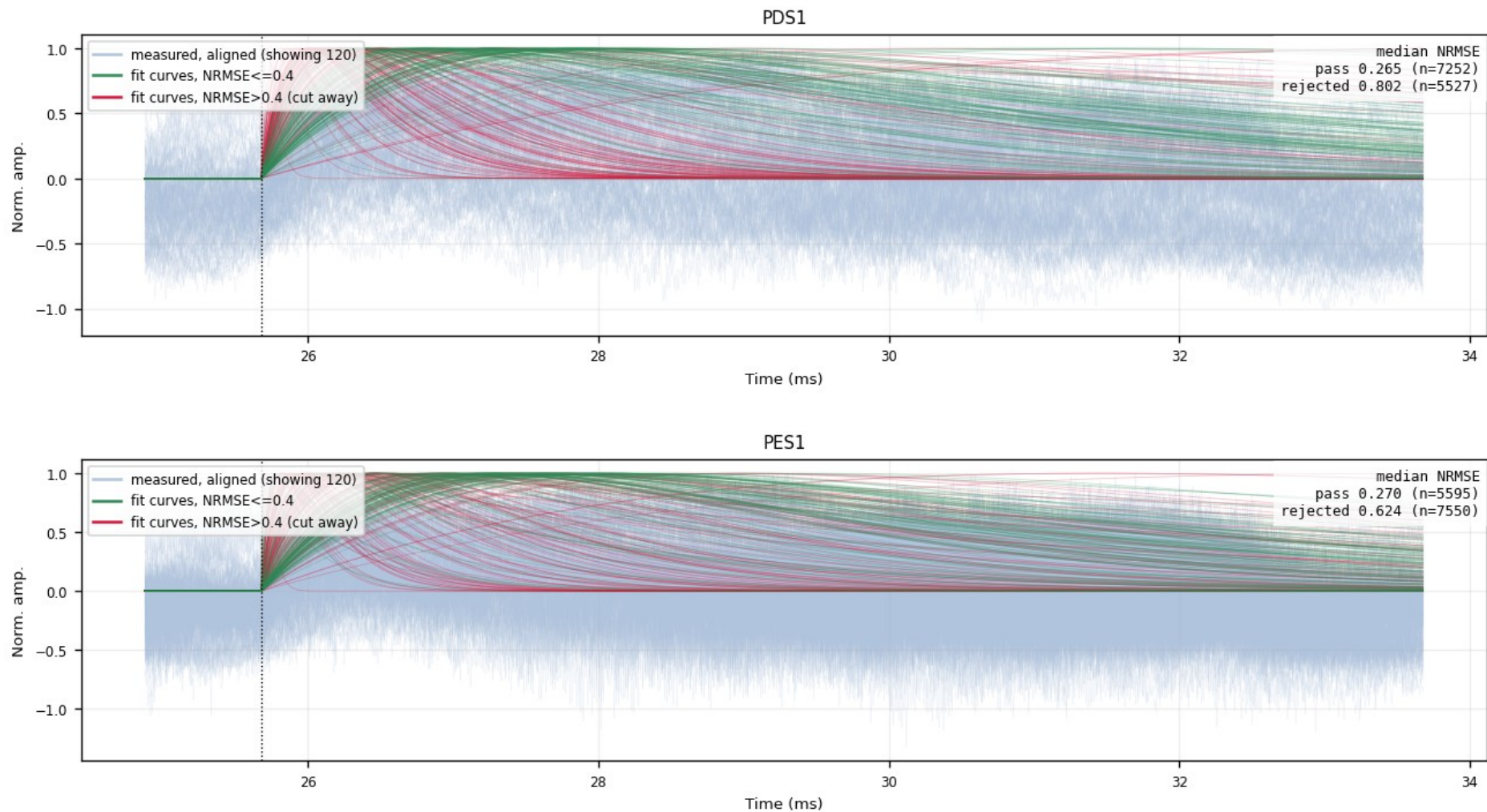
Gray-blue = aligned measured traces; green = fits with  $\text{NRMSE} \leq 0.4$  (kept); red =  $\text{NRMSE} > 0.4$  (cut away). Median NRMSE and counts of both populations stamped per channel.



## Z4 · overlay\_fan\_cut — kept vs rejected fits (2/6)

figure: zip4\_overlay\_fan\_cut\_nrmse0.4.png — channel panels 3–4 of 11, top to bottom

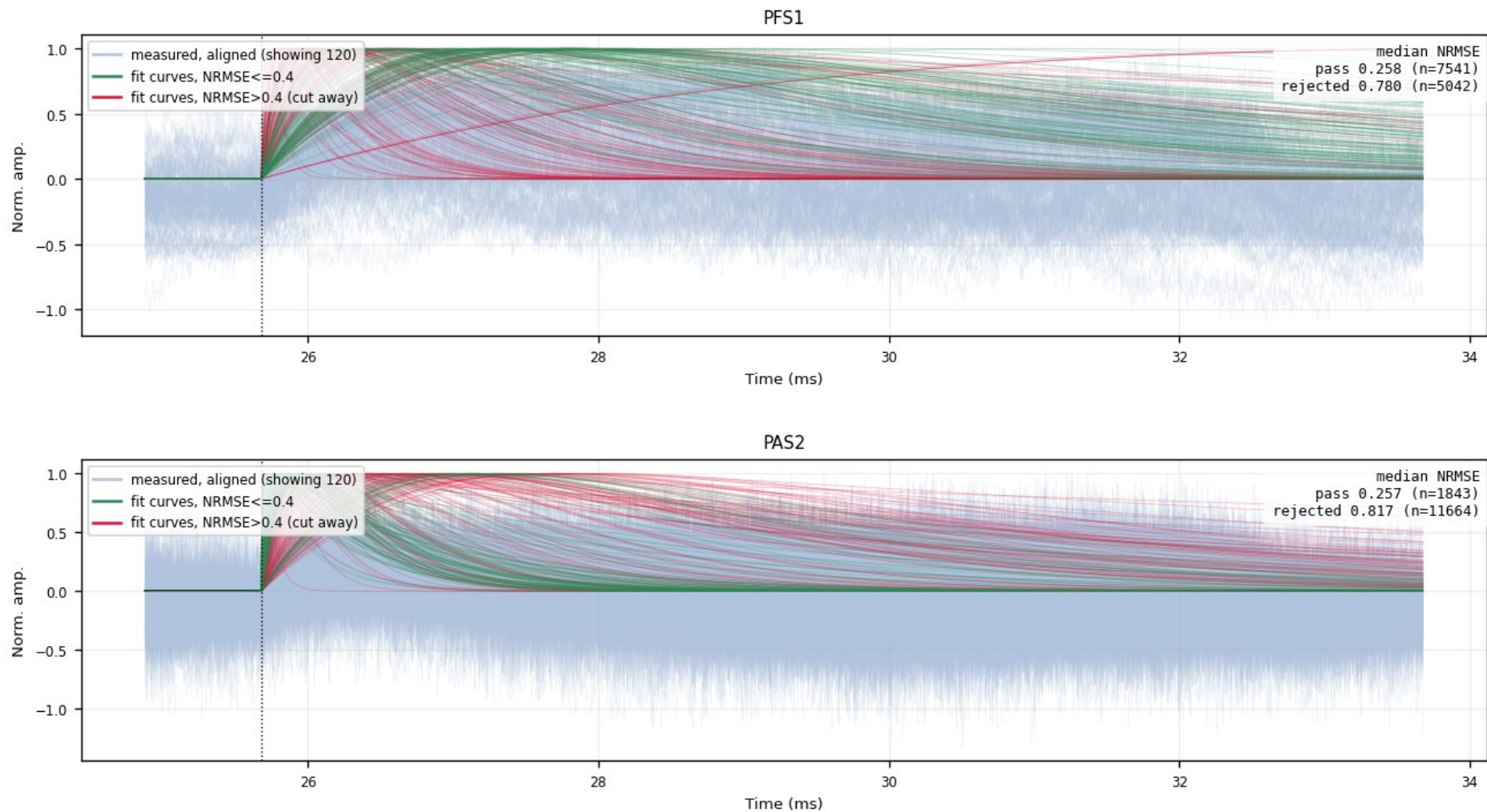
Gray-blue = aligned measured traces; green = fits with  $\text{NRMSE} \leq 0.4$  (kept); red =  $\text{NRMSE} > 0.4$  (cut away). Median NRMSE and counts of both populations stamped per channel.



## Z4 · overlay\_fan\_cut — kept vs rejected fits (3/6)

figure: zip4\_overlay\_fan\_cut\_nrmse0.4.png — channel panels 5–6 of 11, top to bottom

Gray-blue = aligned measured traces; green = fits with  $\text{NRMSE} \leq 0.4$  (kept); red =  $\text{NRMSE} > 0.4$  (cut away). Median NRMSE and counts of both populations stamped per channel.

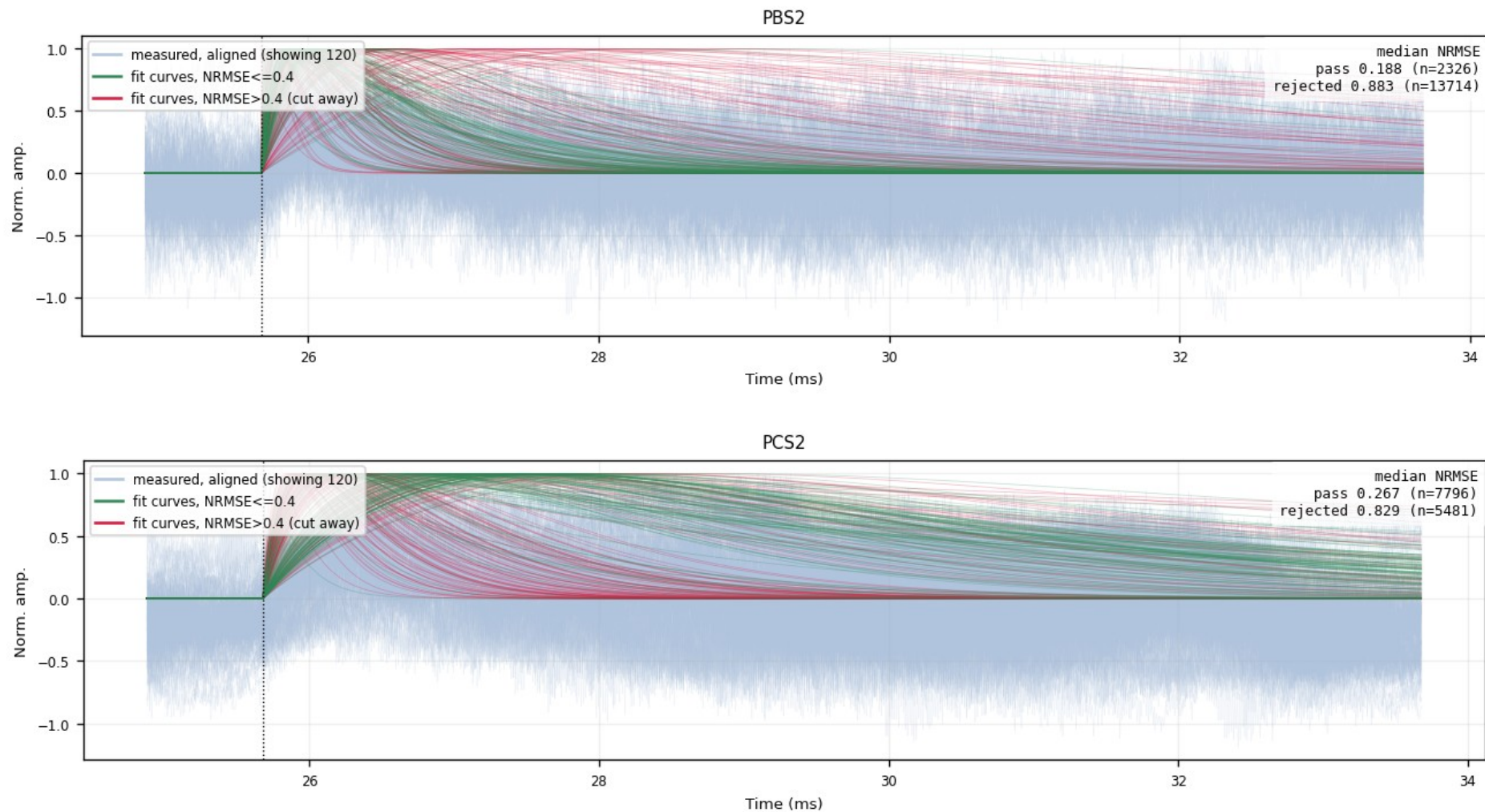




## Z4 · overlay\_fan\_cut — kept vs rejected fits (4/6)

figure: zip4\_overlay\_fan\_cut\_nrmse0.4.png — channel panels 7–8 of 11, top to bottom

Gray-blue = aligned measured traces; green = fits with  $\text{NRMSE} \leq 0.4$  (kept); red =  $\text{NRMSE} > 0.4$  (cut away). Median NRMSE and counts of both populations stamped per channel.

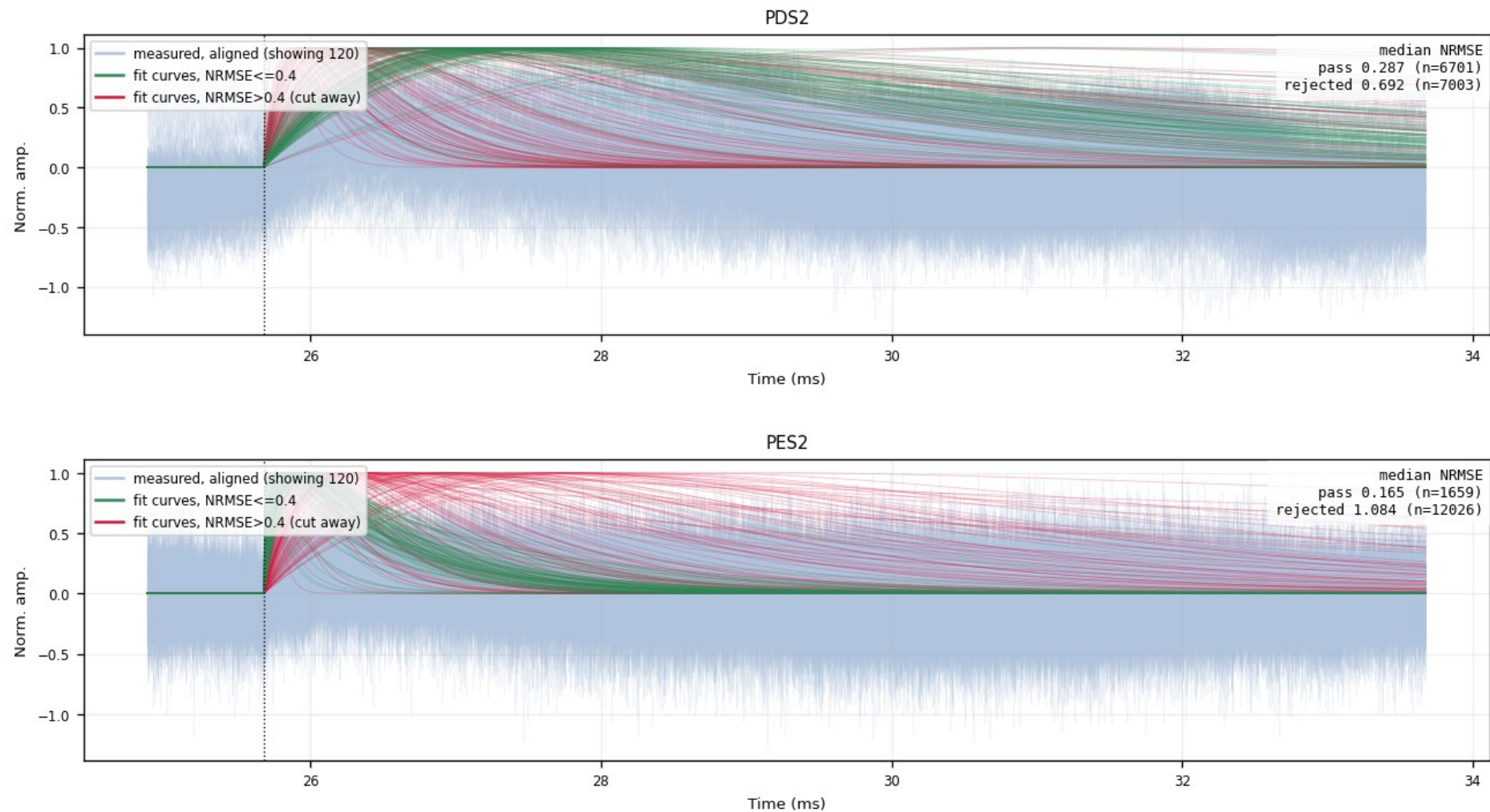




## Z4 · overlay\_fan\_cut — kept vs rejected fits (5/6)

figure: zip4\_overlay\_fan\_cut\_nrmse0.4.png — channel panels 9–10 of 11, top to bottom

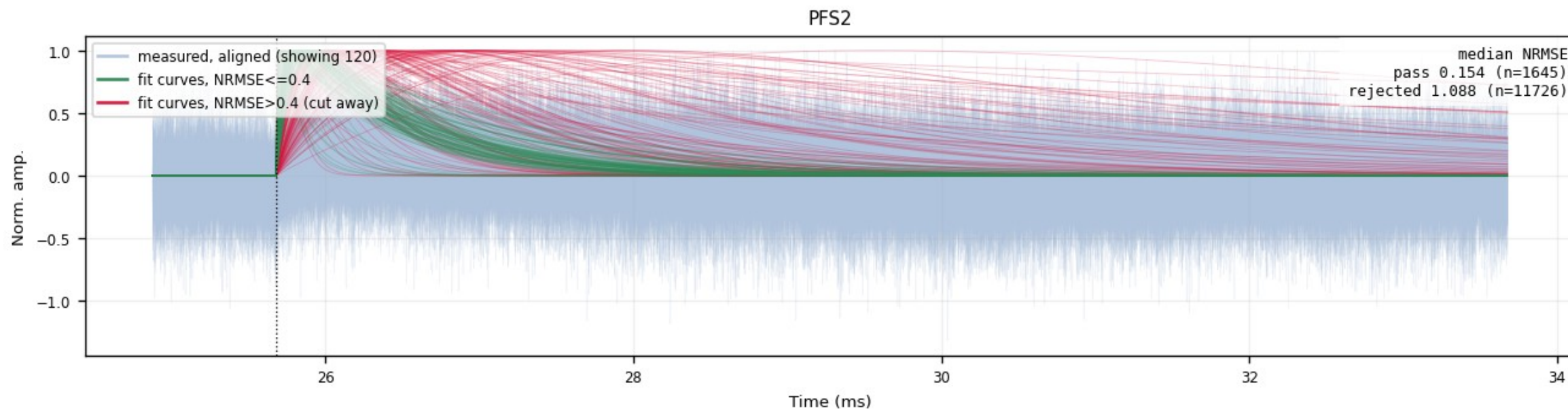
Gray-blue = aligned measured traces; green = fits with  $\text{NRMSE} \leq 0.4$  (kept); red =  $\text{NRMSE} > 0.4$  (cut away). Median NRMSE and counts of both populations stamped per channel.



## Z4 · overlay\_fan\_cut — kept vs rejected fits (6/6)

figure: zip4\_overlay\_fan\_cut\_nrmse0.4.png — channel panels 11–11 of 11, top to bottom

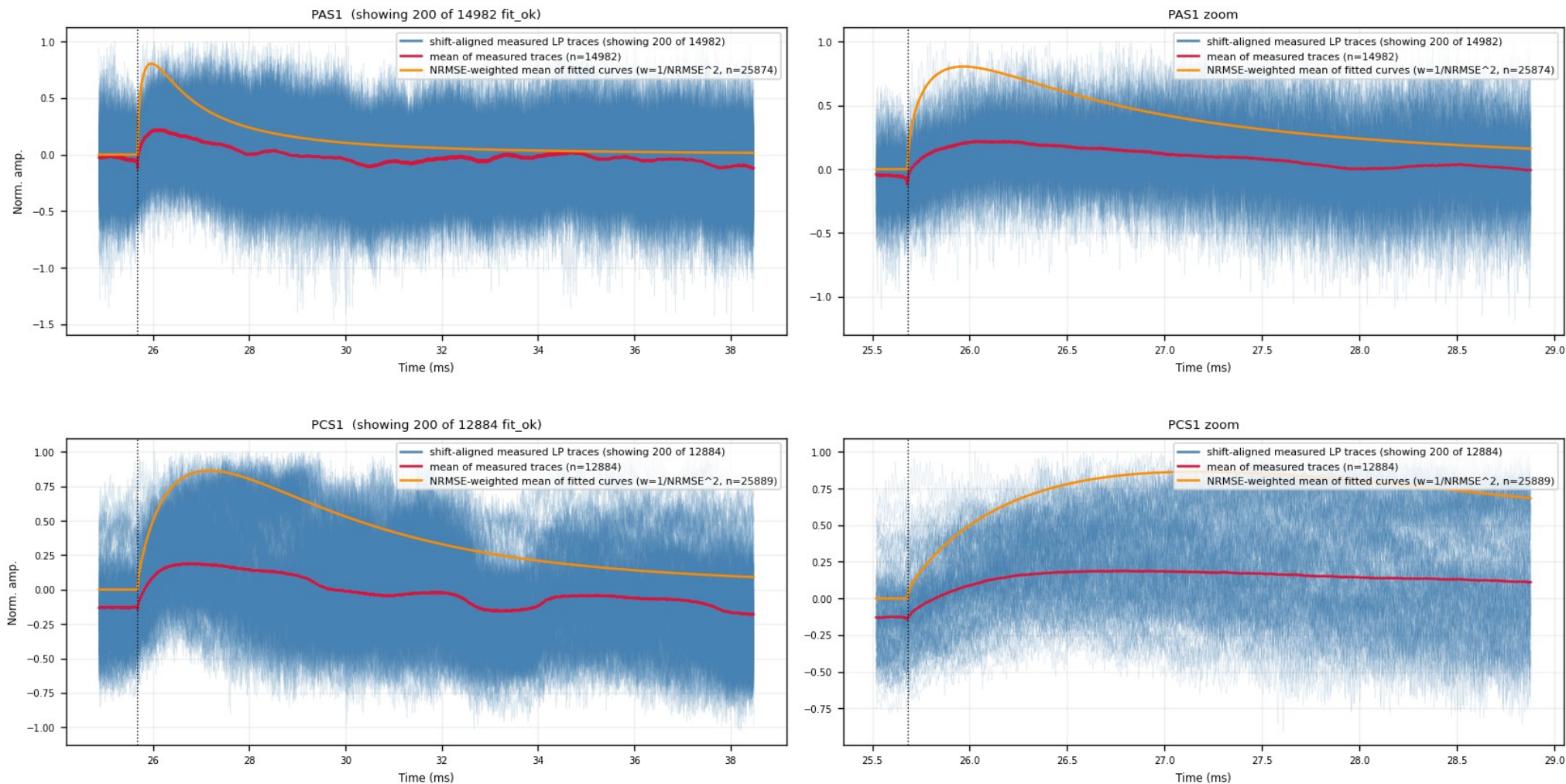
Gray-blue = aligned measured traces; green = fits with  $\text{NRMSE} \leq 0.4$  (kept); red =  $\text{NRMSE} > 0.4$  (cut away). Median NRMSE and counts of both populations stamped per channel.



# Z4 · aligned\_overlay — aligned measured traces (1/6)

figure: zip4\_lp\_aligned\_overlay.png — channel panels 1–2 of 11, top to bottom

Blue = measured LP traces shifted by (fitted pretrigger – 16050); red = mean of all fit\_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.

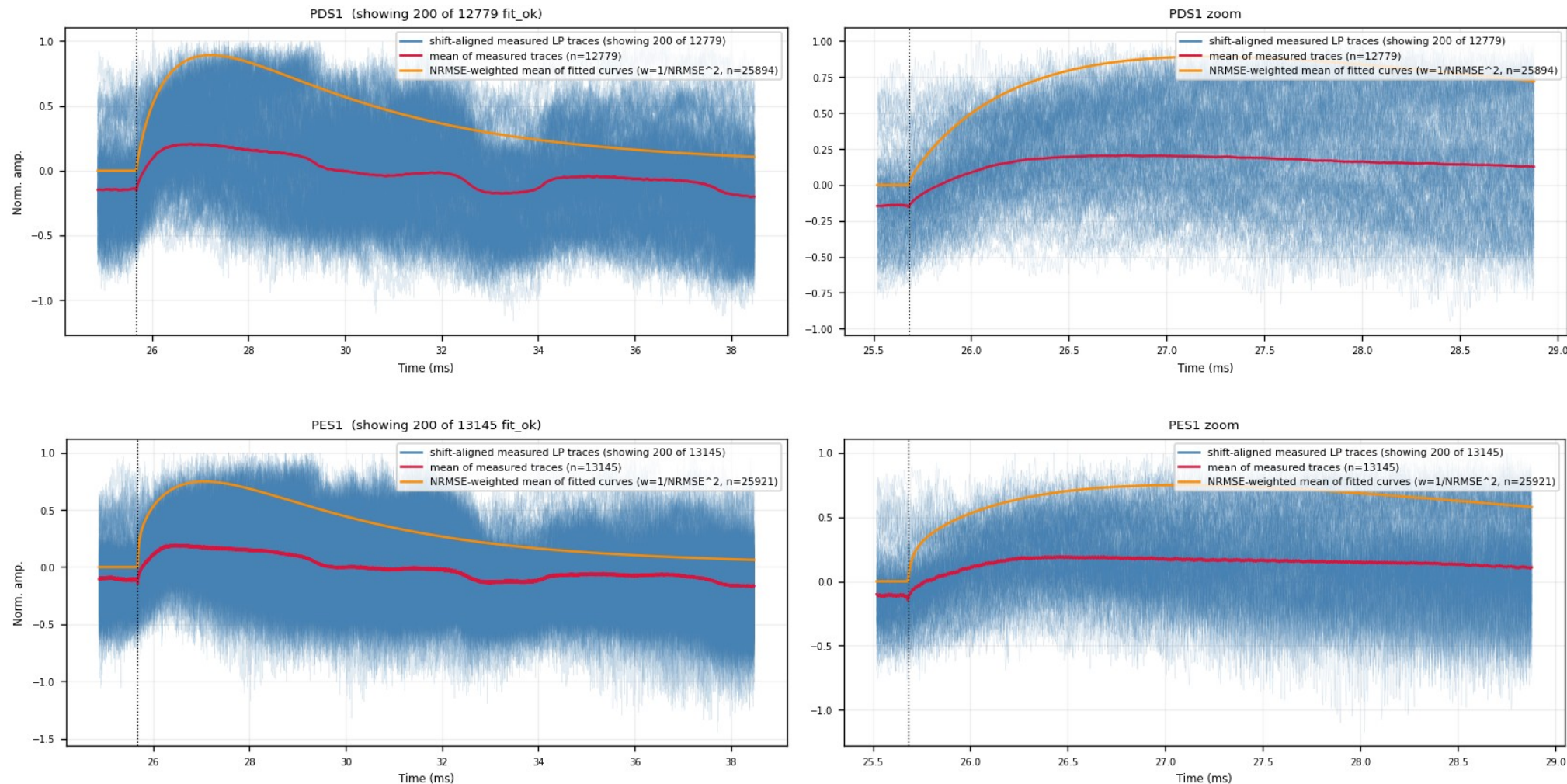




## Z4 · aligned\_overlay — aligned measured traces (2/6)

figure: zip4\_lp\_aligned\_overlay.png — channel panels 3–4 of 11, top to bottom

Blue = measured LP traces shifted by (fitted pretrigger – 16050); red = mean of all fit\_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.

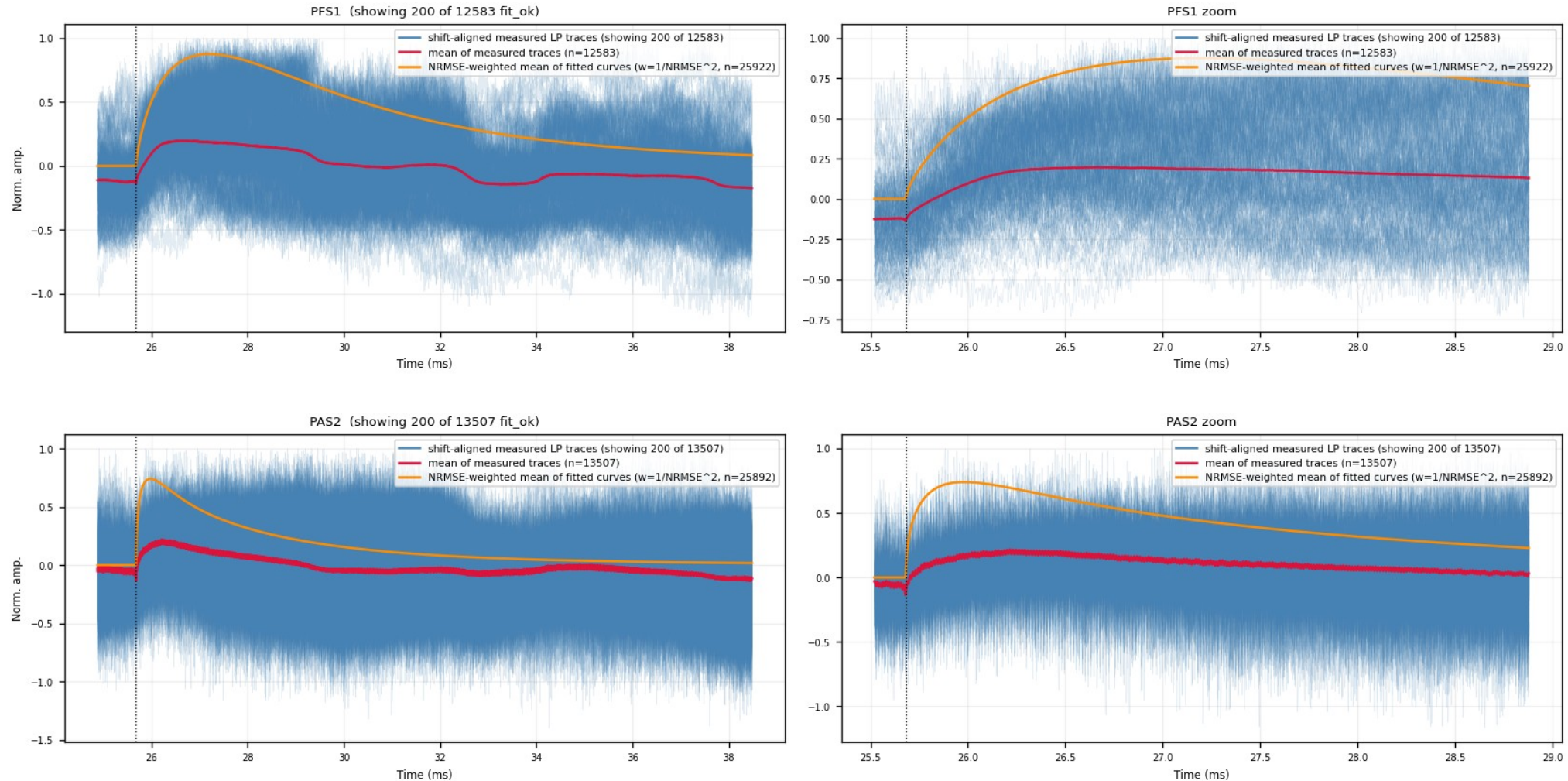




# Z4 · aligned\_overlay — aligned measured traces (3/6)

figure: zip4\_lp\_aligned\_overlay.png — channel panels 5–6 of 11, top to bottom

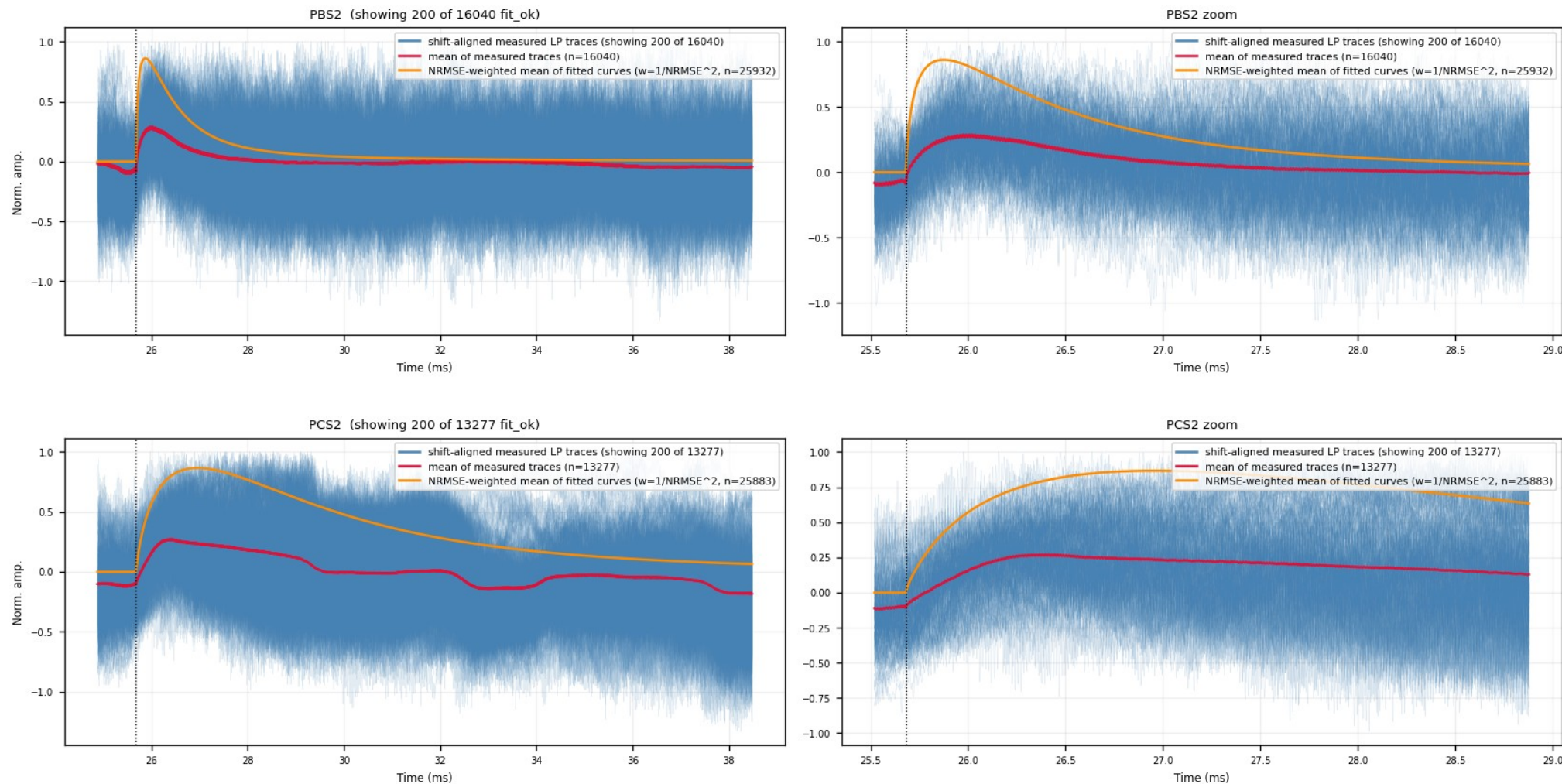
Blue = measured LP traces shifted by (fitted pretrigger - 16050); red = mean of all fit\_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



# Z4 · aligned\_overlay — aligned measured traces (4/6)

figure: zip4\_lp\_aligned\_overlay.png — channel panels 7–8 of 11, top to bottom

Blue = measured LP traces shifted by (fitted pretrigger – 16050); red = mean of all fit\_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.

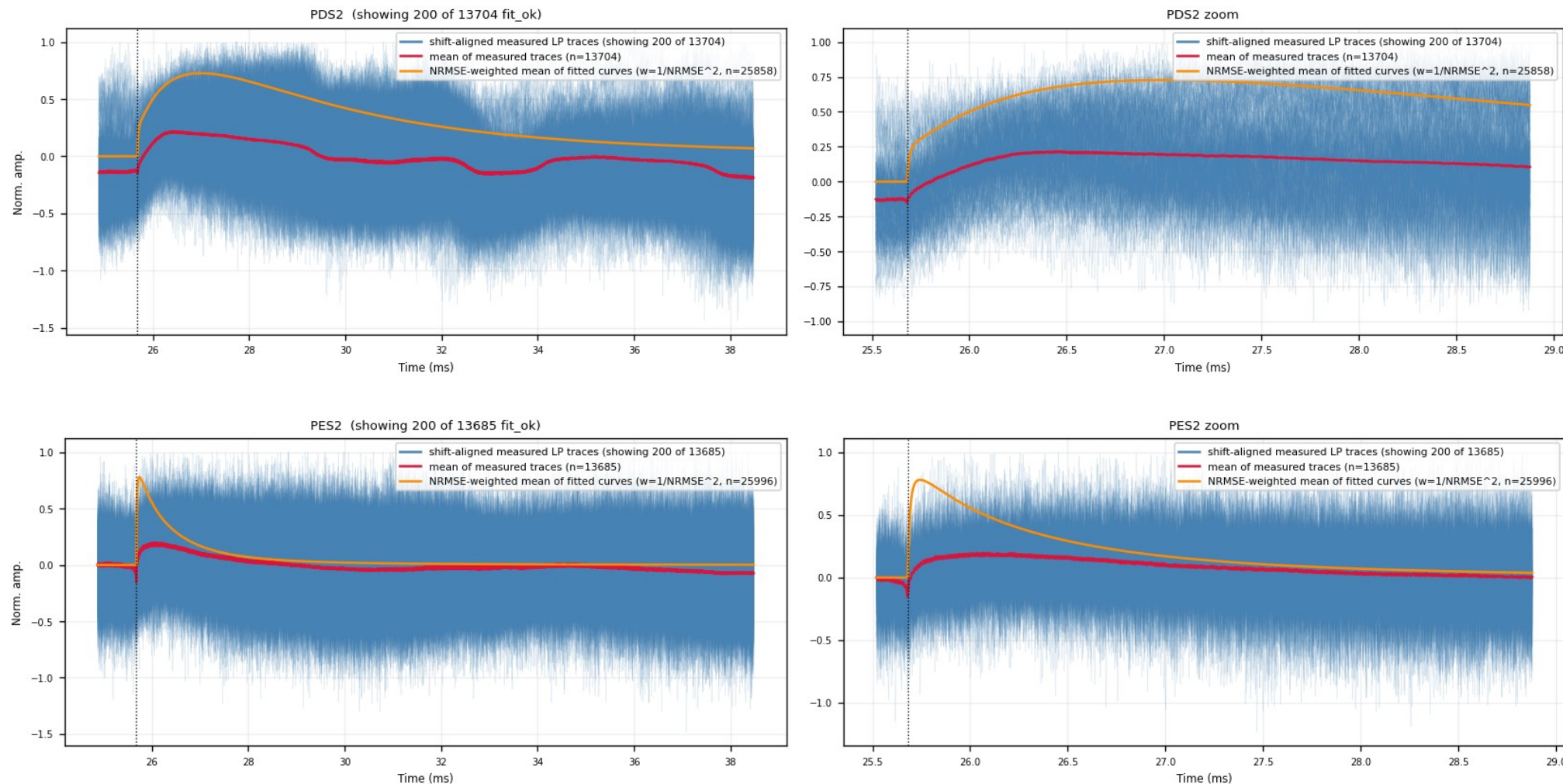




## Z4 · aligned\_overlay — aligned measured traces (5/6)

figure: zip4\_lp\_aligned\_overlay.png — channel panels 9–10 of 11, top to bottom

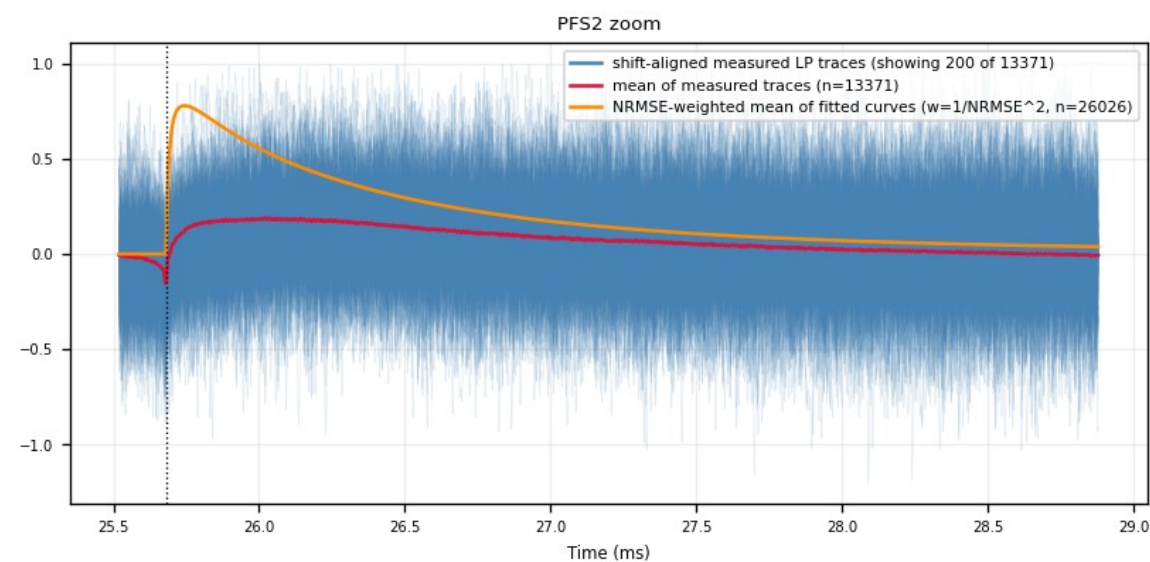
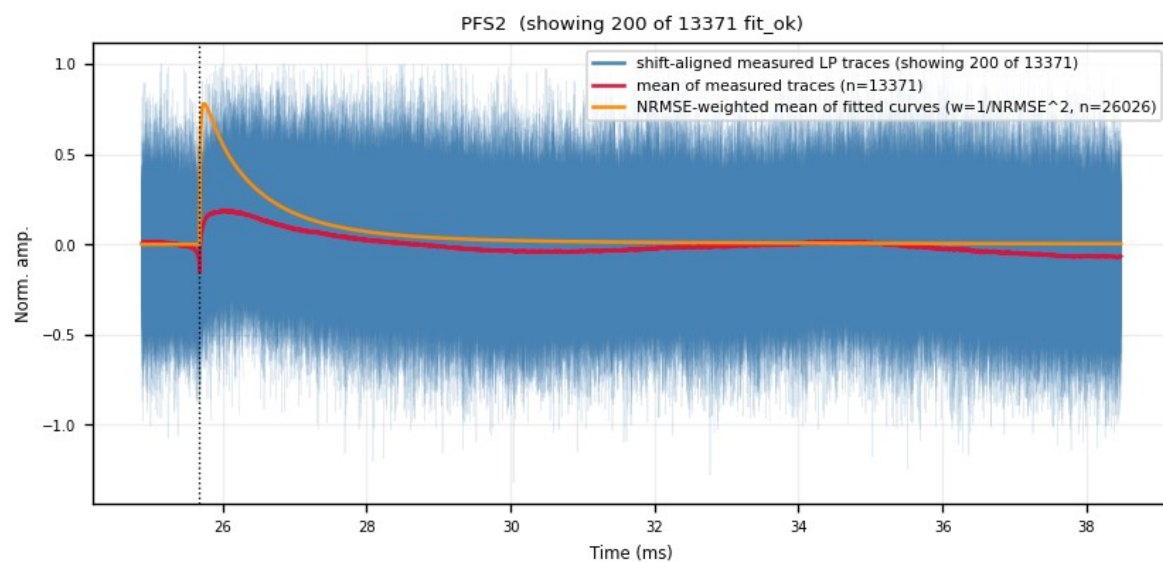
Blue = measured LP traces shifted by (fitted pretrigger – 16050); red = mean of all fit\_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



## Z4 · aligned\_overlay — aligned measured traces (6/6)

figure: zip4\_lp\_aligned\_overlay.png — channel panels 11–11 of 11, top to bottom

Blue = measured LP traces shifted by (fitted pretrigger – 16050); red = mean of all fit\_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.

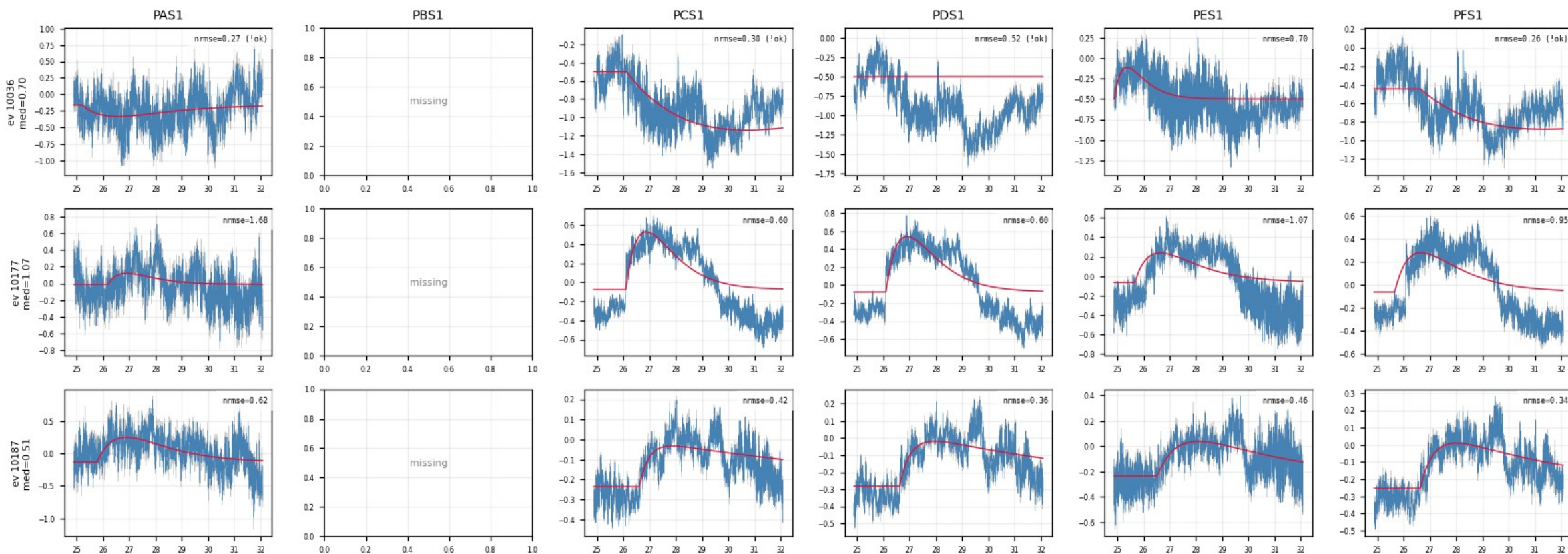




# Z4 · slow\_rise\_events — NRMSE-rejected events (1/10)

figure: zip4\_slow\_rise\_events.png — event rows 1–3 of 15, left half of the channels

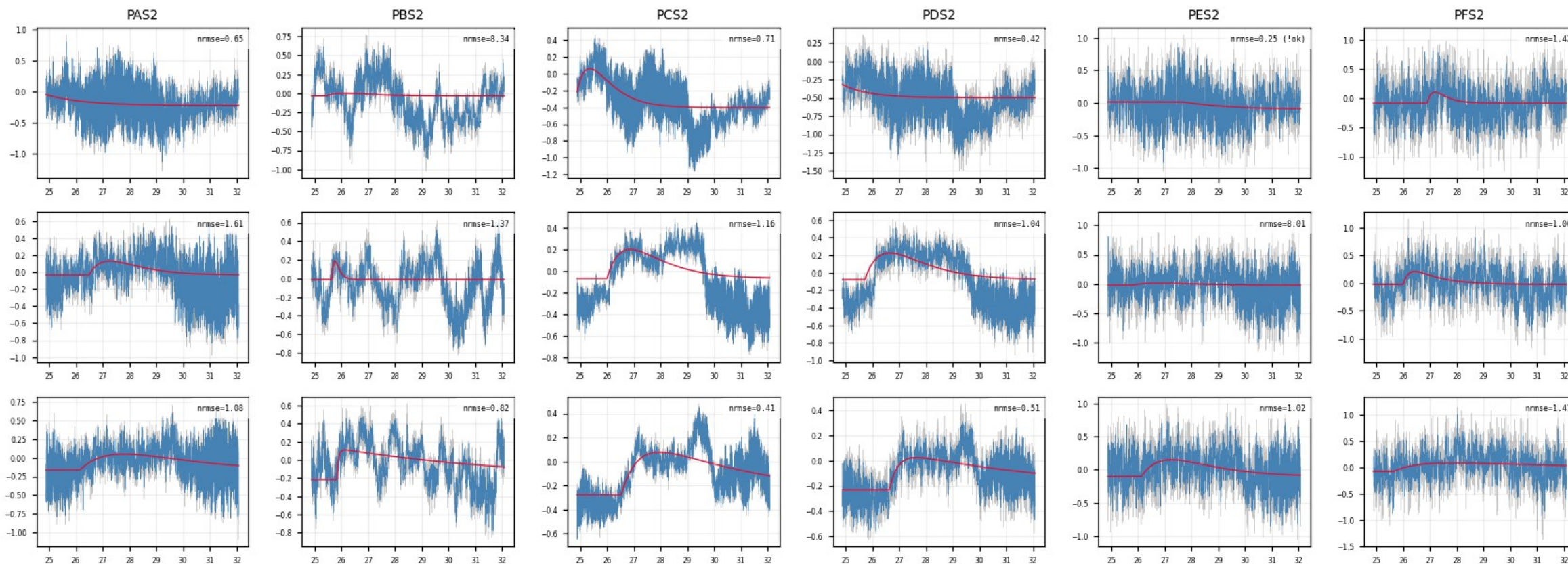
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel — the rejected population is noise triggers, not physics.



# Z4 · slow\_rise\_events — NRMSE-rejected events (2/10)

figure: zip4\_slow\_rise\_events.png — event rows 1–3 of 15, right half of the channels

Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel — the rejected population is noise triggers, not physics.

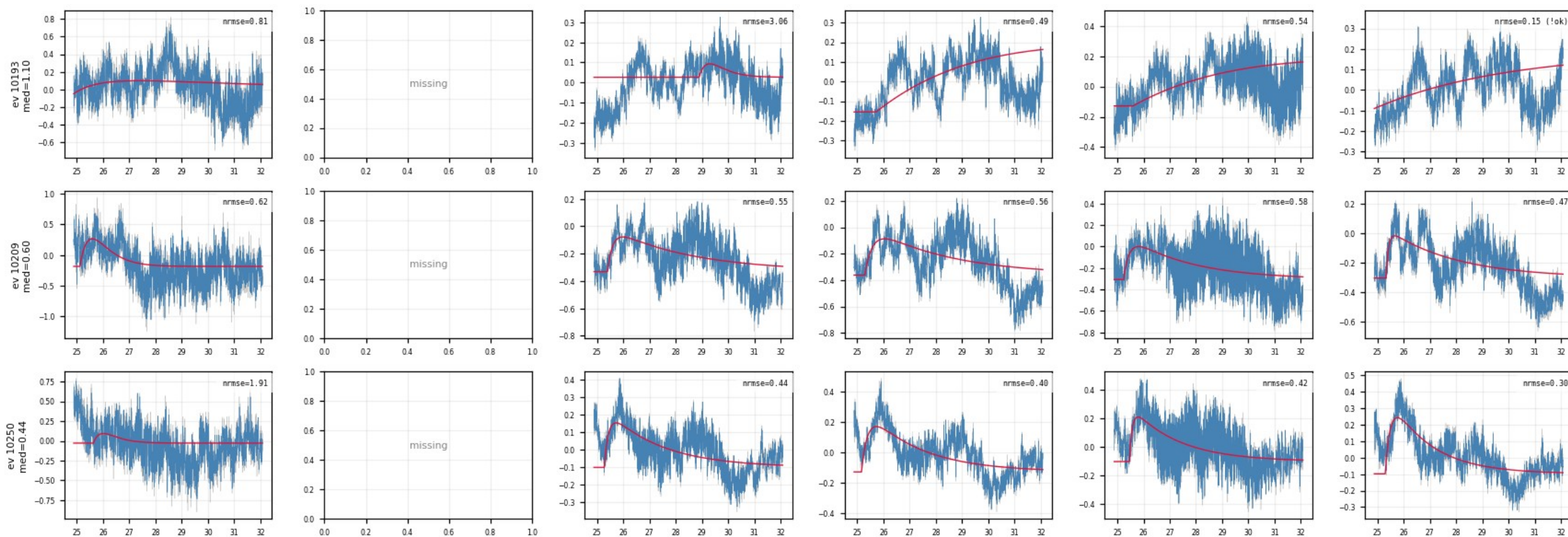




# Z4 · slow\_rise\_events — NRMSE-rejected events (3/10)

figure: zip4\_slow\_rise\_events.png — event rows 4–6 of 15, left half of the channels

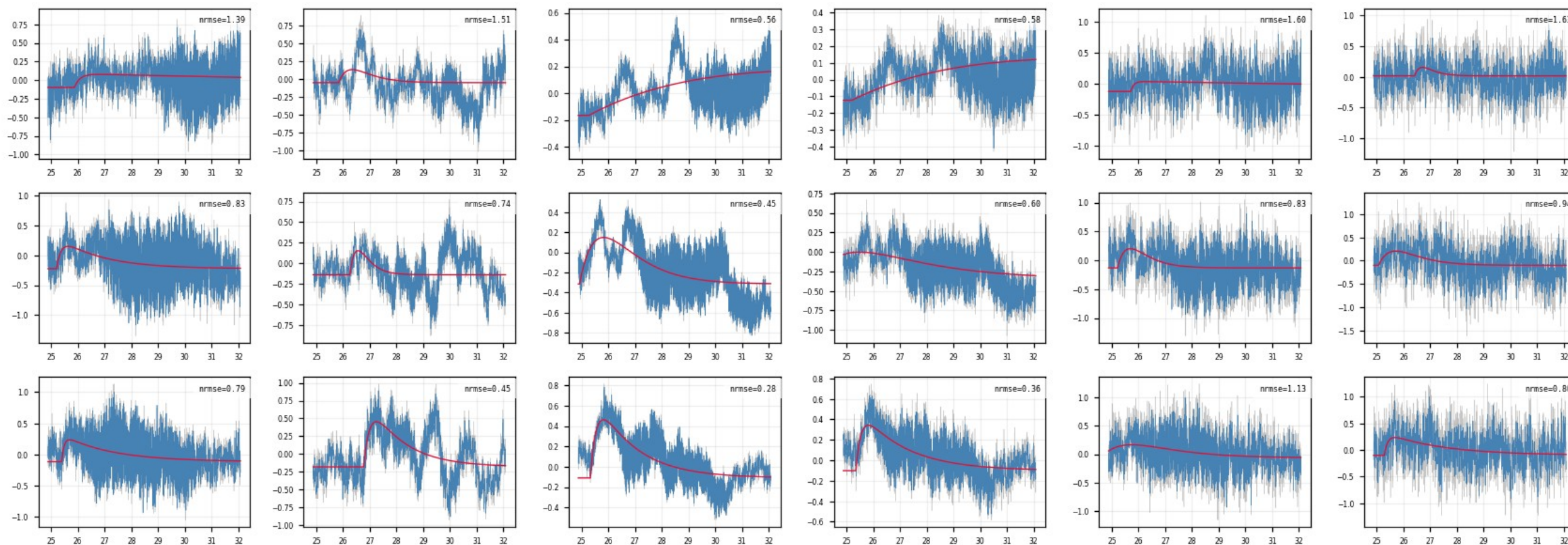
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel — the rejected population is noise triggers, not physics.



# Z4 · slow\_rise\_events — NRMSE-rejected events (4/10)

figure: zip4\_slow\_rise\_events.png — event rows 4–6 of 15, right half of the channels

Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel — the rejected population is noise triggers, not physics.

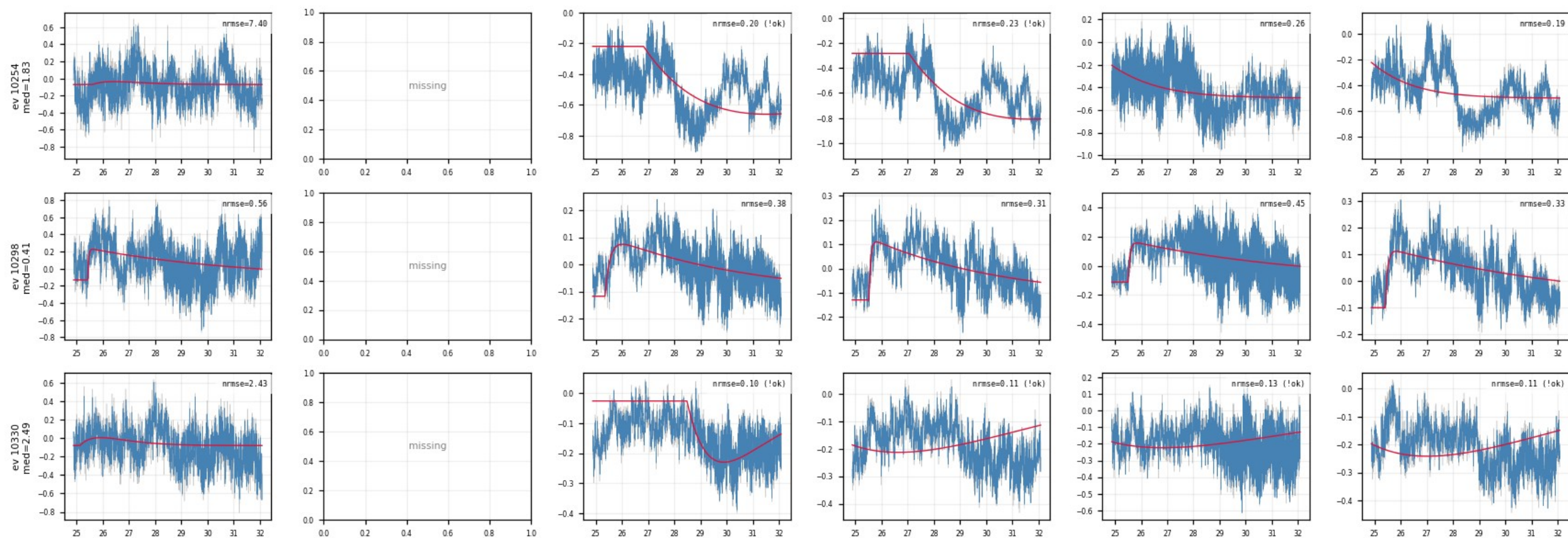




# Z4 · slow\_rise\_events — NRMSE-rejected events (5/10)

figure: zip4\_slow\_rise\_events.png — event rows 7–9 of 15, left half of the channels

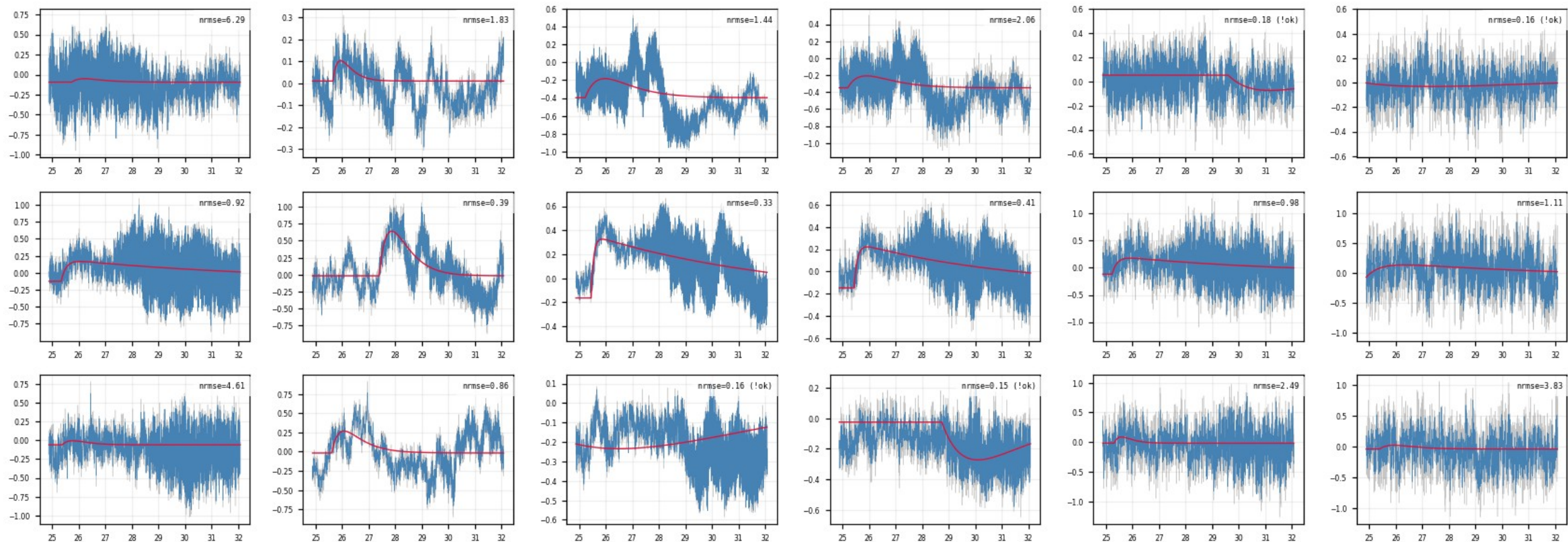
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel — the rejected population is noise triggers, not physics.



# Z4 · slow\_rise\_events — NRMSE-rejected events (6/10)

figure: zip4\_slow\_rise\_events.png — event rows 7–9 of 15, right half of the channels

Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel — the rejected population is noise triggers, not physics.

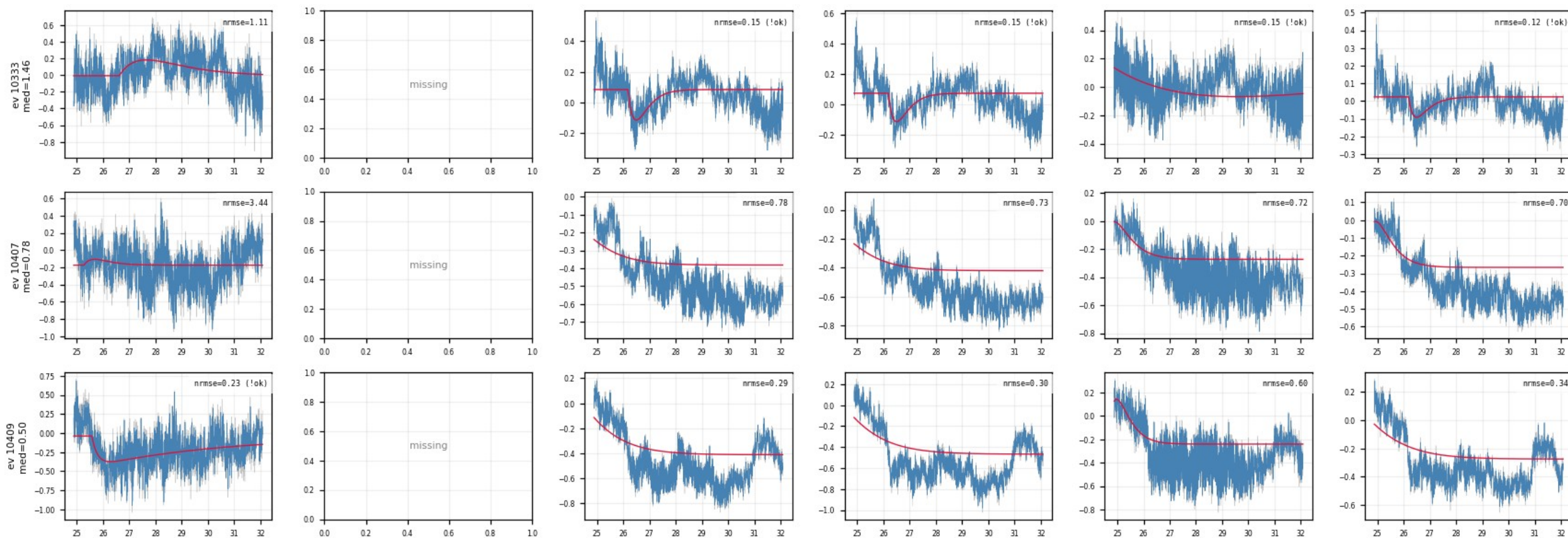




# Z4 · slow\_rise\_events — NRMSE-rejected events (7/10)

figure: zip4\_slow\_rise\_events.png — event rows 10–12 of 15, left half of the channels

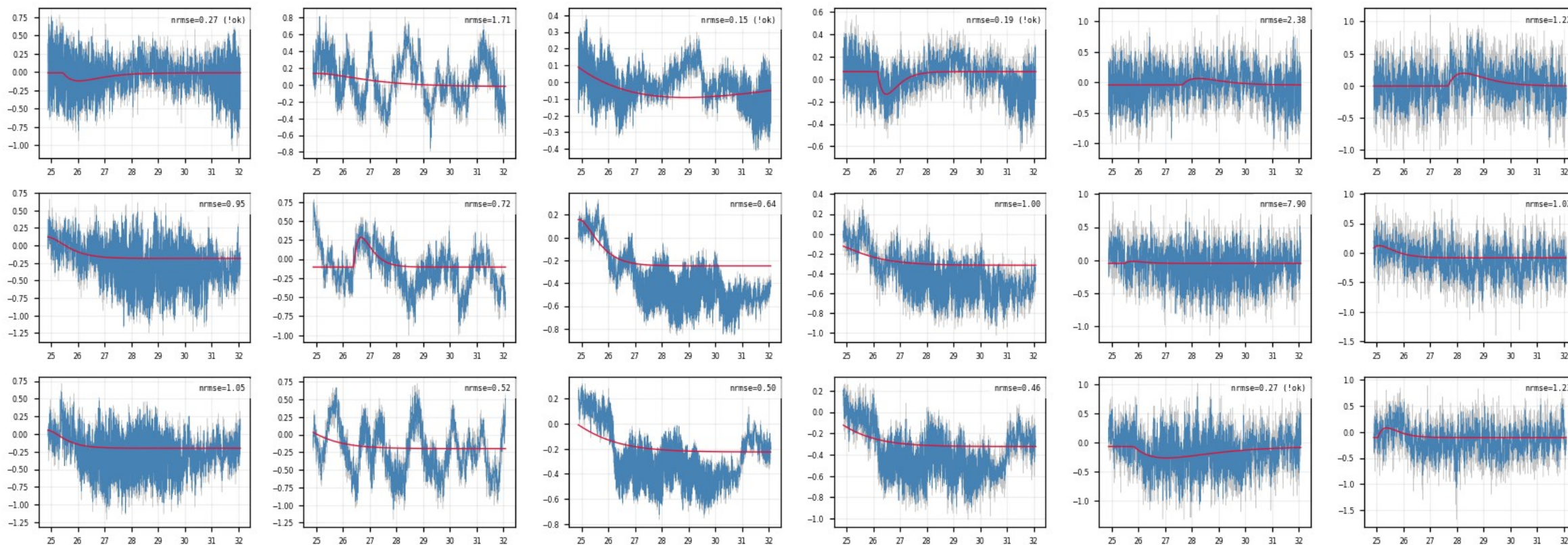
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel — the rejected population is noise triggers, not physics.



# Z4 · slow\_rise\_events — NRMSE-rejected events (8/10)

figure: zip4\_slow\_rise\_events.png — event rows 10–12 of 15, right half of the channels

Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel — the rejected population is noise triggers, not physics.

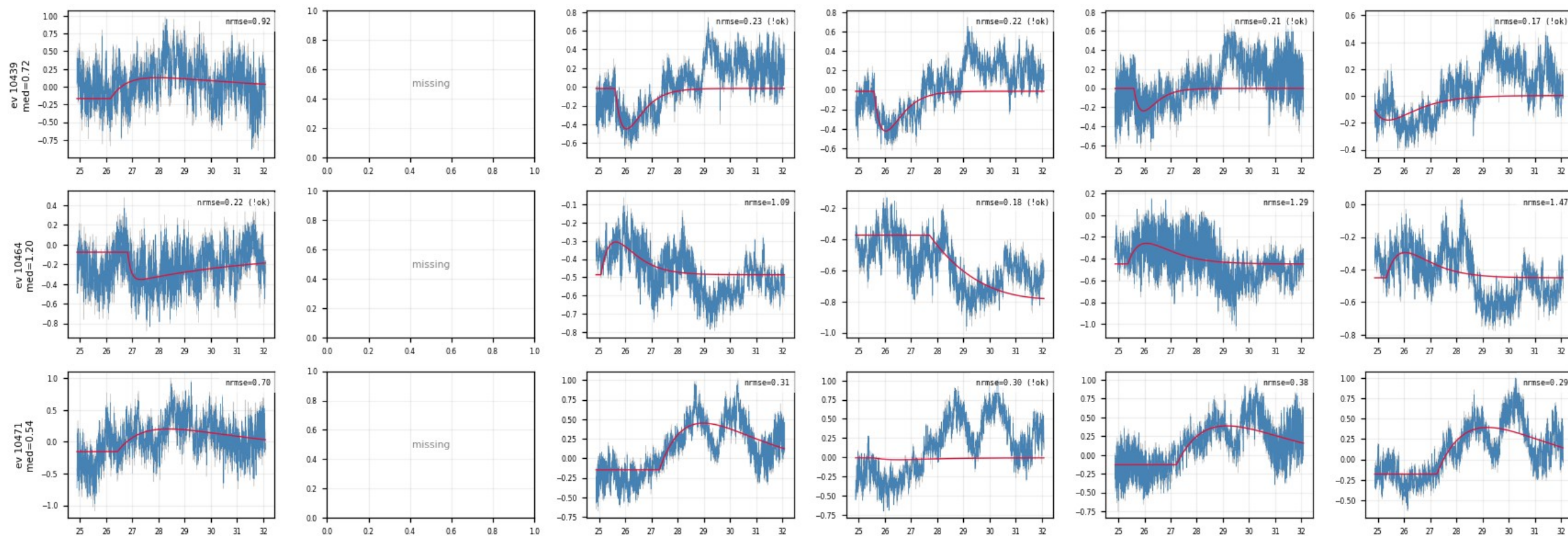




# Z4 · slow\_rise\_events — NRMSE-rejected events (9/10)

figure: zip4\_slow\_rise\_events.png — event rows 13–15 of 15, left half of the channels

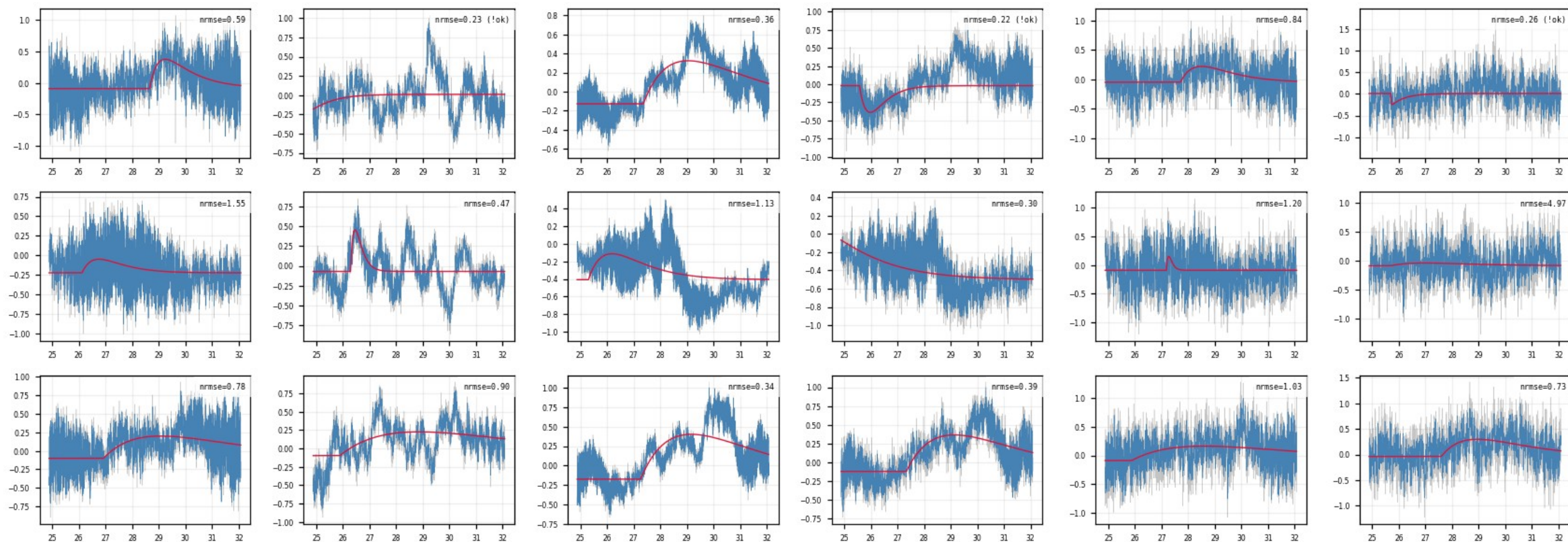
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel — the rejected population is noise triggers, not physics.



# Z4 · slow\_rise\_events — NRMSE-rejected events (10/10)

figure: zip4\_slow\_rise\_events.png — event rows 13–15 of 15, right half of the channels

Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel — the rejected population is noise triggers, not physics.

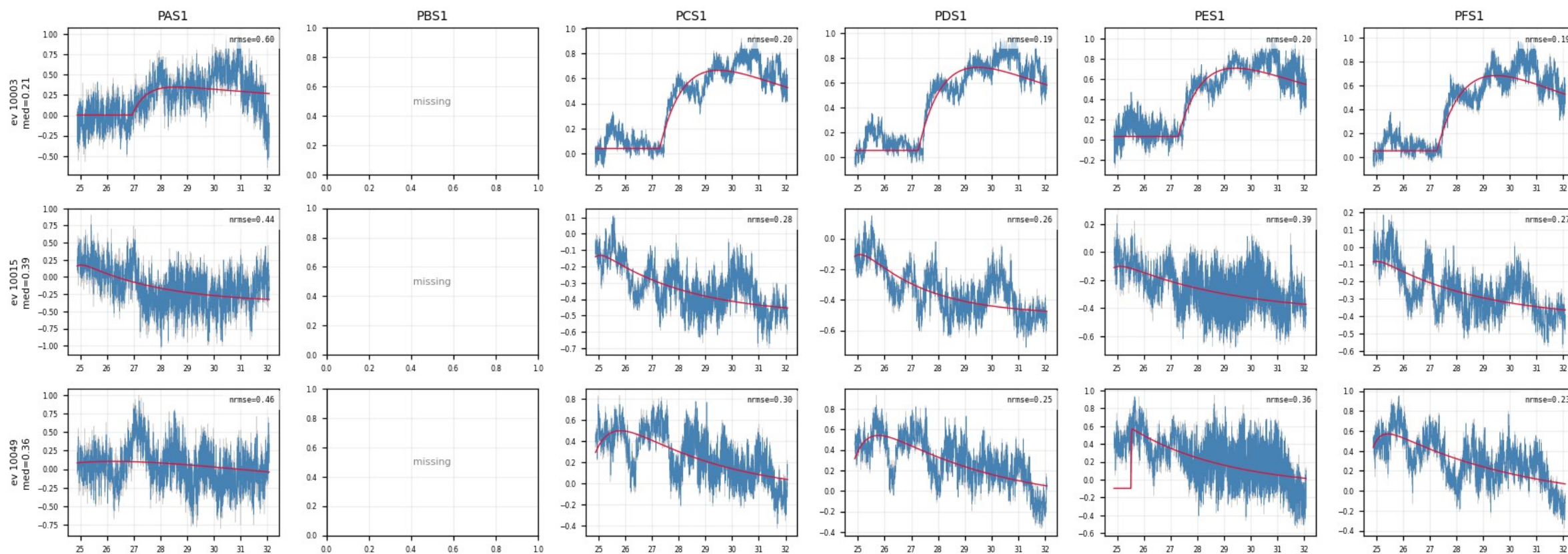




# Z4 · shadow\_events — well-fit slow-rise events (1/10)

figure: zip4\_shadow\_events.png — event rows 1–3 of 15, left half of the channels

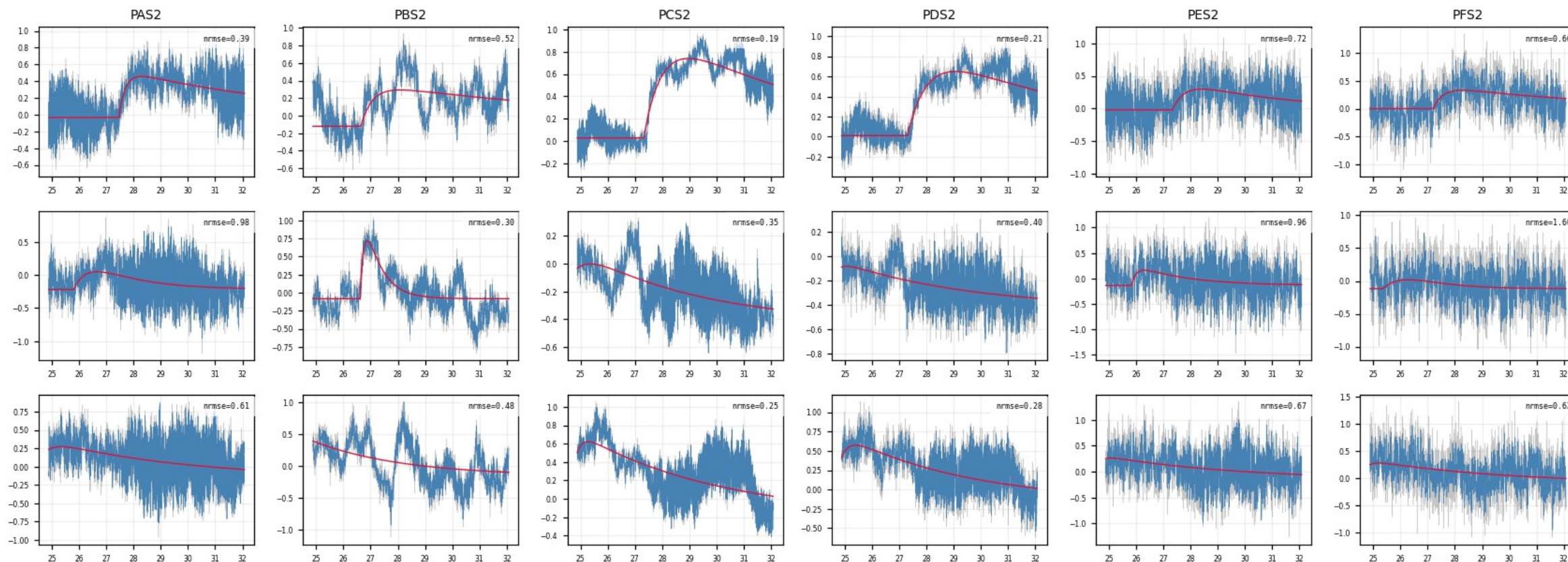
Median NRMSE  $\leq 0.4$  AND median  $\tau_{\text{rise}} > 0.2$  ms: genuinely slow, well-fit pulses — kept, and exactly what the NxM multi-templates are for.



# Z4 · shadow\_events — well-fit slow-rise events (2/10)

figure: zip4\_shadow\_events.png — event rows 1–3 of 15, right half of the channels

Median NRMSE  $\leq 0.4$  AND median  $\tau_{\text{rise}} > 0.2$  ms: genuinely slow, well-fit pulses — kept, and exactly what the NxM multi-templates are for.

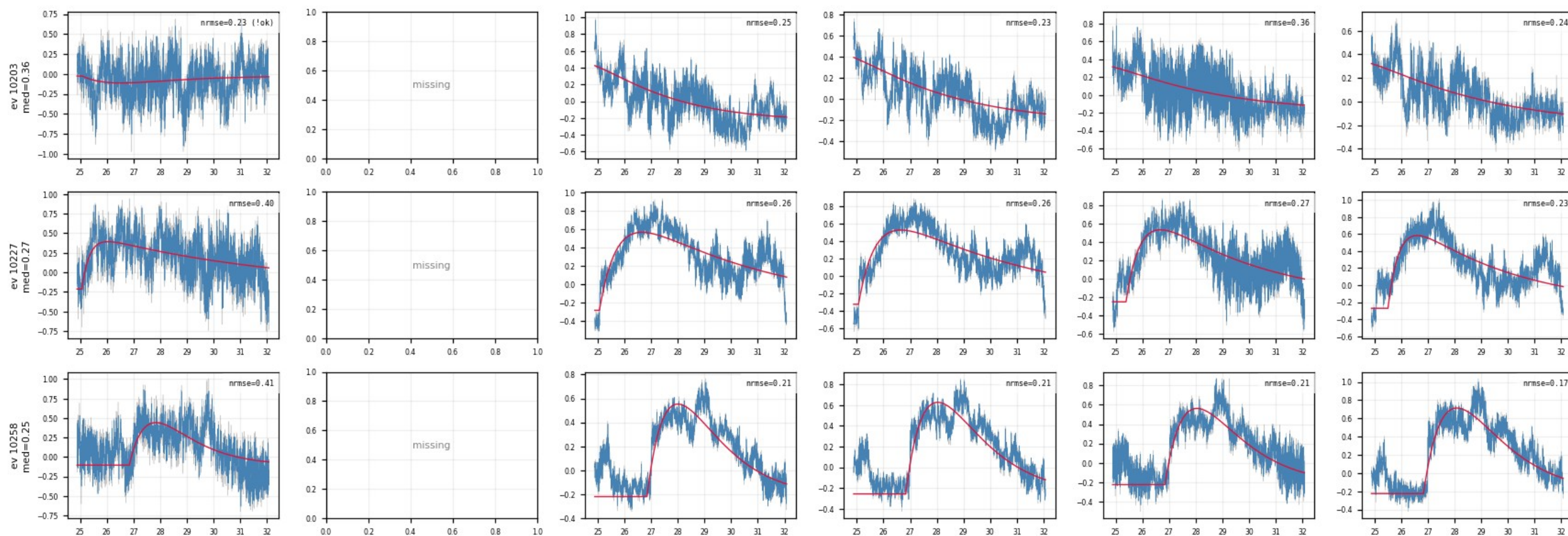




# Z4 · shadow\_events — well-fit slow-rise events (3/10)

figure: zip4\_shadow\_events.png — event rows 4–6 of 15, left half of the channels

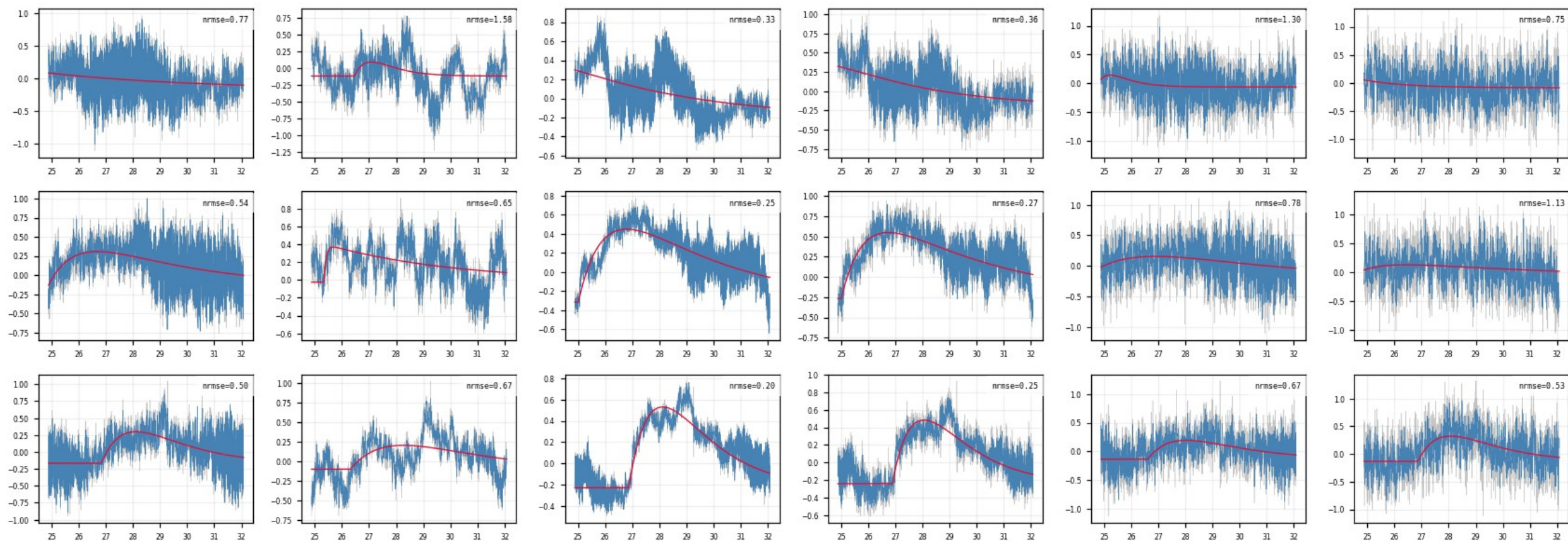
Median NRMSE  $\leq 0.4$  AND median  $\tau_{\text{rise}} > 0.2$  ms: genuinely slow, well-fit pulses — kept, and exactly what the NxM multi-templates are for.



# Z4 · shadow\_events — well-fit slow-rise events (4/10)

figure: zip4\_shadow\_events.png — event rows 4–6 of 15, right half of the channels

Median NRMSE  $\leq 0.4$  AND median  $\tau_{\text{rise}} > 0.2$  ms: genuinely slow, well-fit pulses — kept, and exactly what the NxM multi-templates are for.

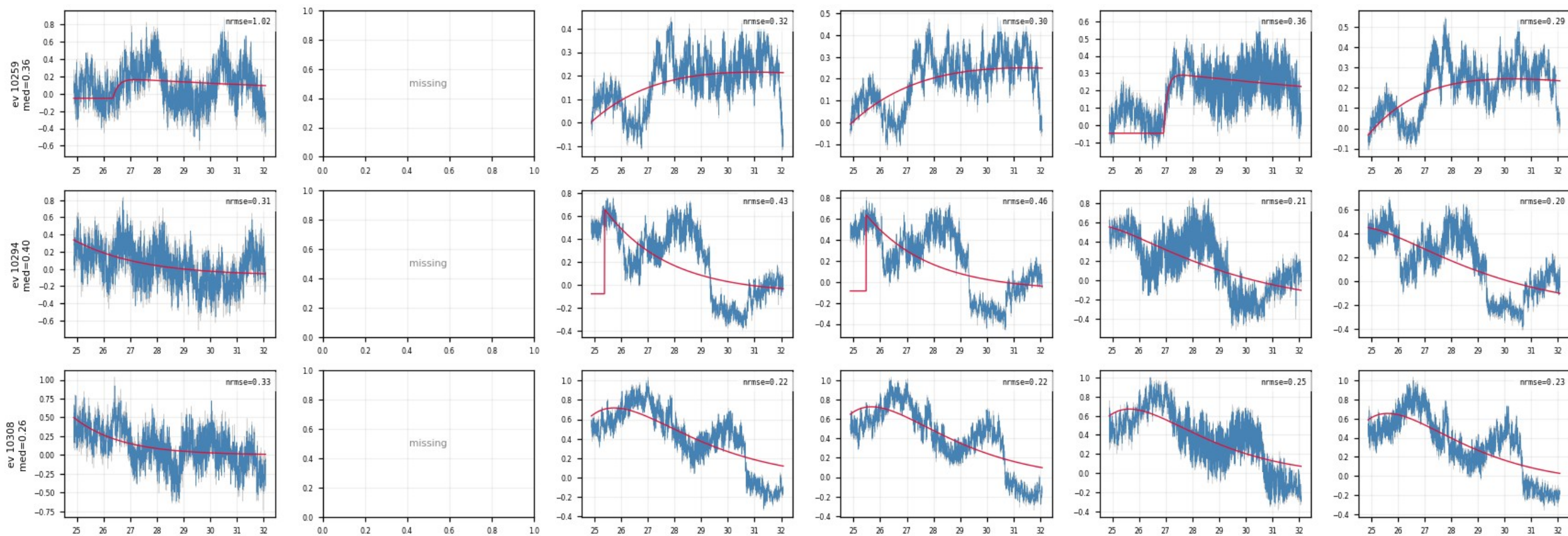




# Z4 · shadow\_events — well-fit slow-rise events (5/10)

figure: zip4\_shadow\_events.png — event rows 7–9 of 15, left half of the channels

Median NRMSE  $\leq 0.4$  AND median  $\tau_{\text{rise}} > 0.2$  ms: genuinely slow, well-fit pulses — kept, and exactly what the NxM multi-templates are for.

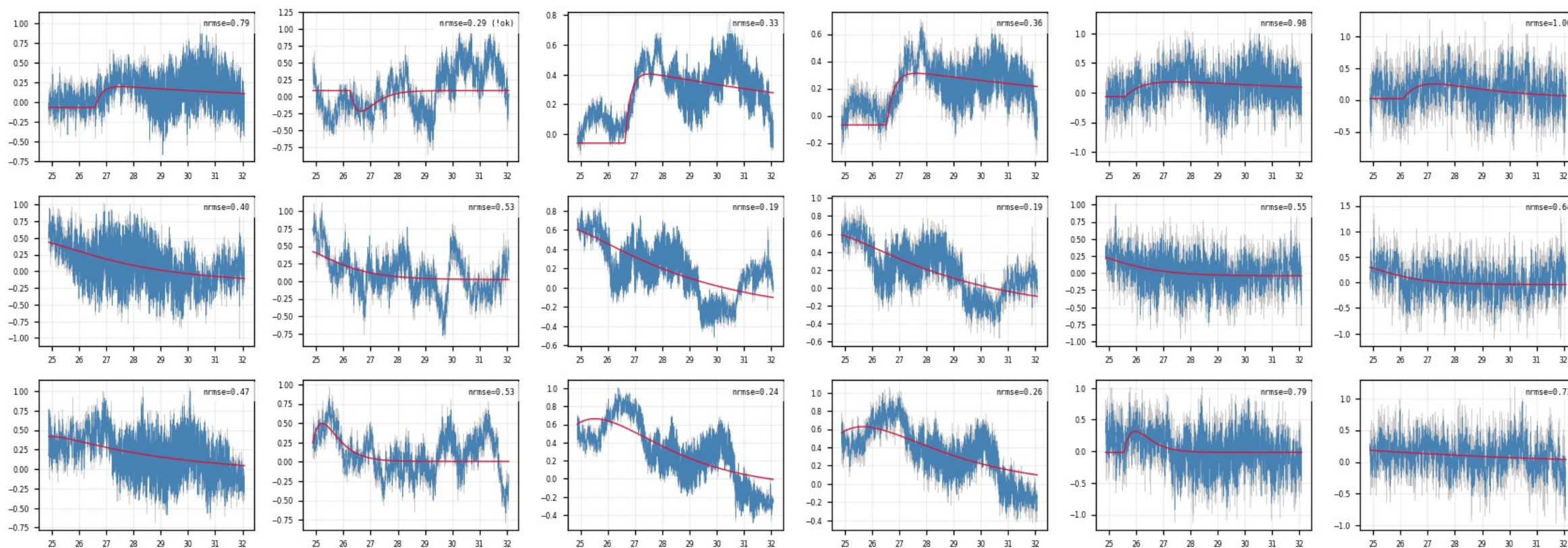




# Z4 · shadow\_events — well-fit slow-rise events (6/10)

figure: zip4\_shadow\_events.png — event rows 7–9 of 15, right half of the channels

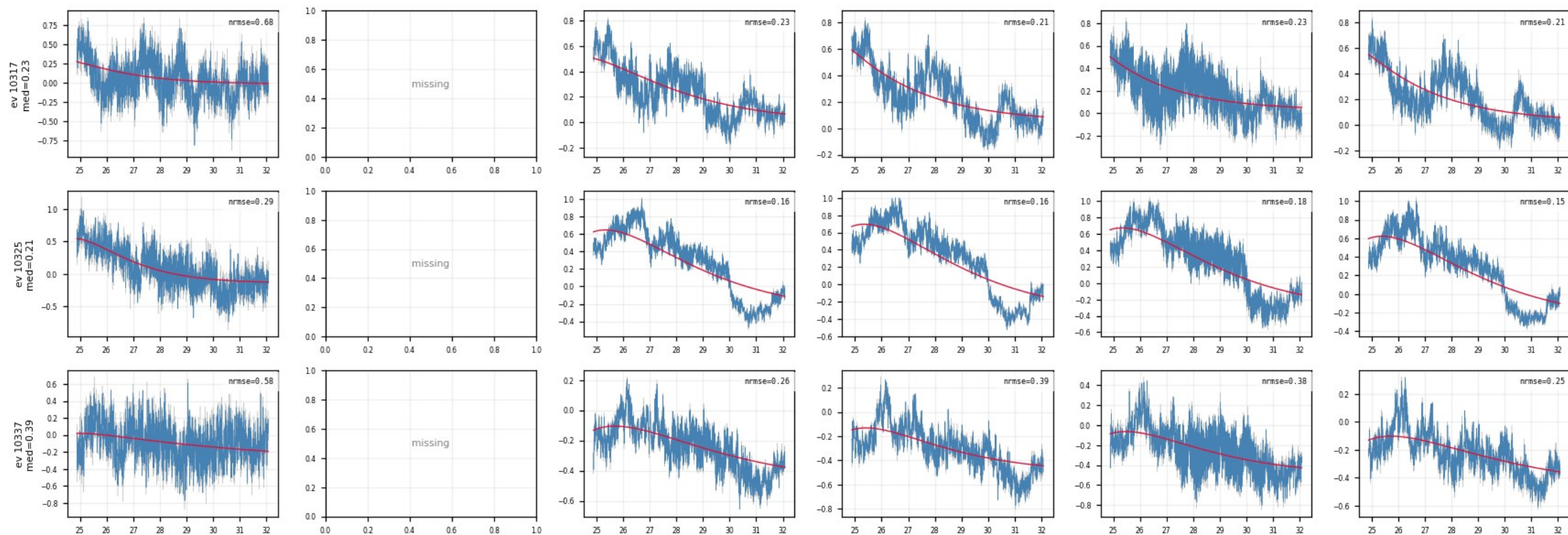
Median NRMSE  $\leq 0.4$  AND median  $\tau_{\text{rise}} > 0.2$  ms: genuinely slow, well-fit pulses — kept, and exactly what the NxM multi-templates are for.



# Z4 · shadow\_events — well-fit slow-rise events (7/10)

figure: zip4\_shadow\_events.png — event rows 10–12 of 15, left half of the channels

Median NRMSE  $\leq 0.4$  AND median  $\tau_{\text{rise}} > 0.2$  ms: genuinely slow, well-fit pulses — kept, and exactly what the NxM multi-templates are for.

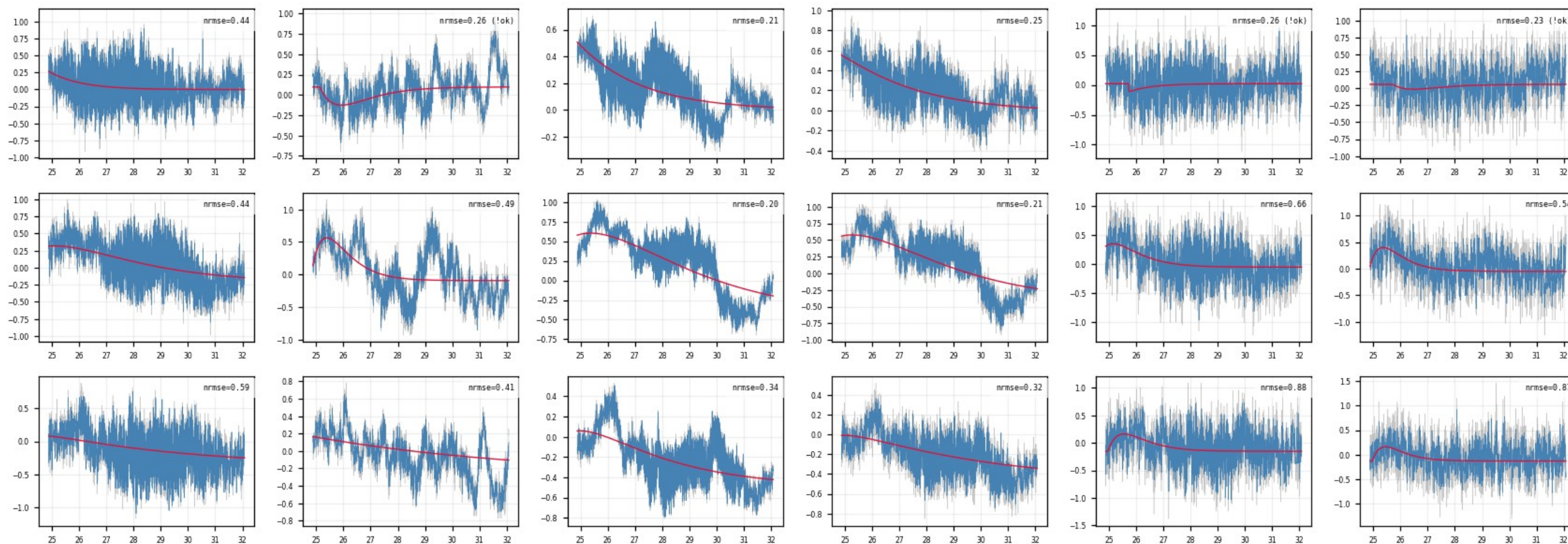




# Z4 · shadow\_events — well-fit slow-rise events (8/10)

figure: zip4\_shadow\_events.png — event rows 10–12 of 15, right half of the channels

Median NRMSE  $\leq 0.4$  AND median  $\tau_{\text{rise}} > 0.2$  ms: genuinely slow, well-fit pulses — kept, and exactly what the NxM multi-templates are for.

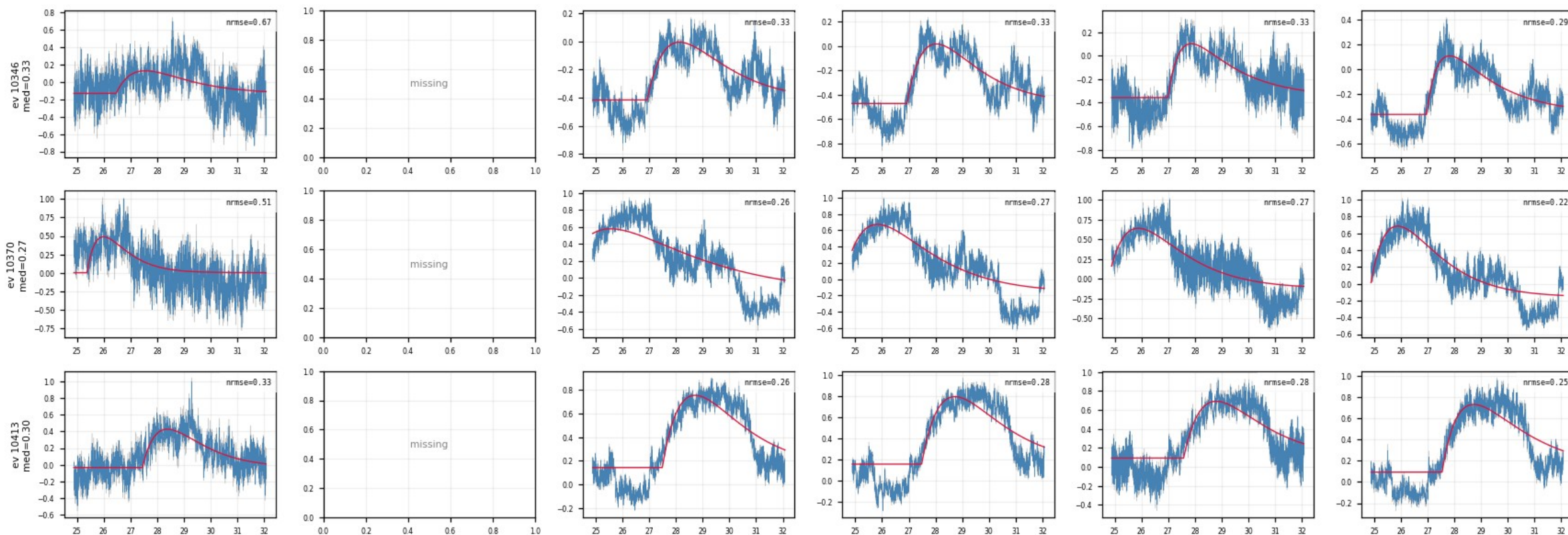




# Z4 · shadow\_events — well-fit slow-rise events (9/10)

figure: zip4\_shadow\_events.png — event rows 13–15 of 15, left half of the channels

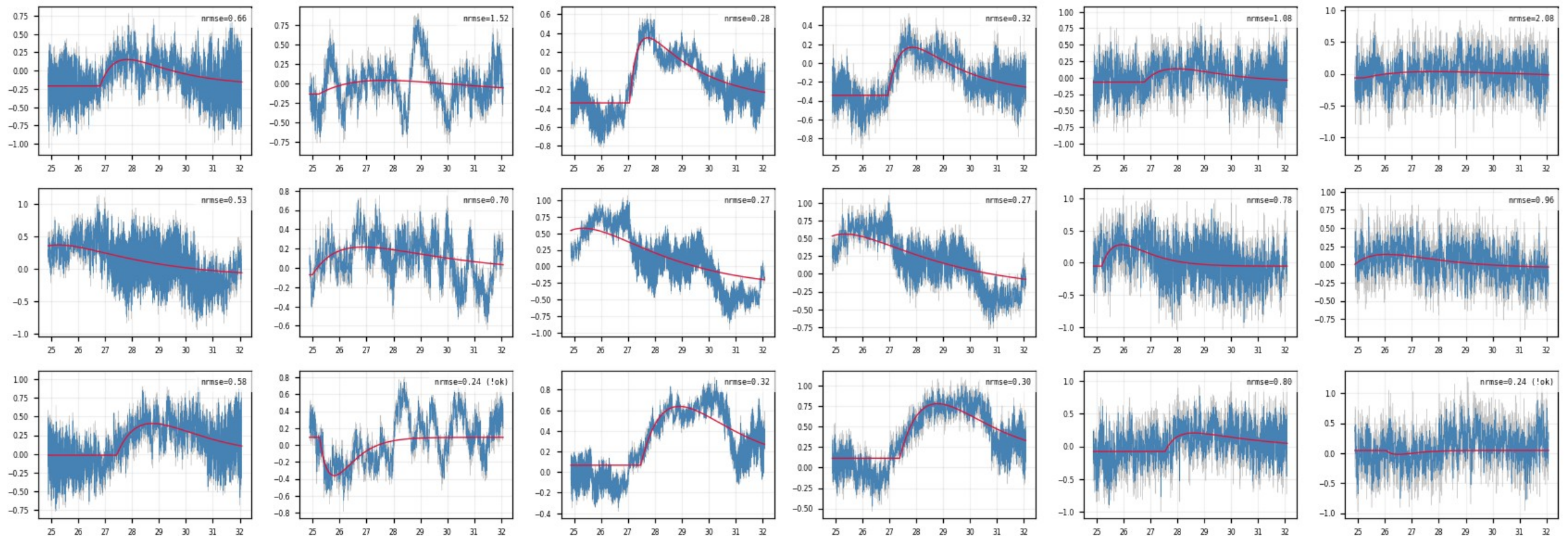
Median NRMSE  $\leq 0.4$  AND median  $\tau_{\text{rise}} > 0.2$  ms: genuinely slow, well-fit pulses — kept, and exactly what the NxM multi-templates are for.



# Z4 · shadow\_events — well-fit slow-rise events (10/10)

figure: zip4\_shadow\_events.png — event rows 13–15 of 15, right half of the channels

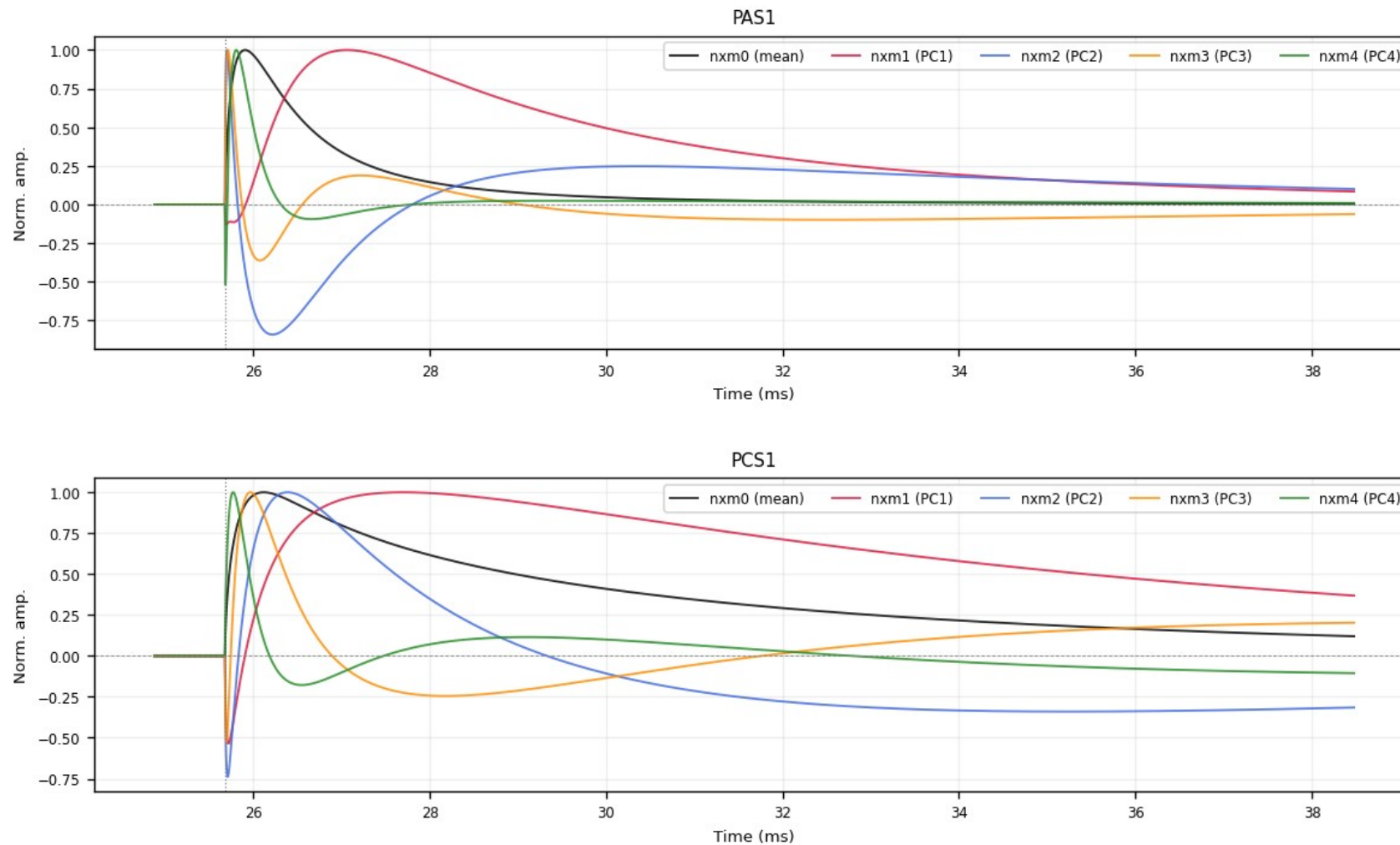
Median NRMSE  $\leq 0.4$  AND median  $\tau_{\text{rise}} > 0.2$  ms: genuinely slow, well-fit pulses — kept, and exactly what the NxM multi-templates are for.



## Z4 · deliverables/nxm — PCA templates nxm0–4 (1/6)

figure: zip4\_pca\_templates.png — channel panels 1–2 of 11, top to bottom

nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; population =  $\text{fit\_ok} + \text{NRMSE} \leq 0.4 + \tau_{\text{rise}} \leq 0.3 \text{ ms}$ .

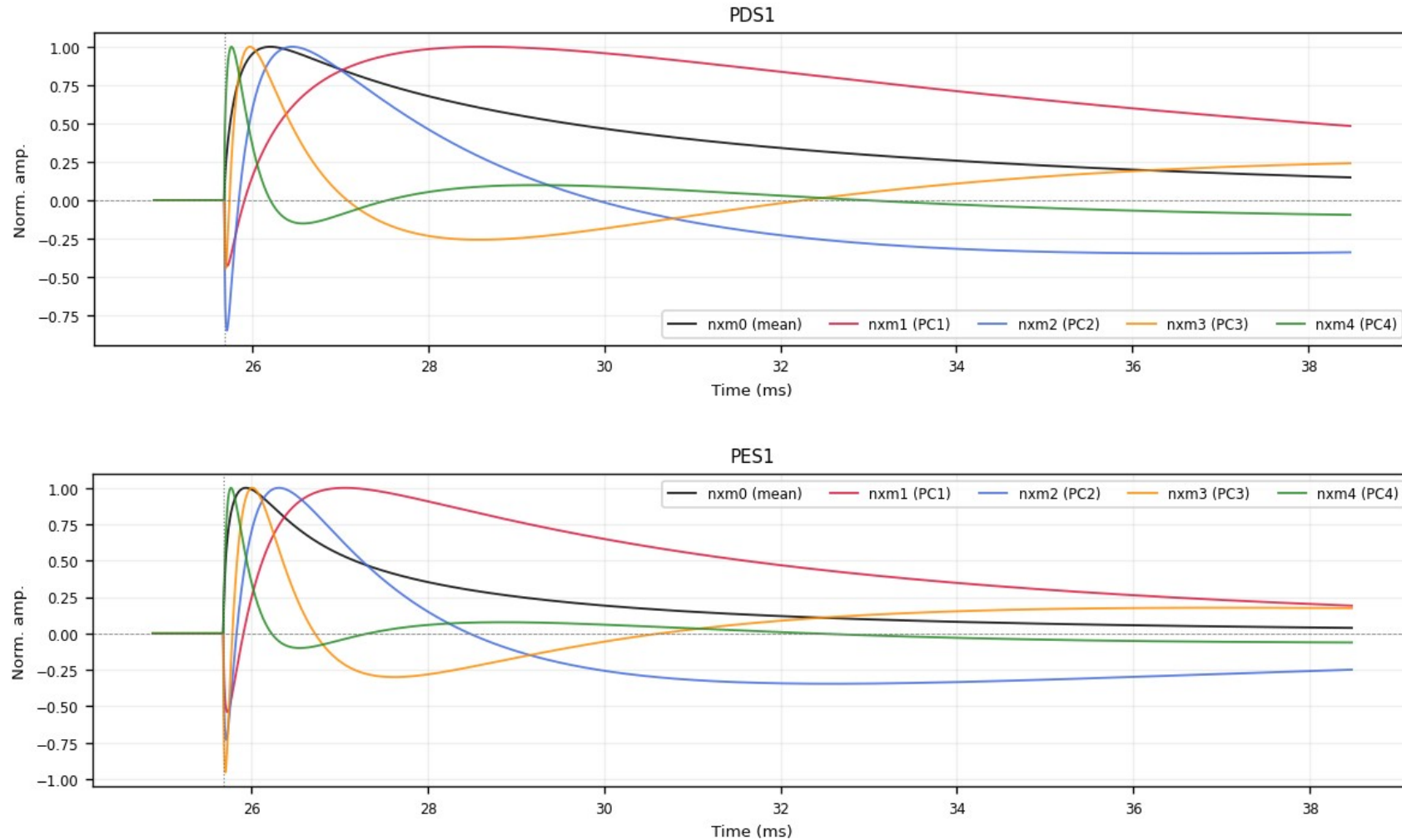




## Z4 · deliverables/nxm — PCA templates nxm0–4 (2/6)

figure: zip4\_pca\_templates.png — channel panels 3–4 of 11, top to bottom

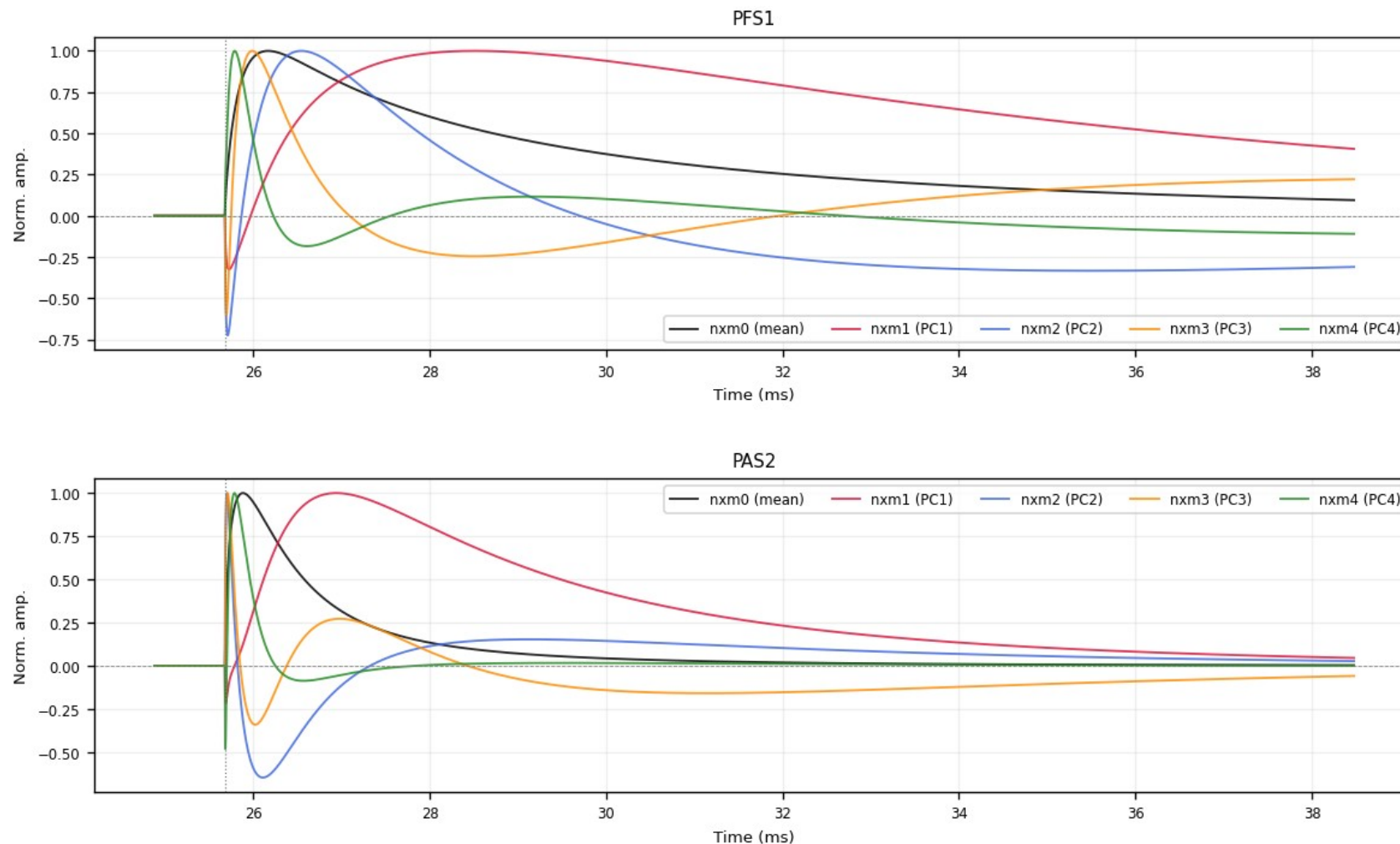
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; population =  $\text{fit\_ok} + \text{NRMSE} \leq 0.4 + \tau_{\text{rise}} \leq 0.3 \text{ ms}$ .



## Z4 · deliverables/nxm — PCA templates nxm0–4 (3/6)

figure: zip4\_pca\_templates.png — channel panels 5–6 of 11, top to bottom

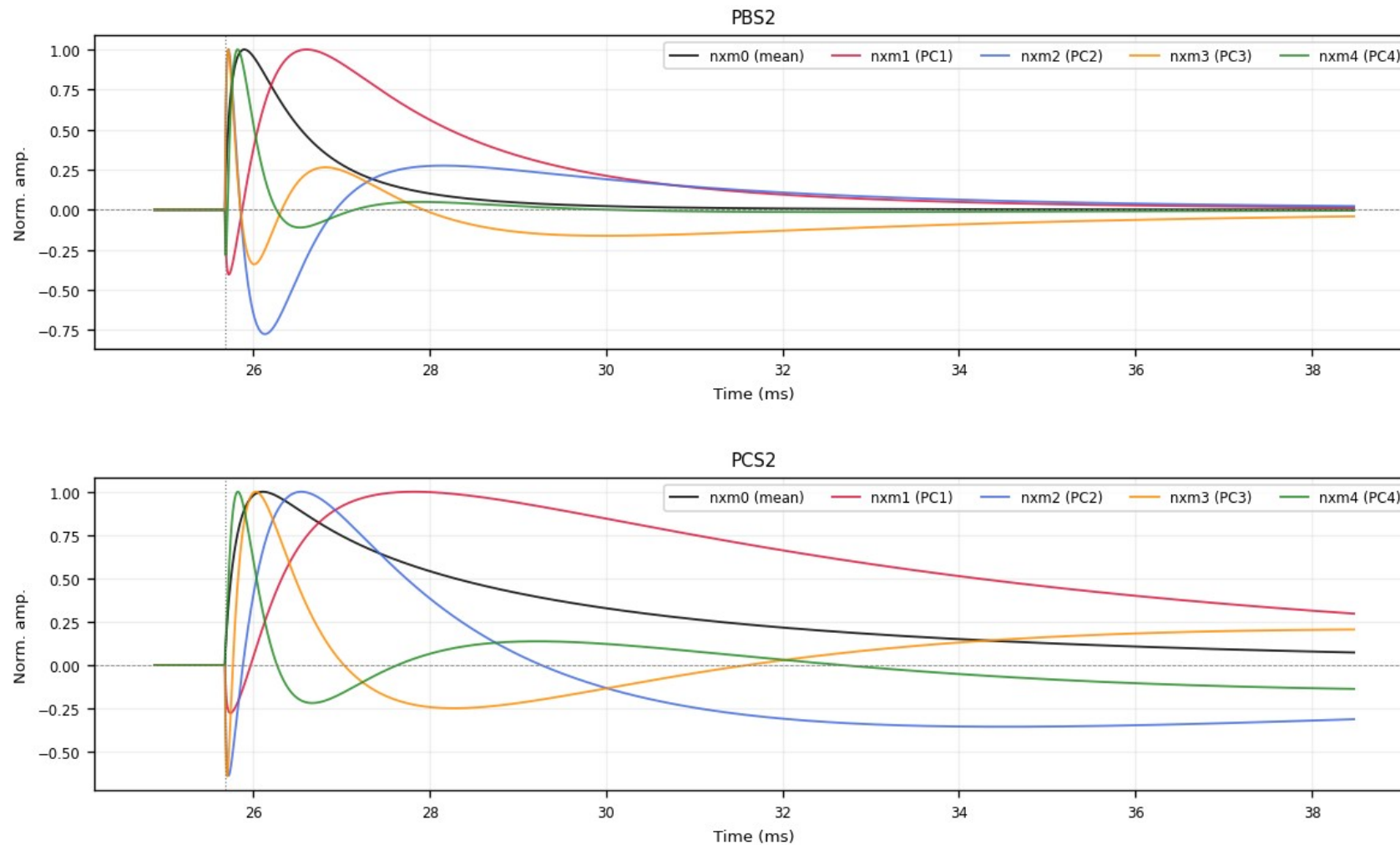
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; population =  $\text{fit\_ok} + \text{NRMSE} \leq 0.4 + \tau_{\text{rise}} \leq 0.3 \text{ ms}$ .



## Z4 · deliverables/nxm — PCA templates nxm0–4 (4/6)

figure: zip4\_pca\_templates.png — channel panels 7–8 of 11, top to bottom

nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; population =  $\text{fit\_ok} + \text{NRMSE} \leq 0.4 + \tau_{\text{rise}} \leq 0.3 \text{ ms}$ .

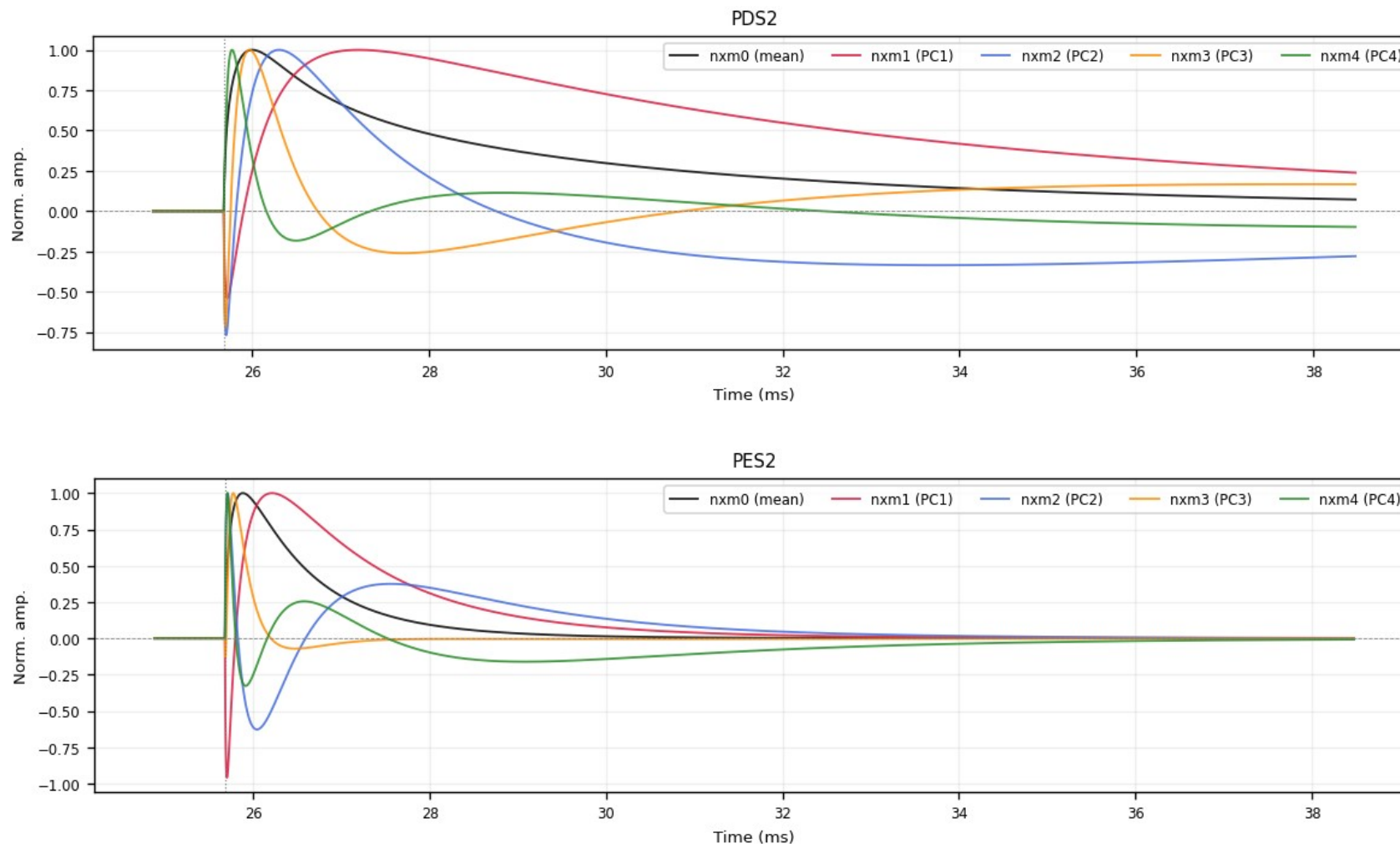




## Z4 · deliverables/nxm — PCA templates nxm0–4 (5/6)

figure: zip4\_pca\_templates.png — channel panels 9–10 of 11, top to bottom

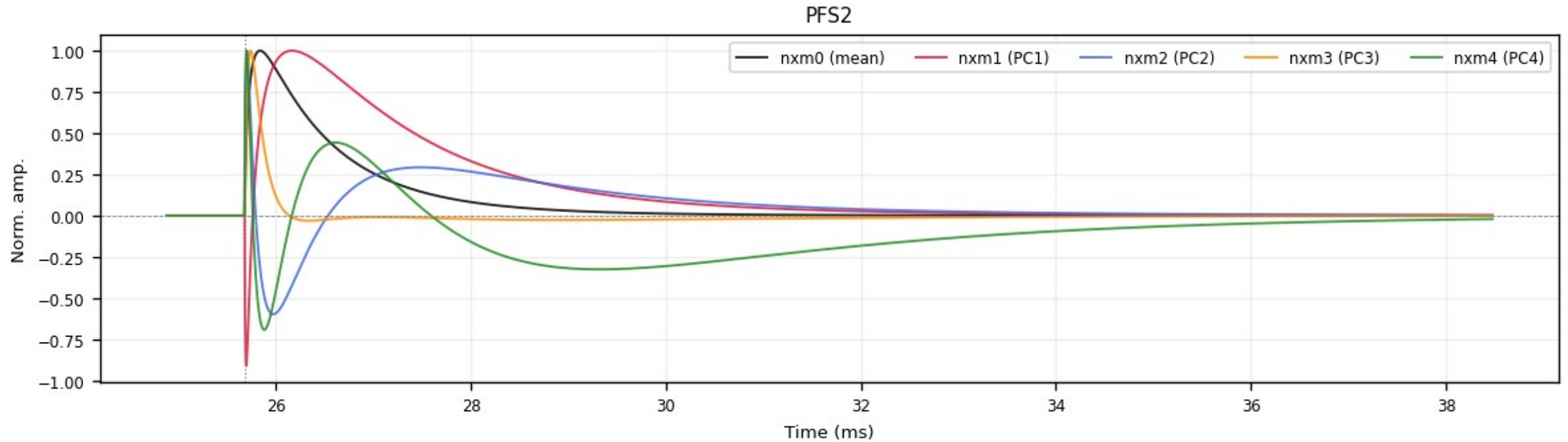
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; population =  $\text{fit\_ok} + \text{NRMSE} \leq 0.4 + \tau_{\text{rise}} \leq 0.3 \text{ ms}$ .



## Z4 · deliverables/nxm — PCA templates nxm0–4 (6/6)

figure: zip4\_pca\_templates.png — channel panels 11–11 of 11, top to bottom

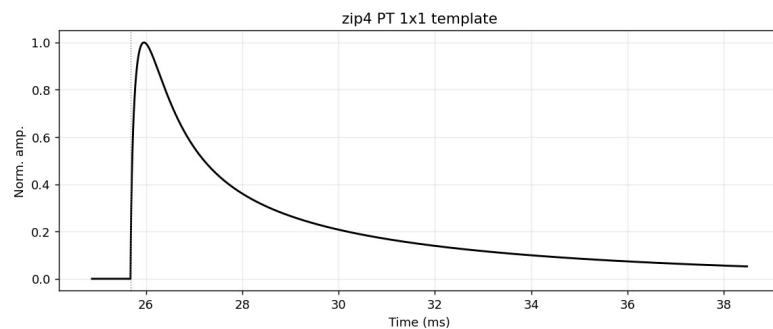
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; population = fit\_ok + NRMSE  $\leq$  0.4 +  $\tau_{\text{rise}} \leq$  0.3 ms.



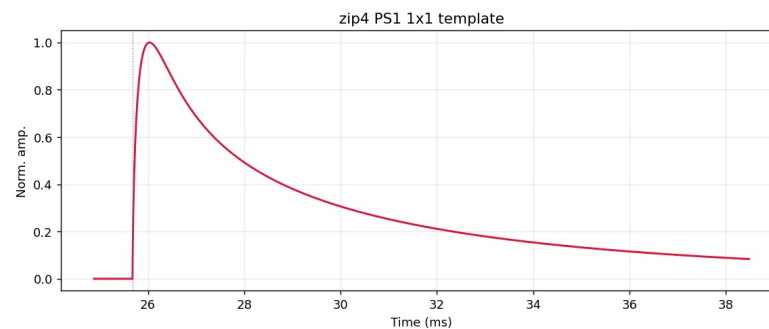
# Z4 · 1x1 summed templates PT / PS1 / PS2

figures: deliverables/1x1/plots/{PT,PS1,PS2}/zip4\_\*.png

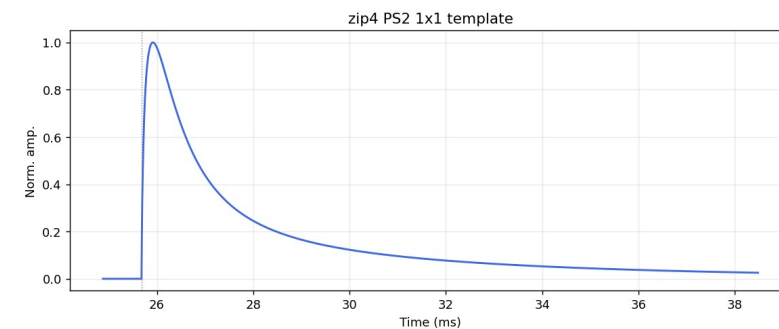
Single-template (1x1) summed curves: peak-normalized average of the per-channel nxm0 templates — PT = all channels, PS1 / PS2 = side-1 / side-2 only.



Z4 PT



Z4 PS1



Z4 PS2